MASTER OF COMMERCE

COMMERCE

(MCOM)

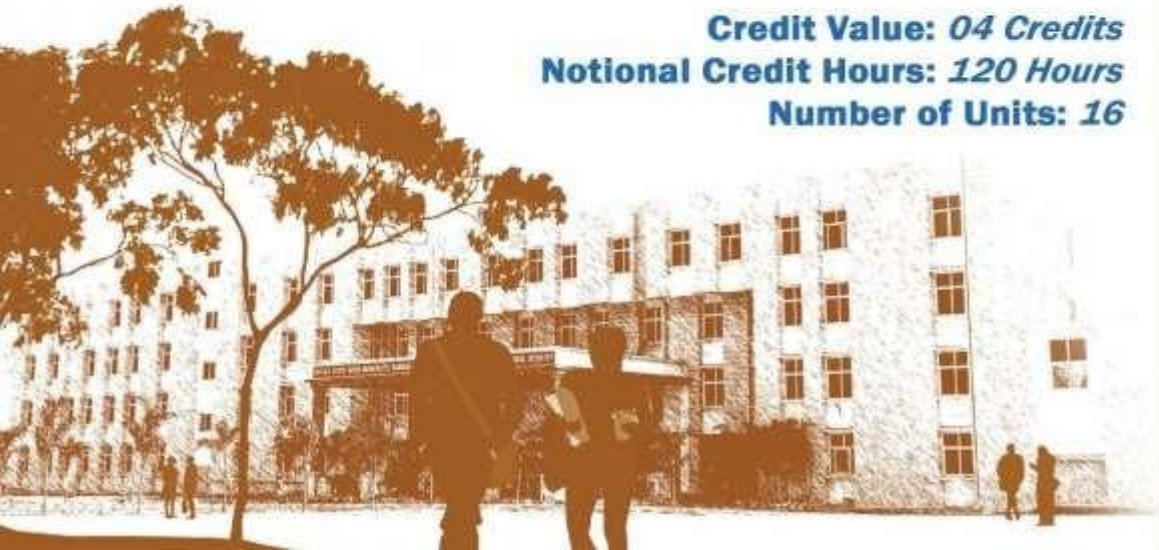
SEMESTER-I

SUBJECT SPECIFIC CORE COURSE

PCO-105:

MANAGERIAL ACCOUNTING

Credit Value: *04 Credits*
Notional Credit Hours: *120 Hours*
Number of Units: *16*



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ODISHA STATE OPEN UNIVERSITY, SAMBALPUR

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ଓଡ଼ିଶା ରାଜ୍ୟ ମୁକ୍ତ ବିଶ୍ୱବିଦ୍ୟାଳୟ, ସମ୍ବଲପୁର, ଓଡ଼ିଶା
Odisha State Open University, Sambalpur, Odisha
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SLM Structure

Name of the Programme	MASTER OF COMMERCE (COMMERCE)
Credit Value	80 Credits
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First Semester

Course Code	**PCO-105**
Course Title	Managerial Accounting

Credit Value	04 Credits
Notional Credit Hours	120 Hours

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ODISHA STATE OPEN UNIVERSITY, SAMBALPUR

Programme Name: MASTER OF COMMERCE (COMMERCE)

Course Name: Managerial Accounting **Course Code:** PCO-105

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PREFACE

Managerial Accounting serves as a vital tool for informed decision-making by providing relevant financial insights to managers. In a rapidly evolving business environment, the need for accurate cost control, performance analysis, and strategic financial planning is more important than ever. This book is designed to equip M.Com. students with a clear and comprehensive understanding of managerial accounting concepts, techniques, and applications.

The content begins with the fundamentals of **Management Accounting**, along with its objectives, nature, scope, and relationship with **Financial**, **Cost**, and **International Accounting**. It further discusses **Financial Statement Analysis**, **Ratio Analysis**, and their practical significance in business decision-making.

Key areas such as **Cost Control**, **Cost Reduction**, **Cost Accounting Methods**, **Marginal Costing**, **Absorption Costing**, **Activity-Based Costing**, and **Cost-Volume-Profit Analysis** are covered in detail. Special focus is given to **Standard Costing**, **Variance Analysis**, **Budgetary Control**, and **Responsibility Accounting**, offering tools for performance management and operational control.

The book concludes with contemporary topics like **Transfer Pricing**, addressing pricing methods, implementation rules, and associated challenges in decentralized organizations. This text aims to build both conceptual clarity and practical skills, enabling students to analyze, plan, and control financial resources effectively in real business scenarios.

Additionally, the book emphasizes the importance of integrating accounting data with managerial functions such as planning, organizing, and controlling. It highlights how accounting information supports forecasting, resource optimization, and competitive strategy formulation. Through real-world applications and illustrations, it enhances students' ability to apply theoretical knowledge to business problems. This comprehensive approach prepares learners to contribute effectively to organizational decision-making and financial management in a globalized economy.

UNIT-1: INTRODUCTION TO MANAGEMENT ACCOUNTING

STRUCTURE

1.1 Introduction

1.2 Management accounting as an area of accounting

1.3 Objective, Nature and Scope of Financial accounting

1.4 Objective, Nature and Scope of Cost accounting

1.5 Objective, Nature and Scope of Management accounting

1.6 Objective, Nature and Scope of International accounting

1.7 Limitations of Management accounting

1.8 Change in Management accounting

1.9 Let's sum up

1.10 Review Questions

LEARNING OUTCOME

1. Explain the concept of management accounting and its distinct role in the broader field of accounting.
2. Understand the specific objectives, characteristics, and areas of application for financial accounting, cost accounting, and management accounting.
3. Identify the key functions and constraints associated with management accounting.
4. Recognize the dynamic forces that are reshaping management accounting practices.
5. Compare and contrast the scope and function of management accounting with financial and cost accounting.
6. Apply the principles of management accounting in real-world decision-making scenarios.

Bloom's Level of Learning:

The provided outline primarily corresponds to the **Remembering** and **Understanding** levels of Bloom's Taxonomy of educational objectives. It emphasizes the identification

and recall of fundamental concepts, definitions, and classifications pertaining to various branches of accounting, including financial, cost, management, and international accounting. Furthermore, it requires comprehension of the objectives, nature, and scope associated with these areas, along with the ability to recognize the limitations of management accounting and the changes it has undergone over time. These learning activities focus on acquiring foundational knowledge and developing a clear understanding of accounting principles and practices, thereby situating them within the lower cognitive levels of Bloom's hierarchical framework.

1.1 INTRODUCTION

Management Accounting, in simple terms, is designed to meet the informational needs of managers. It begins where Financial Accounting concludes, focusing on supporting managerial functions at all levels. By providing essential information, Management Accounting enables managers to operate the business efficiently, formulate policies, exercise control, and make informed decisions. This field of accounting also plays a critical role in evaluating performance, managing costs, and determining costs for financial reporting purposes. Essentially, Management Accounting involves collecting data from both internal and external sources, then analyzing, processing, interpreting, and communicating this information. The objective is to aid management in planning, decision-making, and organizational control.

To clarify the concept further, definitions can serve as a helpful tool. The Chartered Institute of Management Accountants (CIMA), UK, defines Management Accounting as: "An integral part of management that focuses on identifying, presenting, and interpreting information for purposes such as:

1. Formulating strategies,
2. Planning and controlling activities,
3. Making decisions,
4. Optimizing resource utilization,
5. Disclosing information to shareholders and external stakeholders,
6. Sharing information with employees, and
7. Safeguarding assets."

Another concise definition by Robert Anthony describes Management Accounting as:

"The branch of accounting concerned with information that is useful for management."

1.2 MANAGEMENT ACCOUNTING AS AN AREA OF ACCOUNTING

Accounting is a broad field that encompasses several specialized branches, each with its own unique focus and purpose. Financial accounting primarily deals with the preparation of financial statements for external stakeholders, providing an overview of a company's financial performance and position. Management accounting, on the other hand, focuses on providing internal decision-makers with data and analyses to help in planning, controlling, and making informed business decisions. Cost accounting is concerned with determining and analyzing the costs of producing goods or services, which helps in cost control and pricing decisions. Auditing involves the independent examination and verification of financial records to ensure accuracy, transparency, and compliance with regulations. Tax accounting focuses on preparing tax returns and ensuring compliance with tax laws, while also assisting in tax planning. Forensic accounting is used to investigate financial discrepancies and fraud, often for legal purposes. Governmental accounting is employed by public sector entities to manage finances, ensure accountability, and comply with public regulations. International accounting addresses the complexities of financial reporting and transactions in different countries, considering various international standards and currencies. Lastly, environmental accounting focuses on tracking and reporting the financial impact of a company's environmental activities and ensuring sustainability practices are properly documented. Each branch of accounting serves a distinct role, but they all work together to provide a comprehensive understanding of an organization's financial standing and operations.

Management Accounting focuses on collecting data from internal and external sources to create a comprehensive pool of information. This data pool serves as a critical resource for management, enabling them to plan, monitor, and make informed decisions. By addressing the broader operational challenges of the organization, Management Accounting provides valuable support to management.

To achieve its objectives, Management Accounting incorporates information from various disciplines, including financial accounting, economics, statistics, and operational research. The integration of these diverse sources allows for thorough analysis and informed decision-making.

For instance, consider a business that produces two products and seeks to expand one of them. A management accountant would analyze the financial data and other relevant information to decide whether to expand one product, expand both equally, or discontinue one of the products. This process ensures that decisions are based on accurate and comprehensive insights.

1.3 OBJECTIVE, NATURE AND SCOPE OF FINANCIAL ACCOUNTING

Financial Accounting plays a crucial role in providing essential financial information about a business. Its primary focus is on offering information to external stakeholders, such as shareholders, creditors, bankers, government agencies, and tax authorities. Financial Accounting is commonly defined as: *"The art of recording, classifying, and summarizing transactions in a significant manner and in monetary terms, along with interpreting the results thereof."*

1. **Objectives**

The key objective of Financial Accounting is to deliver information that helps ascertain the profitability and financial health of the business. It aims to determine the profits or losses incurred during a specific period. By systematically recording, summarizing, and classifying transactions, Financial Accounting facilitates the preparation of:

1. A profit and loss account or income and expenditure account, and
2. A balance sheet at the end of the financial period.

2. **Nature of Financial Accounting**

The nature of Financial Accounting is characterized by the following features:

- Double-Entry System: It is grounded in the principles of double-entry bookkeeping.
- Generally Accepted Accounting Principles (GAAP): Governed by established accounting standards and principles.
- Monetary Expression: Transactions are recorded in monetary terms, such as rupees, dollars, etc.
- Historical Data: It deals with past financial transactions.
- Precision and Accuracy: The process is systematic and precise, aiming to

provide accurate monetary information.

- Annual Cycle: It is typically prepared on an annual basis.
- Independent: The financial accounting process functions independently and is standardized for external reporting purposes.

3. Scope of Financial Accounting

The scope of Financial Accounting extends far beyond the business domain. It is widely applied in various sectors of society and professions. In the modern era, Financial Accounting is utilized in:

1. Business Entities: It is an integral part of all types of businesses.
2. Non-Trading Institutions: Schools, colleges, hospitals, and charitable organizations also employ accounting methods to manage their finances.
3. Professional Practices: Professionals like doctors, lawyers, and chartered accountants use tailored accounting methods suitable to their needs.
4. Financial Transactions: Any individual or entity engaged in financial transactions benefits from accounting practices.

The scope of Financial Accounting has significantly evolved over time. With the dynamic nature of the subject, its application has expanded to accommodate socio-economic changes. Continuous research and development in the field have led to new areas of accounting, broadening its scope and relevance in the modern world.

1.4 OBJECTIVE, NATURE AND SCOPE OF COST ACCOUNTING

The terms ‘costing’ and ‘cost accounting’ are frequently used interchangeably. Cost accounting is a branch of accounting dedicated to providing detailed information about the costs associated with producing goods or services. According to the CIMA (UK), cost accounting is defined as: *"The technique and process of determining costs."*

1. Objectives of Cost Accounting

The primary objectives of cost accounting include:

- Determining the cost of products, processes, or services.
- Analyzing, classifying, and recording expenditures related to the cost of products, processes, or services to determine their overall cost.

- Providing timely information to management for decision-making.
- Supporting the preparation of profit and loss accounts and balance sheets periodically.
- Serving as a reference by offering actual data for comparison.
- Facilitating price determination and providing quotations.
- Assisting in budgetary control.
- Supporting cost control and cost reduction initiatives.
- Recording the performance of production units to evaluate efficiency.
- Offering a basis for production planning and minimizing material and store wastage.
- Providing data for various periods and production volumes, which aids in planning and future business expansion.
- Informing decisions such as:
    - Whether to shut down or continue operations.
    - Whether to make or buy a product.
    - Whether to continue using existing plant/machinery or replace them.
- Determining the cost-volume-profit relationship.
- Helping management devise appropriate policies for other critical areas.

2. Nature of Cost Accounting

Cost accounting focuses on determining the costs of products, processes, divisions, departments, or services. It deals primarily with quantitative data expressed in monetary terms. Various techniques, such as standard costing, marginal costing, and budgetary control, are used to ascertain and control costs. In certain industries mandated by the government, cost accounting is a compulsory practice. The system can be independently implemented and serves as a foundation for accounting.

3. Scope of Cost Accounting

While the scope of cost accounting is narrower than that of management accounting, it involves a range of techniques, tools, and procedures for calculating and controlling costs. Its scope can be divided into three primary areas:

a. Cost Ascertainment: The process of calculating and determining the total costs.
b. Cost Records: The maintenance of detailed records of costs for analysis and decision-making purposes.

1.5 OBJECTIVE, NATURE AND SCOPE OF MANAGEMENT ACCOUNTING

Objectives of Management Accounting

The objectives of management accounting are often referred to as its functions, and they play a vital role in organizational success. These objectives include:

1.3.1 **Planning:** The information provided by management accounting helps managers forecast and prepare for long-term, medium-term, and short-term plans. These plans form the foundation for developing appropriate strategies.

1.3.2 **Controlling:** Effective control is crucial in both business operations and daily life. Management accounting supports control through various techniques such as standard costing, budgetary control, control ratios, and internal audits.

1.3.3 **Coordinating:** Coordination is key to achieving success. By implementing a well-structured Management Information System (MIS), management accounting facilitates coordination across various business activities, departments, and units.

1.3.4 **Qualitative Information:** Management accounting offers qualitative data that aids decision-making. This includes information about the quality of goods, customer satisfaction, employee performance, legal judgments, public opinion, and other factors that guide effective management.

1.3.5 **Financial Analysis and Interpretation:** Techniques like ratio analysis, cash flow statements, and trend analysis are used in management accounting to evaluate financial performance and support decision-making.

1.3.6 **Communication:** The MIS of management accounting produces reports for all levels of management, keeping them updated on key developments and helping them make informed decisions to steer the business towards success.

1.3.7 **Decision Making:** Accurate and timely decision-making is vital for organizational success. Management accounting provides tools like marginal costing, differential costing, discounted cash flows, and capital expenditure decisions to guide critical decisions such as product pricing, whether to make or

buy a product, whether to continue or discontinue a product, and decisions regarding repair or replacement of assets.

1.3.8 **Tax Policies:** Well-structured tax policies are possible only when an organization's long-term goals are clearly defined. Management reports are used by tax experts to devise effective tax strategies.

1.3.9 **Nature of Management Accounting:** The nature of management accounting is shaped by its purpose and objectives, which can be summarized as:

1.3.10 **Future-Oriented:** While financial accounting provides historical data, management accounting uses this information to make informed decisions and future plans.

1.3.11 **Enhances Efficiency:** Dynamic concepts like responsibility accounting and **cost centers** help improve efficiency and enhance the performance of personnel.

1.3.12 **Cause-and-Effect Relationship:** Management accounting analyzes both causes and effects when evaluating reasons for profit or loss.

1.3.13 **Flexibility:** Unlike traditional accounting systems, management accounting does not follow rigid rules or formats. The methods and analysis are flexible and tailored to the needs of the users.

1.3.14 **Provides Information, Not Decisions:** Management accounting provides essential information, but the final decision-making process rests with management, who interprets the data.

1.3.15 **Use of Tools and Techniques:** Management accounting utilizes various tools and techniques from statistics and mathematics to present data in a meaningful way.

1.3.16 **Utilizes Accounting Information:** Management accounting leverages financial information to meet the managerial needs of planning, controlling, and decision-making.

1.3.17 **Scope of Management Accounting:** Management accounting has a broad scope, as it integrates aspects of both cost and financial accounting while covering multiple areas such as:

1.3.18 **Financial Accounting:** A well-structured financial accounting system is fundamental to the success of management accounting.

1.3.19 **Cost Accounting:** Management accounting utilizes several cost accounting tools, including marginal costing, standard costing, and budgeting, to support decision-making.

1.3.20 **Cost Control and Procedures:** Effective cost control mechanisms are an essential component of any management accounting system.

1.3.21 **Statistical Tools:** Tools such as tables, graphs, charts, and other statistical methods are used to prepare detailed reports and analysis.

1.3.22 **Reporting to Management:** Management accounting provides a robust reporting system to top-level management for better decision-making.

1.3.23 **Tax Planning:** A sound management accounting system is necessary for effective tax planning and tax management.

1.3.24 **Budgeting and Forecasting:** Budgeting and budgetary control play a crucial role in guiding decision-making processes within the organization.

1.3.25 **Internal Audit:** The internal audit report by management accounting are used to assess the performance of various departments and functions.

1.6 OBJECTIVE, NATURE AND SCOPE OF INTERNATIONAL ACCOUNTING

1. Introduction to International Accounting

International accounting refers to the application of accounting principles and practices across countries, focusing on the management and reporting of financial transactions that involve multiple countries. It deals with the challenges posed by differences in national laws, tax systems, accounting norms, and business practices. As global business operations continue to expand, international accounting provides the necessary tools and frameworks to ensure uniformity, transparency, and comparability in financial reporting. This includes the use of global standards such as International Financial Reporting Standards (IFRS), allowing businesses to comply with regulatory requirements in different regions while ensuring the accuracy and consistency of their financial statements.

2. Objectives of International Accounting

2.1 Standardization of Financial Reporting:

The primary goal of international accounting is to create consistency in financial reporting across different countries, making it easier for investors, regulators, and other stakeholders to compare financial statements.

2.2 Support for Global Business Operations:

International accounting facilitates cross- border transactions, allowing multinational

corporations to effectively manage their financial data and comply with various legal and tax requirements in multiple jurisdictions.

2.3 Enhancing Transparency and Accountability:

By following recognized international standards, companies can provide transparent and reliable financial information, which helps build trust with international investors, creditors, and regulators.

2.4 Promoting Global Investment: Standardized accounting practices make it easier for investors to make decisions based on consistent and comparable financial data from companies operating in various countries.

2.5 Effective Financial Decision Making:

It aids in informed decision-making by providing the necessary data about financial performance and positioning for businesses operating internationally.

3. Nature of International Accounting

The nature of international accounting is shaped by the need to manage financial reporting across different jurisdictions with varying regulations and practices. Key characteristics include:

3.1 Diverse Regulations: International accounting works within a framework of multiple accounting systems, such as IFRS, US GAAP, and local standards, which vary from country to country.

3.2 Currency and Exchange Rates: It involves managing foreign currency transactions, handling the risks of exchange rate fluctuations, and translating foreign financial statements into a common reporting currency.

3.3 Cross-Border Consolidation: International accounting supports the consolidation of financial statements from subsidiaries located in different countries, considering the need to adjust for different local accounting treatments.

3.4 Multinational Taxation: It also addresses the challenges of tax systems across borders, including transfer pricing rules, tax treaties, and the avoidance of double taxation.

3.5 Harmonization of Standards: One of the central aspects of international accounting is the ongoing effort to harmonize accounting practices worldwide, making financial reporting more consistent and comparable.

4. Scope of International Accounting

International accounting encompasses several areas that go beyond traditional domestic accounting:

4.1 Global Financial Reporting Standards: The use of international standards like IFRS ensures that financial statements are comparable across countries, providing a foundation for global businesses.

4.2 Cross-Border Transactions and Trade: It involves the accounting for and reporting of international transactions, including imports, exports, and global investments.

4.3 Foreign Currency Transactions:

International accounting handles the complexities of foreign currency translation, including managing exchange rate risks and adjusting financial statements for fluctuations in currency values.

4.4 Taxation in Multiple Jurisdictions: Companies must navigate complex multinational tax environments, handling issues like tax planning, transfer pricing, and the impact of double taxation agreements.

4.5 Consolidation of Global Financial Statements: International accounting facilitates the consolidation of financial data from subsidiaries and foreign branches, ensuring uniformity across different countries and currencies.

4.6 Global Auditing Standards: Auditors in international accounting must apply global standards and guidelines to ensure that financial statements comply with international regulations.

4.7 Capital Market Considerations: International accounting deals with the impact of global capital markets, including the raising of capital from foreign investors and the reporting of investments in other countries.

4.8 Regulatory Compliance: Companies must adhere to both local regulations and international guidelines, such as those set by the International Accounting Standards Board (IASB), ensuring consistent and compliant financial reporting.

DIFFERENCE BETWEEN FINANCIAL ACCOUNTING, COST ACCOUNTING, AND MANAGEMENT ACCOUNTING

Comparison between Financial Accounting, Cost Accounting, and Management Accounting:

Aspect	**Financial Accounting**	**Cost Accounting**	**Management Accounting**

Purpose	To provide financial information to external parties.	To ascertain and control the cost of producing goods or services.	To aid internal decision-making, planning, and control processes.
Primary Users	External stakeholders like investors, creditors, and regulatory authorities.	Internal management and production teams.	Internal managers and decision-makers.
Scope of Reporting	Focuses on overall financial performance and position.	Concentrates on tracking and controlling costs related to production or service.	Covers a wide range of financial analysis, cost analysis, and performance evaluation for management.
Time Orientation	Historical – deals with past financial transactions.	Both historical and current – focuses on actual and projected costs.	Primarily future-oriented, providing insights for future planning and decision-making.
Legal Requirements	Governed by legal standards (e.g., GAAP, IFRS) and statutory regulations.	Not legally required, but often essential for operational control.	No legal obligation; used primarily for internal purposes.
Financial Statements	Income statement, balance sheet, and cash flow statement.	Cost sheets, cost control reports, and profit analysis.	Budgets, forecasts, variance reports, and management performance evaluations.
Information Type	Quantitative, standardized financial data.	Detailed cost breakdowns, including labor, material, and overhead.	Both qualitative and quantitative data for decision support.

Metho d ology	Follows established accounting standards and principles.	Utilizes costing techniques like standard costing and job order costing.	Employs tools such as ratio analysis, break- even analysis, and budgeting.
Role in Decisio n- Makin g	Provides financial reports for external analysis and regulatory compliance.	Assists in internal pricing, cost control, and efficiency improvements.	Provides actionable insights and data for strategic management decisions.

1.7 LIMITATIONS OF MANAGEMENT ACCOUNTING

Despite its usefulness, management accounting has several limitations that can impact its effectiveness. Some of the key limitations include:

- Dependence on Historical Data: Management accounting often relies on historical data, which may not always reflect current or future conditions. As a result, decisions based on past performance may not be entirely accurate or relevant.
- Quality Depends on the Knowledge of the Accountant: The effectiveness of management accounting is directly influenced by the knowledge and expertise of the management accountant. Inadequate skills or understanding can lead to incorrect interpretations of data and flawed decision-making.
- Relatively New Concept: Management accounting is a relatively modern discipline that is still evolving. As such, its methods and techniques may not yet be fully developed or standardized across industries.
- High Installation Costs: Setting up a management accounting system can be expensive. The cost of software, training, and maintenance may be prohibitive for smaller organizations.
- Complex Methods: The techniques used in management accounting, such as marginal costing, budgeting, and standard costing, can be complex. These methods may sometimes lead to unscientific or overly complicated approaches to decision-making, making it harder to implement practical solutions.

1.8 CHANGE IN MANAGEMENT ACCOUNTING

In today's rapidly changing business environment, management accounting systems must continuously adapt and innovate to meet the evolving needs of organizations. Several factors are driving the change in management accounting practices, including:

1. **Rapid Technological Changes and Innovations:** Advances in technology, such as automation, artificial intelligence, and data analytics, are transforming how data is collected, processed, and analyzed in management accounting.
2. **Ever-Increasing Global Competition:** The globalization of markets has intensified competition, requiring businesses to adopt more efficient and effective management accounting practices to stay ahead.
3. **Fast-Changing Consumer Demand, Awareness, and Taste:** Shifting consumer preferences and demands require businesses to be more agile and responsive, and management accounting systems need to provide timely and relevant information to support this adaptability.
4. **Changing Governmental Regulations and Stricter Rules:** Governments are continuously updating their regulations and financial reporting standards. As a result, management accounting must adjust to comply with these changes and ensure accurate and transparent reporting.
5. **More Uniformity in Presentation of Financial Data:** There is a growing emphasis on standardizing financial data and reporting formats to enhance comparability and transparency, which is driving changes in how financial information is presented and analyzed.

   These factors illustrate the dynamic nature of management accounting and the need for businesses to evolve their accounting practices to remain competitive and compliant in a fast-changing world.

1.9 LET'S SUM UP

In this chapter, we have covered the fundamental concepts that form the backbone of financial management in businesses.

a. **Management accounting** provides vital insights for internal decision-making by helping managers plan, control, and optimize operations. It ensures that the organization can achieve its strategic goals by offering timely and relevant information.

b. **Financial accounting** focuses on generating standardized financial statements for external stakeholders, aiding in the evaluation of a company's financial health.

c. **Cost accounting** deals with the calculation and analysis of production costs, helping businesses determine pricing strategies and improve operational efficiency.

Through a detailed exploration of the objectives, nature, and scope of each accounting type, we highlighted their specific roles and how they interrelate. Additionally, we examined the challenges and limitations of management accounting, along with the changing trends influenced by technological progress and globalization.

Mastering these concepts equips students with the tools to make informed financial decisions, manage resources effectively, and guide organizational growth.

KEYWORDS

Management Accounting, Area of Accounting, Financial Accounting, Cost Accounting, International Accounting, Changes in Management Accounting, Accounting Techniques, Decision Making, Planning and Control, Global Accounting Standards.

1.10 REVIEW QUESTIONS

1. What are the core goals of financial accounting?
2. What is the nature of cost accounting and how does it contribute to determining the cost of products or services?
3. Explain the wide-ranging scope of management accounting and its importance in modern business operations.
4. What are the limitations of management accounting, and how might these challenges influence decision-making processes?
5. How have advances in technology and global competition contributed to the evolution of management accounting?
6. What does "cost ascertainment" mean, and why is it crucial in the context of cost accounting?
7. Identify the key tools and techniques used in management accounting for effective decision-making.

8. Discuss the interrelationship between financial accounting, cost accounting, and management accounting.

9. Why is management accounting future-focused, and how does it support business planning and strategic decisions?

10. Define management accounting. How does it differ from traditional financial accounting?

UNIT 2: TOOLS FOR FINANCIAL STATEMENT ANALYSIS

STRUCTURE

2.1 Introduction

2.2 Meaning of financial statement analysis

2.3 Objective of financial statement analysis

2.4 Procedure of financial statement analysis

2.5 Type of financial statement analysis

2.6 Tools of financial statement analysis

2.7 Limitations of financial statement analysis

2.8 Let's Sum up

2.9 Review questions

LEARNING OUTCOMES

After studying this module, you shall be able to:

1. Learners will be able to explain the meaning, objectives, and significance of financial statement analysis within the context of business decision-making.
2. Learners will be able to describe the procedure for conducting financial statement analysis, differentiate between its various types, and apply appropriate analytical tools for evaluating the financial performance of an organization.
3. Learners will be able to critically assess the limitations of financial statement analysis and understand its constraints when used as a decision-making tool.

Bloom's Level of Learning:

This Chapter corresponds to the **Remembering**, **Understanding**, and **Applying** levels of Bloom's Taxonomy. Learners are first expected to **recall and define** the basic concepts, meaning, and objectives of financial statement analysis, which falls under the **Remembering** level. They are then required to **explain and interpret** the procedures, types, and tools of financial statement analysis, representing the **Understanding** level. Additionally, by identifying appropriate tools and applying procedures to analyze financial statements, learners engage in the **Applying** level of learning, where theoretical

knowledge is put into practice. The identification of limitations also involves understanding the constraints of financial analysis in decision-making, reinforcing comprehension and practical application.

2.1 INTRODUCTION

Business is mainly concerned with the financial activities. In order to ascertain the financial status of the business every enterprise prepares certain statements, known as financial statements. Financial statements are mainly prepared for decision making purposes. But the information as is provided in the financial statements is not adequately helpful in drawing a meaningful conclusion. Thus, an effective analysis and interpretation of financial statements is required. Analysis means establishing a meaningful relationship between various items of the two financial statements with each other in such a way that a conclusion is drawn. By financial statements we mean two statements:

- Profit and loss Account or Income Statement
- Balance Sheet or Position Statement

These are prepared at the end of a given period of time. They are the indicators of profitability and financial soundness of the business concern.

Financial statement analysis is an exceptionally powerful tool for a variety of users of financial statements, each having different objectives in learning about the financial circumstances of the entity.

2.2 MEANING OF FINANCIAL STATEMENT ANALYSIS

Properly comparing a balance sheet with the corresponding profit and loss account to determine the strengths and weaknesses of a business describes financial statement analysis. Financial analysis determines a company's health and stability. The data gives you an intuitive understanding of how the company conducts business. Stockholders can find out how management employs resources and whether they use them properly. Governments and regulatory authorities use financial statements to determine the legality of a company's fiscal decisions and whether the firm is following correct accounting procedures. Finally, government agencies, such as the Internal Revenue Service, use financial statement analysis to decide the correct taxation for the company.

2.3 OBJECTIVE OF FINANCIAL STATEMENT ANALYSIS

The term financial analysis is also known as analysis and interpretation of financial

statements. It refers to the establishing meaningful relationship between various items of the two financial statements i.e. Income statement and position statement. It determines financial strength and weaknesses of the firm.

Analysis of financial statements is an attempt to assess the efficiency and performance of an enterprise. Thus, the analysis and interpretation of financial statements is very essential to measure the efficiency, profitability, financial soundness and future prospects of the business units. Financial analysis serves the following purposes:

- Measuring the profitability

  The main objective of a business is to earn a satisfactory return on the funds invested in it. Financial analysis helps in ascertaining whether adequate profits are being earned on the capital invested in the business or not. It also helps in knowing the capacity to pay the interest and dividend.

- Indicating the trend of Achievements

  Financial statements of the previous years can be compared and the trend regarding various expenses, purchases, sales, gross profits and net profit etc. can be ascertained. Value of assets and liabilities can be compared and the future prospects of the business can be envisaged.

- Assessing the growth potential of the business

  The trend and other analysis of the business provide sufficient information indicating the growth potential of the business.

- Comparative position in relation to other firms

  The purpose of financial statements analysis is to help the management to make a comparative study of the profitability of various firms engaged in similar businesses. Such comparison also helps the management to study the position of their firm in respect of sales, expenses, profitability and utilizing capital, etc.

- Assess overall financial strength

  The purpose of financial analysis is to assess the financial strength of the business. Analysis also helps in taking decisions, whether funds required for the purchase of new machines and equipment are provided from internal sources of the business or not if yes, how much? And also, to assess how much funds have been received

from external sources.

- Assess solvency of the firm

The different tools of an analysis tell us whether the firm has sufficient funds to meet its short term and long-term liabilities or not.

2.4 PROCEDURE OF FINANCIAL STATEMENT ANALYSIS

A common procedure is followed for financial statement analysis. Such procedure is briefly explained below:

i. Objective of Analysis: The objective of analysis is differing from one interested party to another. In other words, the user of financial statement analysis fixes or determines the objectives of analysis.

ii. Decide the Extent of Analysis: The extent of analysis is also decided by the interested party. For example: Shareholder considers long term solvency of the business concern. The debenture holder considers short term solvency of the business concern.

iii. Scope of Analysis: It means that an analyst should determine the depth of the analysis. This can be decided depending upon the nature of problem.

iv. Going through the Financial Statements: The analyst should go through every item of the financial statements. If not so, the hidden facts cannot be found out through.

v. Pooling of Relevant Data: The analyst should collect relevant data from the financial statements. If not so, he/she can get relevant information from the published financial statements.

vi. Rearrangement of Financial Data: The contents of the financial statements are rearranged before making actual analysis and interpretation. Under this step, approximation of figures, consolidation of items etc. is done.

vii. Understanding: The analyst should go through financial documents and other documents for clearly understand the problem.

viii. Classification: After understanding the problem, the collected relevant data are to be classified according to the needs of the problem to find out a correct solution.

ix. Analysis: After making above preparation, actual analysis is done. Any one of the tools or techniques of financial statement analysis can be used.

x. Interpretation and Conclusion: The interpretation is made and the inferences are drawn only on the basis of analysis.

xi. Report Form: All the inferences and interpretation should be presented in a report form to the management.

2.5 TYPES OF FINANCIAL STATEMENT ANALYSIS

There are two key methods for analyzing financial statements. The first method is the use of horizontal and vertical analysis. Horizontal analysis is the comparison of financial information over a series of reporting periods, while vertical analysis is the proportional analysis of a financial statement, where each line item on a financial statement is listed as a percentage of another item. Typically, this means that every line item on an income statement is stated as a percentage of gross sales, while every line item on a balance sheet is stated as a percentage of total assets. Thus, horizontal analysis is the review of the results of multiple time periods; while vertical analysis is the review of the proportion of accounts to each other within a single period. The following links will direct you to more information about horizontal and vertical analysis:

- Horizontal analysis
- Vertical analysis

The second method for analyzing financial statements is the use of many kinds of ratios. You use ratios to calculate the relative size of one number in relation to another. After you calculate a ratio, you can then compare it to the same ratio calculated for a prior period, or that is based on an industry average, to see if the company is performing in accordance with expectations. In a typical financial statement analysis, most ratios will be within expectations, while a small number will flag potential problems that will attract the attention of the reviewer.

There are several general categories of ratios, each designed to examine a different aspect of a company's performance. The general groups of ratios are:

i. Liquidity ratios. This is the most fundamentally important set of ratios, because they measure the ability of a company to remain in business.

ii. Activity ratios. These ratios are a strong indicator of the quality of management, since they reveal how well management is utilizing company resources.

iii. Leverage ratios. These ratios reveal the extent to which a company is relying upon debt to fund its operations, and its ability to pay back the debt.

iv. Profitability ratios. These ratios measure how well a company performs in generating a profit.

2.6 TOOLS FOR FINANCIAL STATEMENT ANALYSIS

Important tools or techniques of financial statement analysis are as follows.

1. Comparative Statement or Comparative Financial and Operating Statements.
2. Common Size Statements.
3. Trend Ratios or Trend Analysis.
4. Average Analysis.
5. Statement of Changes in Working Capital.
6. Fund Flow Analysis.
7. Cash Flow Analysis.
8. Ratio Analysis.
9. Cost Volume Profit Analysis

A brief explanation of the tools or techniques of financial statement analysis presented below:

1. Comparative Statements

Comparative statements deal with the comparison of different items of the Profit and Loss Account and Balance Sheets of two or more periods. Separate comparative statements are prepared for Profit and Loss Account as Comparative Income Statement and for Balance Sheets.

As a rule, any financial statement can be presented in the form of comparative statement such as comparative balance sheet, comparative profit and loss account, comparative cost of production statement, comparative statement of working capital and the like.

2. Comparative Income Statement

Three important information are obtained from the Comparative Income Statement. They are Gross Profit, Operating Profit and Net Profit. The changes or the

improvement in the profitability of the business concern is find out over a period of time. If the changes or improvement is not satisfactory, the management can find out the reasons for it and some corrective action can be taken.

3. Comparative Balance Sheet

The financial condition of the business concern can be found out by preparing comparative balance sheet. The various items of Balance sheet for two different periods are used. The assets are classified as current assets and fixed assets for comparison. Likewise, the liabilities are classified as current liabilities, long term liabilities and shareholders' net worth. The term shareholders' net worth includes Equity Share Capital, Preference Share Capital, Reserves and Surplus and the like.

1. Common Size Statements

A vertical presentation of financial information is followed for preparing common-size statements. Besides, the rupee value of financial statement contents are not taken into consideration. But, only percentage is considered for preparing common size statement.

The total assets or total liabilities or sales is taken as 100 and the balance items are compared to the total assets, total liabilities or sales in terms of percentage. Thus, a common size statement shows the relation of each component to the whole. Separate common size statement is prepared for profit and loss account as Common Size Income Statement and for balance sheet as Common Size Balance Sheet.

2. Trend Analysis

The ratios of different items for various periods are find out and then compared under this analysis. The analysis of the ratios over a period of years gives an idea of whether the business concern is trending upward or downward. This analysis is otherwise called as Pyramid Method.

3. Average Analysis

Whenever, the trend ratios are calculated for a business concern, such ratios are compared with industry average. These both trends can be presented on the graph paper also in the shape of curves. This presentation of facts in the shape of pictures makes the analysis and comparison more comprehensive and impressive.

4. Statement of Changes in Working Capital

The extent of increase or decrease of working capital is identified by preparing the statement of changes in working capital. The amount of net working capital is calculated by subtracting the sum of current liabilities from the sum of current assets. It does not detail the reasons for changes in working capital.

5. Fund Flow Analysis

Fund flow analysis deals with detailed sources and application of funds of the business concern for a specific period. It indicates where funds come from and how they are used during the period under review. It highlights the changes in the financial structure of the company.

6. Cash Flow Analysis

Cash flow analysis is based on the movement of cash and bank balances. In other words, the movement of cash instead of movement of working capital would be considered in the cash flow analysis. There are two types of cash flows. They are actual cash flows and notional cash flows.

7. Ratio Analysis

Ratio analysis is an attempt of developing meaningful relationship between individual items (or group of items) in the balance sheet or profit and loss account. Ratio analysis is not only useful to internal parties of business concern but also useful to external parties. Ratio analysis highlights the liquidity, solvency, profitability and capital gearing.

8. Cost Volume Profit Analysis

This analysis discloses the prevailing relationship among sales, cost and profit. The cost is divided into two. They are fixed cost and variable cost. There is a constant relationship between sales and variable cost. Cost analysis enables the management for better profit planning.

2.7 LIMITATIONS OF FINANCIAL STATEMENT ANALYSIS

Although analysis of financial statement is essential to obtain relevant information for making several decisions and formulating corporate plans and policies, it should be carefully performed as it suffers from a number of the following limitations.

i. Mislead the user

The accuracy of financial information largely depends on how accurately financial statements are prepared. If their preparation is wrong, the information obtained from their analysis will also be wrong which may mislead the user in making decisions.

ii. Not useful for planning

Since financial statements are prepared by using historical financial data, therefore, the information derived from such statements may not be effective in corporate planning, if the previous situation does not prevail.

iii. Qualitative aspects

Then financial statement analysis provides only quantitative information about the company's financial affairs. However, it fails to provide qualitative information such as management labor relation, customer's satisfaction, and management’s skills and so on which are also equally important for decision making.

iv. Comparison not possible

The financial statements are based on historical data. Therefore comparative analysis of financial statements of different years cannot be done as inflation distorts the view presented by the statements of different years.

v. Wrong judgment

The skills used in the analysis without adequate knowledge of the subject matter may lead to negative direction. Similarly, biased attitude of the analyst may also lead to wrong judgment and conclusion.

vi. Dependence on historical costs

Transactions are initially recorded at their cost. This is a concern when reviewing the balance sheet, where the values of assets and liabilities may change over time. Some items, such as marketable securities, are altered to match changes in their market values, but other items, such as fixed assets, do not change. Thus, the balance sheet could be misleading if a large part of the amount presented is based on historical costs.

vii. Inflationary effects

If the inflation rate is relatively high, the amounts associated with assets and liabilities in the balance sheet will appear inordinately low, since they are not

being adjusted for inflation. This mostly applies to long-term assets.

viii. Intangible assets not recorded

Many intangible assets are not recorded as assets. Instead, any expenditures made to create an intangible asset are immediately charged to expense. This policy can drastically underestimate the value of a business, especially one that has spent a large amount to build up a brand image or to develop new products. It is a particular problem for startup companies that have created intellectual property, but which have so far generated minimal sales.

ix. Subject to fraud

The management team of a company may deliberately skew the results presented. This situation can arise when there is undue pressure to report excellent results, such as when a bonus plan calls for payouts only if the reported sales level increases. One might suspect the presence of this issue when the reported results spike to a level exceeding the industry norm.

x. No predictive value

The information in a set of financial statements provides information about either historical results or the financial status of a business as of a specific date. The statements do not necessarily provide any value in predicting what will happen in the future. For example, a business could report excellent results in one month, and no sales at all in the next month, because a contract on which it was relying has ended.

The limitations mentioned above about financial statement analysis make it clear that the analysis is a means to an end and not an end to itself. The users and analysts must understand the limitations before analyzing the financial statements of the company.

2.8 LET'S SUM UP

Financial analysis is a process of selecting, evaluating, and interpreting financial data, along with other pertinent information, in order to formulate an assessment of a company's present and future financial condition and performance. To meet their financial reporting obligations and to assist in strategic decision-making, firms prepare financial statements. However, the information provided in the financial statements is not

an end in itself as no meaningful conclusions can be drawn from these statements alone. Firms employ financial analysts to read, compare and interpret the data as necessary for quantitative analysis and decision-making.

KEYWORDS

Financial Statement Analysis, Steps in Analysis, Horizontal Analysis, Vertical Analysis, Trend Analysis, Ratio Analysis, Cash Flow Analysis, Tools of Financial Statement Analysis, Limitations of Financial Statement Analysis, Interpretation of Financial Data, Comparative Financial Statements, Financial Ratios, Decision-Making Tool, Financial Performance Evaluation

2.9 REVIEW QUESTIONS

1. What is the primary objective of financial statement analysis?
   A) To prepare accounting records
   B) To interpret the financial data for decision-making
   C) To collect tax-related information
   D) To issue company shares
2. Which of the following is **NOT** a type of financial statement analysis?
   A) Horizontal Analysis
   B) Vertical Analysis
   C) Diagonal Analysis
   D) Ratio Analysis
3. Which of the following is a tool of financial statement analysis?
   A) Ratio Analysis
   B) Budgeting
   C) Ledger Posting
   D) Trial Balance
4. One major limitation of financial statement analysis is:
   A) It considers qualitative factors
   B) It is based on historical data
   C) It always gives future forecasts
   D) It ignores accounting records
5. The process of reviewing and evaluating a company's financial statements to make better economic decisions is known as _____.

6. The comparison of financial data over a period of time to identify trends and changes is called _____ **analysis**.
7. What is financial statement analysis? Compare horizontal and vertical analysis.
8. What are the limitations of financial statement analysis?
9. Enumerate the procedure of financial statement analysis.
10. Explain the various tools of financial statement analysis

UNIT 3: RATIO ANALYSIS

STRUCTURE

3.1 Introduction

3.2 Meaning of financial Ratios

3.3 Procedure for Computation of Ratios

3.4 Objective of Ratio analysis

3.5 Importance of Ratio analysis

3.6 Types of Ratios

3.7 Advantages and Limitations of Ratio analysis

3.8 Let us Sum up

3.9 Review Questions

LEARNING OUTCOME

After reading this chapter, students should be able to know:

1. Learners will be able to define financial ratios, explain the objectives of ratio analysis, and describe its significance in assessing an organization's financial performance.
2. Learners will be able to follow the standard procedure for computing various financial ratios and categorize them according to their types, such as liquidity, profitability, solvency, and activity ratios.
3. Learners will be able to critically assess the advantages and limitations of using financial ratio analysis as a financial decision-making and performance evaluation tool.

Bloom's Level of Learning:

This chapter corresponds to the **Remembering**, **Understanding**, and **Applying** levels of Bloom's Taxonomy. At the **Remembering** level, learners are expected to recall and define the basic concepts, such as the meaning, objectives, and types of financial ratios. Moving to the **Understanding** level, learners are required to explain the importance of ratio analysis, describe its procedures, and interpret different types of financial ratios in the context of business performance. Finally, at the **Applying** level, learners will be able to compute various financial ratios using the appropriate procedure and apply them to

analyze and interpret financial data for decision-making purposes. The discussion of advantages and limitations further enhances comprehension, enabling learners to recognize both the strengths and constraints of ratio analysis in practical situations.

3.1 INTRODUCTION

Ratio analysis refers to the analysis and interpretation of the figures appearing in the financial statements (i.e., Profit and Loss Account, Balance Sheet and Fund Flow statement etc.).

It is a process of comparison of one figure against another. It enables the users like shareholders, investors, creditors, Government, and analysts etc. to get better understanding of financial statements.

Ratio analysis is a very powerful analytical tool useful for measuring performance of an organisation. Accounting ratios may just be used as symptom like blood pressure, pulse rate, body temperature etc. The physician analyses this information to know the causes of illness. Similarly, the financial analyst should also analyse the accounting ratios to diagnose the financial health of an enterprise.

3.2 MEANING OF FINANCIAL RATIOS

As stated earlier, accounting ratios are an important tool of financial statements analysis. A ratio is a mathematical number calculated as a reference to relationship of two or more numbers and can be expressed as a fraction, proportion, percentage and a number of times. When the number is calculated by referring to two accounting numbers derived from the financial statements, it is termed as accounting ratio.

It needs to be observed that accounting ratios exhibit relationship, if any, between accounting numbers extracted from financial statements. Ratios are essentially derived numbers and their efficacy depends a great deal upon the basic numbers from which they are calculated. Hence, if the financial statements contain some errors, the derived numbers in terms of ratio analysis would also present an erroneous scenario. Further, a ratio must be calculated using numbers which are meaningfully correlated. A ratio calculated by using two unrelated numbers would hardly serve any purpose. For example, the furniture of the business is Rs.1, 00,000 and Purchases are Rs.3, 00,000. The ratio of purchases to furniture is 3 (3, 00,000/1, 00,000) but it hardly has any relevance. The reason is that there is no relationship between these two aspects.

Metcalf and Tigard have defined financial statement analysis and interpretations as a process of evaluating the relationship between component parts of a financial statement to obtain a better understanding of a firm's position and performance.

Khan and Jain define the term ratio analysis as “the systematic use of ratios to interpret the financial statements so that the strengths and weaknesses of a firm as well as its historical performance and current financial conditions can be determined.”

3.3 PROCEDURE FOR COMPUTATION OF RATIOS

Generally, ratio analysis involves four steps:

(i) Collection of relevant accounting data from financial statements.

(ii) Constructing ratios of related accounting figures.

(iii) Comparing the ratios thus constructed with the standard ratios which may be the corresponding past ratios of the firm or industry average ratios of the firm or ratios of competitors.

(iv) Interpretation of ratios to arrive at valid conclusions.

3.4 OBJECTIVE OF RATIO ANALYSIS

Ratio analysis is indispensable part of interpretation of results revealed by the financial statements. It provides users with crucial financial information and points out the areas which require investigation. Ratio analysis is a technique which involves regrouping of data by application of arithmetical relationships, though its interpretation is a complex matter. It requires a fine understanding of the way and the rules used for preparing financial statements. Once done effectively, it provides a lot of information which helps the analyst:

(i) To know the areas of the business which need more attention;

(ii) To know about the potential areas which can be improved with the effort in the desired direction;

(iii) To provide a deeper analysis of the profitability, liquidity, solvency and efficiency levels in the business;

(iv) To provide information for making cross-sectional analysis by comparing the performance with the best industry standards; and

(v) To provide information derived from financial statements useful for making projections and estimates for the future.

3.5 IMPORTANCE OF RATIO ANALYSIS

Ratio analysis is an important tool for analyzing the company's financial performance. The following are the importance of the accounting ratios in the corporate world.

1. Analyzing Financial Statements

Ratio analysis is an important technique of financial statement analysis. Accounting ratios are useful for understanding the financial position of the company. Different users such as investors, management. bankers and creditors use the ratio to analyze the financial situation of the company for their decision making purpose.

2. Judging Efficiency

Accounting ratios are important for judging the company's efficiency in terms of its operations and management. They help judge how well the company has been able to utilize its assets and earn profits.

3. Locating Weakness

Accounting ratios can also be used in locating weakness of the company's operations even though its overall performance may be quite good. Management can then pay attention to the weakness and take remedial measures to overcome them.

4. Formulating Plans

Although accounting ratios are used to analyze the company's past financial performance, they can also be used to establish future trends of its financial performance. As a result, they help formulate the company's future plans.

5. Comparing Performance

It is essential for a company to know how well it is performing over the years and as compared to the other firms of the similar nature. Besides, it is also important to know how well its different divisions are performing among themselves in different years. Ratio analysis facilitates such comparison ratio analysis involves four steps:

(i) Collection of relevant accounting data from financial statements.

(ii) Constructing ratios of related accounting figures.

(iii) Comparing the ratios thus constructed with the standard ratios which may be the corresponding past ratios of the firm or industry average ratios of the firm or

ratios of competitors.

(iii) Interpretation of ratios to arrive at valid conclusions.

6. Forecasting and Planning

The trend in costs, sales, profits and other facts can be known by computing ratios of relevant accounting figures of last few years. This trend analysis with the help of ratios may be useful for forecasting and planning future business activities.

7. Budgeting

Budget is an estimate of future activities on the basis of past experience. Accounting ratios help to estimate budgeted figures. For example, sales budget may be prepared with the help of analysis of past sales.

8. Measurement of Operating Efficiency

Ratio analysis indicates the degree of efficiency in the management and utilisation of its assets. Different activity ratios indicate the operational efficiency. In fact, solvency of a firm depends upon the sales revenues generated by utilizing its assets.

9. Communication:

Ratios are effective means of communication and play a vital role in informing the position of and progress made by the business concern to the owners or other parties.

10. Control of Performance and Cost

Ratios may also be used for control of performances of the different divisions or departments of an undertaking as well as control of costs.

11. Inter-firm Comparison

Comparison of performance of two or more firms reveals efficient and inefficient firms, thereby enabling the inefficient firms to adopt suitable measures for improving their efficiency. The best way of inter-firm comparison is to compare the relevant ratios of the organisation with the average ratios of the industry.

12. Indication of Liquidity Position

Ratio analysis helps to assess the liquidity position i.e., short-term debt paying ability of a firm. Liquidity ratios indicate the ability of the firm to pay and help in credit analysis by banks, creditors and other suppliers of short-term loans.

13. Indication of Long-term Solvency Position:

Ratio analysis is also used to assess the long-term debt-paying capacity of a firm.

Long- term solvency position of a borrower is a prime concern to the long-term creditors, security analysts and the present and potential owners of a business. It is measured by the leverage/capital structure and profitability ratios which indicate the earning power and operating efficiency. Ratio analysis shows the strength and weakness of a firm in this respect.

14. Indication of Overall Profitability:

The management is always concerned with the overall profitability of the firm. They want to know whether the firm has the ability to meet its short-term as well as long-term obligations to its creditors, to ensure a reasonable return to its owners and secure optimum utilisation of the assets of the firm. This is possible if all the ratios are considered together.

15. Signal of Corporate Sickness:

A company is sick when it fails to generate profit on a continuous basis and suffers a severe liquidity crisis. Proper ratio analysis can give signal of corporate sickness in advance so that timely measures can be taken to prevent the occurrence of such sickness.

16. Aid to Decision-making:

Ratio analysis helps to take decisions like whether to supply goods on credit to a firm, whether bank loans will be made available etc.

17. Simplification of Financial Statements:

Ratio analysis makes it easy to grasp the relationship between various items and helps in understanding the financial statements.

3.6 TYPES OF RATIOS

There is a two way classification of ratios: (1) traditional classification, and (2) functional classification. The traditional classification has been on the basis of financial statements to which the determinants of ratios belong. On this basis the ratios are classified as follows:

(ii) ‘Statement of Profit and Loss Ratios: A ratio of two variables from the statement of profit and loss is known as statement of profit and loss ratio. For example, ratio of gross profit to revenue from operations is known as gross profit ratio. It is calculated using both figures from the statement of profit and loss.

(iii) Balance Sheet Ratios: In case both variables are from the balance sheet, it is

classified as balance sheet ratios. For example, ratio of current assets to current liabilities known as current ratio. It is calculated using both figures from balance sheet.

(iv) Composite Ratios: If a ratio is computed with one variable from the statement of profit and loss and another variable from the balance sheet, it is called composite ratio. For example, ratio of credit revenue from operations to trade receivables (known as trade receivables turnover ratio) is calculated using one figure from the statement of profit and loss (credit revenue from operations) and another figure (trade receivables) from the balance sheet.

Although accounting ratios are calculated by taking data from financial statements but classification of ratios on the basis of financial statements is rarely used in practice. It must be recalled that basic purpose of accounting is to throw light on the financial performance (profitability) and financial position (its capacity to raise money and invest them wisely) as well as changes occurring in financial position (possible explanation of changes in the activity level). As such, the alternative classification (functional classification) based on the purpose for which a ratio is computed, is the most commonly used classification which is as follows:

A. Profitability Ratios

B. Liquidity Ratios

C. Activity (or Turnover) Ratios

D. Solvency Ratios

Profitability Ratios

Net profit ratio (NP ratio) is a popular profitability ratio that shows relationship between net profit after tax and net sales. It is computed by dividing the net profit (after

$$\text{Net profit (NP) ratio} = \frac{\text{Net profit after tax}}{\text{Net sales}}$$

tax) by net sales.

For the purpose of this ratio, net profit is equal to gross profit minus operating expenses and income tax. All non-operating revenues and expenses are not taken into

account because the purpose of this ratio is to evaluate the profitability of the business from its primary operations.

Net profit (NP) ratio is a useful tool to measure the overall profitability of the business. A high ratio indicates the efficient management of the affairs of business.

Gross profit ratio (GP ratio) is a profitability ratio that shows the relationship between gross profit and total net sales revenue. It is a popular tool to evaluate the operational performance of the business. The ratio is computed by dividing the gross profit figure by net sales.

The following formula/equation is used to compute gross profit ratio:

$$\text{Gross profit Ratio} = \frac{\text{Gross profit}}{\text{Net sales}}$$

When gross profit ratio is expressed in percentage form, it is known as gross profit margin or gross profit percentage. The formula of gross profit margin or percentage is given below:

$$\text{Gross profit margin} = \frac{\text{Gross profit}}{\text{Net sales}} \times 100$$

The basic components of the formula of **gross profit ratio (GP ratio)** are gross profit and net sales. Gross profit is equal to net sales minus cost of goods sold. Net sales are equal to total gross sales less returns inwards and discount allowed. The information about gross profit and net sales is normally available from income statement of the company.

Price earnings ratios (P/E ratio) measures how many times the earnings per share (EPS) has been covered by current market price of an ordinary share. It is computed by dividing the current market price of an ordinary share by earnings per share.

The formula of price earnings ratio is given below:

$$\text{Price earnings ratio (P/E ratio)} = \frac{\text{Market price per equity share}}{\text{Earnings per share (EPS)}}$$

A higher P/E ratio is the indication of strong position of the company in the market and a fall in ratio should be investigated.

Operating ratio is computed by dividing operating expenses by net sales. It is expressed in percentage.

Operating ratio is computed as follows:

$$\text{Operating ratio} = \frac{\text{Operating cost}}{\text{Net sales}} \times 100$$

The basic components of the formula are operating cost and net sales. Operating cost is equal to cost of goods sold plus operating expenses. Non-operating expenses such as interest charges, taxes etc., are excluded from the computations.

This ratio is used to measure the operational efficiency of the management. It shows whether the cost component in the sales figure is within normal range. A low operating ratio means high net profit ratio i.e., more operating profit.

The ratio should be compared: (1) with the company's past years ratio, (2) with the ratio of other companies in the same industry. An increase in the ratio should be investigated and brought to attention of management. The operating ratio varies from industry to industry.

Expense ratio (expense to sales ratio) is computed to show the relationship between an individual expense or group of expenses and sales. It is computed by dividing a particular expense or group of expenses by net sales. Expense ratio is expressed in percentage.

$$\text{Expense ratio} = \frac{\text{Particular expense}}{\text{Net sales}} \times 100$$

The numerator may be an individual expense or a group of expenses such as administrative expenses, sales expenses or cost of goods sold.

Expense ratio shows what percentage of sales is an individual expense or a group of expenses. A lower ratio means more profitability and a higher ratio means less profitability.

Return on shareholders' investment ratio is a measure of overall profitability of the business and is computed by dividing the *net income after interest and tax* by average stockholders' equity. It is also known as return on equity (ROE)

ratio and return on net worth ratio. The ratio is usually expressed in percentage.

$$\text{Return on shareholders' investment or return on equity ratio} = \frac{\text{Net income after interest and tax}}{\text{Average stockholders' equity}} \times 100$$

The numerator consists of net income after interest and tax because it is the amount of income available for common and preference stockholders. The denominator is the average of stockholders' equity (preference and common stock). The information about net income after interest and tax is normally available from <u>income statement</u> and the information about preference and common stock is available from <u>balance sheet.</u>

Return on equity (ROE) is widely used to measure the overall profitability of the company from preference and common stockholders' view point. The ratio also indicates the efficiency of the management in using the resources of the business.

Return on common stockholders' equity ratio measures the success of a company in generating income for the benefit of common stockholders. It is computed by dividing the net income available for common stockholders by common stockholders' equity. The ratio is usually expressed in percentage.

$$\text{Return on common stockholders' equity} = \frac{\text{Net income - Preferred dividend}}{\text{Average common stockholders' equity}} \times 100$$

The numerator in the above formula consists of net income available for common stockholders which are equal to net income less dividend on preferred stock. The denominator consists of average common stockholders' equity which is equal to average total stockholders' equity less average preferred stockholders equity. If preferred stock is not present, the net income is simply divided by the average common stockholders' equity to compute the common stock equity ratio. Like return on equity (ROE) ratio, a higher common stock equity ratio indicates high profitability and strong financial position of the company and can convert potential investors into actual common stockholders.

Earnings per share (EPS) ratio measures how many dollars of net income have been earned by each share of common stock. It is computed by dividing net income less preferred dividend by the number of shares of common stock outstanding during the period. It is a popular measure of overall profitability of the company and is usually expressed in dollars. Earnings per share ratio (EPS ratio) is computed by the following formula:

The numerator is the net income available for common stockholders' (net income less preferred dividend) and the denominator is the average number of shares of common stock outstanding during the year.

The formula of EPS ratio is similar to the formula of return on common stockholders' equity ratio except the denominator of EPS ratio formula is the number of average shares of common stock outstanding rather than the average common stockholders' equity. The higher the EPS figure, the better it is. A higher EPS is the sign of higher earnings, strong financial position and, therefore, a reliable company to invest money.

Return on capital employed ratio is computed by dividing the net income before interest and tax by capital employed. It measures the success of a business in generating satisfactory profit on capital invested. The ratio is expressed in percentage.

Formula:

$$\text{Return on capital employed ratio} = \frac{\text{Net income before interest and tax}}{\text{Capital employed}} \times 100$$

The basic components of the formula of return on capital employed ratio are net income before interest and tax and capital employed.

Net income before the deduction of interest and tax expenses is frequently referred to as operating income. Here, interest means interest on long term loans. If company pays interest expenses on short-term borrowings, that is deducted to arrive at operating income.

Return on capital employed ratio measures the efficiency with which the investment made by shareholders and creditors is used in the business. Managers use this ratio for various financial decisions. It is a ratio of overall profitability and a higher ratio is, therefore, better.

Dividend yield ratio shows what percentage of the market price of a share a

company annually pays to its stockholders in the form of dividends. It is calculated by dividing the annual dividend per share by market value per share. The ratio is generally expressed in percentage form and is sometimes called dividend yield percentage.

Since dividend yield ratio is used to measure the relationship between the annual amount of dividend per share and the current market price of a share, it is mostly used by investors looking for dividend income on continuous basis. Formula:

The following formula is used to calculated dividend yield ratio:

$$\text{Dividend yield ratio} = \frac{\text{Dividend per share}}{\text{Market value per share}} \times 100$$

Dividend payout ratio discloses what portion of the current earnings the company is paying to its stockholders in the form of dividend and what portion the company is ploughing back in the business for growth in future. It is computed by dividing the dividend per share by the earnings per share (EPS) for a specific period. The formula of dividend payout ratio is given below:

$$\text{Dividend payout ratio} = \frac{\text{Dividend per share}}{\text{Earnings per share}}$$

The numerator in the above formula is the dividend per share paid to common stockholders only. It does not include any dividend paid to preferred stockholders.

Liquidity Ratios

(i) ***Current ratio*** (also known as **working capital ratio**) is a popular tool to evaluate short-term solvency position of a business. Short-term solvency refers to the ability of a business to pay its short-term obligations when they become due. Short term obligations (also known as current liabilities) are the liabilities payable within a short period of time, usually one year.

Current ratio is computed by dividing total current assets by total current liabilities of the business. This relationship can be expressed in the form of following formula or

equation:

$$Current\ ratio = \frac{Current\ assets}{Current\ liabilities}$$

(i) ***Quick ratio*** (also known as "acid test ratio" and "liquid ratio") is used to test the ability of a business to pay its short-term debts. It measures the relationship between liquid assets and current liabilities. Liquid assets are equal to total current assets minus inventories and prepaid expenses.

The formula for the calculation of quick ratio is given below:

$$\text{Quick ratio} = \frac{\text{Liquid assets}}{\text{Current liabilities}}$$

Quick ratio is considered a more reliable test of short-term solvency than current ratio because it shows the ability of the business to pay short term debts immediately.

Inventories and prepaid expenses are excluded from current assets for the purpose of computing quick ratio because inventories may take long period of time to be

converted into cash and prepaid expenses cannot be used to pay current liabilities.

(ii) ***Absolute Liquid ratio***-some analysts also compute **absolute liquid ratio** to test the liquidity of the business. Absolute liquid ratio is computed by dividing the absolute liquid assets by current liabilities.

The formula to compute this ratio is given below:

$$\text{Absolute liquid ratio} = \frac{\text{Absolute liquid assets}}{\text{Current liabilities}}$$

Absolute liquid assets are equal to liquid assets minus accounts receivables (including bills receivables). Some examples of absolute liquid assets are cash, bank balance and marketable securities etc.

Activity Ratios

Inventory turnover ratio (ITR) is an activity ratio is a tool to evaluate the liquidity of inventory. It measures how many times a company has sold and replaced its inventory during a certain period of time.

Inventory turnover ratio is computed by dividing the cost of goods sold by average inventory at cost. The formula/equation is given below:

Two components of the formula of inventory turnover ratio are cost of goods sold and average inventory at cost. Cost of goods sold is equal to cost of goods manufactured (purchases for trading company) plus opening inventory less closing inventory. Average inventory is equal to opening balance of inventory plus closing balance of inventory divided by two.

Inventory turnover ratio varies significantly among industries. A high ratio indicates fast moving inventories and a low ratio, on the other hand, indicates slow moving or obsolete inventories in stock. A low ratio may also be the result of maintaining excessive inventories needlessly. Maintaining excessive inventories unnecessarily indicates poor inventory management because it involves tiding up funds that could have been used in other business operations.

$$\text{Inventory turnover ratio} = \frac{\text{Cost of goods sold}}{\text{Average inventory at cost}}$$

Receivables turnover ratio (also known as **debtors turnover ratio**) is computed by dividing the net credit sales during a period by average receivables. Accounts receivable turnover ratio simply measures how many times the receivables are collected during a particular period. It is a helpful tool to evaluate the liquidity of receivables.

$$\text{Accounts receivable turnover ratio} = \frac{\text{Net credit sales}}{\text{Average trade receivables (net)}}$$

Two components of the formula are "net credit sales" and "average trade accounts receivable". It is clearly mentioned in the formula that the numerator should include only credit sales. But in examination questions, this information may not be given. In that case, the total sales should be used as numerator assuming all the sales are made on credit.

Average receivables are equal to opening receivables (including notes receivables) plus closing receivables (including notes receivables) divided by two. But sometimes opening receivables may not be given in the examination questions. In that case closing balance of receivables should be used as denominator.

Average collection period is computed by dividing the number of working days for a given period (usually an accounting year) by receivables turnover ratio. It is expressed in days and is an indication of the quality of receivables.

The formula is given below:

$$\text{Average collection period} = \frac{\text{Number of working days}}{\text{Debtors turnover ratio}}$$

A short collection period means prompt collection and better management of receivables. A longer collection period may negatively effect the short-term debt paying ability of the business in the eyes of analysts.

Accounts payable turnover ratio (also known as creditors turnover ratio or creditors' velocity) is computed by dividing the net credit purchases by average accounts payable. It measures the number of times, on average, the accounts payable are paid during a period. Like receivables turnover ratio, it is expressed in times.

$$\text{Accounts payable turnover ratio} = \frac{\text{Net credit purchases}}{\text{Average accounts payable}}$$

In above formula, numerator includes only credit purchases. But if credit purchases are not known, the total net purchases should be used.

Average accounts payable are computed by adding opening and closing balances of accounts payable (including notes payable) and dividing by two. If opening balance of accounts payable is not given, the closing balance (including notes payable) should be used.

Accounts payable turnover ratio indicates the creditworthiness of the company. A high ratio means prompt payment to suppliers for the goods purchased on credit and a low ratio may be a sign of delayed payment. Accounts payable turnover ratio also depends on the credit terms allowed by suppliers. Companies who enjoy longer credit periods allowed by creditors usually have low ratio as compared to others.

Average payment period means the average period taken by the company in making payments to its creditors. It is computed by dividing the number of working days in a year by creditors turnover ratio. Some other formulas for its computation are given below:

Formula:

This ratio may be computed in a number of ways:

1. $$\text{Average accounts payables} = \frac{\text{Accounts payables (including notes payable)}}{\text{Average daily credit purchase*}}$$

$$\text{*Average daily credit purchases} = \frac{\text{Credit purchases}}{\text{Number of working days in a year}}$$

2. $$\text{Average payment period} = \frac{\text{(Accounts payable} \times \text{Number of working days)}}{\text{Net credit purchases}}$$

3. $$\text{Averaage payment period} = \frac{\text{Number of working days}}{\text{Payables turnover ratio}}$$

Any of the above formulas may be used to compute average payment period. If credit purchases are unknown, the total purchases may be used. A shorter payment period indicates prompt payments to creditors. Like accounts payable turnover ratio, average payment period also indicates the creditworthiness of the company. But a very short payment period may be an indication that the company is not taking full advantage of the credit terms allowed by suppliers.

Working capital turnover ratio is computed by dividing the cost of goods sold by net working capital. It represents how many times the working capital has been turned over during the period.

$$\text{Working capital turnover ratio} = \frac{\text{Cost of goods sold}}{\text{Net working capital}}$$

The formula consists of two components – cost of goods sold and net working capital. If the cost of goods sold figure is not available or cannot be computed from the available information, the total net sales can be used as numerator. Net working capital is equal to current assets minus current liabilities. This information is available from the balance sheet.

Generally, a high working capital turnover ratio is better. A low ratio indicates inefficient utilization of working capital. The ratio should be carefully interpreted because a very high ratio may also be a sign of insufficient working capital.

Fixed assets turnover ratio (also known as sales to fixed assets ratio) is a commonly used activity ratio that measures the efficiency with which a company uses its fixed assets to generate its sales revenue. It is computed by dividing net sales by average fixed assets.

Generally, a high fixed assets turnover ratio indicates better utilization of fixed assets and a low ratio means inefficient or under-utilization of fixed assets. The usefulness of this ratio can be increased by comparing it with the ratio of other companies, industry standards and past years.

Solvency Ratios

(i) Debt to equity ratio is a long term solvency ratio that indicates the soundness of long-term financial policies of a company. It shows the relation between the portion of assets financed by creditors and the portion of assets financed by stockholders. As the debt to equity ratio expresses the relationship between external equity (liabilities) and internal equity (stockholder's equity), it is also known as "external-internal equity ratio".

Debt to equity ratio is calculated by dividing total liabilities by stockholder's equity.

$$\text{Debt to equity ratio} = \frac{\text{Total liabilities}}{\text{Stockholder's equity}}$$

The numerator consists of the total of current and long term liabilities and the denominator consists of the total stockholders' equity including preferred stock. Both the elements of the formula are obtained from company's balance sheet.

A ratio of 1 (or 1: 1) means that creditors and stockholders equally contribute to the assets of the business. A less than 1 ratio indicates that the portion of assets provided by stockholders is greater than the portion of assets provided by creditors and a greater than1 ratio indicates that the portion of assets provided by creditors is greater than the portion of assets provided by stockholders.

Creditors usually like a low debt to equity ratio because a low ratio (less than 1) is the indication of greater protection to their money. But stockholders like to get benefit from the funds provided by the creditors therefore they would like a high debt to equity ratio.

(ii) *The* ***proprietary ratio*** (also known as net worth ratio or equity ratio) is used to evaluate the soundness of the capital structure of a company. It is computed by dividing the stockholders' equity by total assets.

Formula:

$$\text{Proprietory ratio} = \frac{\text{Stockholders' equity}}{\text{Total assets}} \times 100$$

The proprietary ratio shows the contribution of stockholders' in total capital of the company. A high proprietary ratio, therefore, indicates a strong financial position of the

company and greater security for creditors. A low ratio indicates that the company is already heavily depending on debts for its operations. A large portion of debts in the total capital may reduce creditors interest, increase interest expenses and also the risk of bankruptcy.

(i) ***Fixed assets to equity ratio*** measures the contribution of stockholders and the contribution of debt sources in the fixed assets of the company. It is computed by dividing the fixed assets by the stockholders' equity.

Other names of this ratio are *fixed assets to net worth ratio* and *fixed assets to proprietors fund ratio*.

Formula:

$$\text{Fixed assets to stockholders' equity ratio} = \frac{\text{Fixed assets}}{\text{Stockholders' equity}}$$

The numerator in the above formula is the book value of fixed assets (fixed assets less depreciation) and the denominator is the stockholders' equity that consists of common stock, preferred stock, paid in capital and retained earnings. Information about fixed assets and stockholders' equity is available from balance sheet.

(ii) ***Capital gearing ratio*** is a useful tool to analyze the capital structure of a company and is computed by dividing the common stockholders' equity by fixed

interest or dividend bearing funds.

Analyzing capital structure means measuring the relationship between the funds provided by common stockholders and the funds provided by those who receive a periodic interest or dividend at a fixed rate.

A company is said to be low geared if the larger portion of the capital is composed of common stockholders' equity. On the other hand, the company is said to be highly geared if the larger portion of the capital is composed of fixed interest/dividend bearing funds.

Formula:

$$\text{Capital gearing ratio} = \frac{\text{Common stockholders' equity}}{\text{Fixed interest bearing funds}}$$

Or

$$\text{Capital gearing ratio} = \text{Common stockholders' equity} : \text{Fixed interest bearing funds}$$

In the above formula, the numerator consists of common stockholders' equity that is equal to total stockholders' equity less preferred stock and the denominator consists of fixed interest or dividend bearing funds that usually include long term loans, bonds, debentures and preferred stock etc. All the information required to compute capital gearing ratio is available from the balance sheet.

Leverage Ratio

A **leverage ratio** is meant to evaluate a company's debt levels. The most common leverage ratios are the debt ratio and the debt-to-equity ratio. A debt ratio is simply a company's total debt divided by its total assets. The formula is:

Debt Ratio = Total Debt / Total Assets

For example, if Company XYZ had 10 million of debt on its balance sheet and 15 million of assets, then Company XYZ's debt ratio is:

Debt Ratio = 10,000,000 / 15,000,000 = 0.67 or 67%

This means that for every dollar of Company XYZ assets, Company XYZ had 0.67 of

debt. A ratio above 1.0 indicates that the company has more debt than assets.

The **debt-to-equity ratio** is a measure of the relationship between the capital contributed by creditors and the capital contributed by owners. It also shows the extent to which shareholders' equity can fulfill a company's obligations to creditors in the event of a liquidation.

Here is the formula for the debt-to-equity ratio:

Debt-to-Equity Ratio = Total Debt/Total Equity

For example, if Company XYZ had 10 million of debt on its balance sheet and 10million of total equity, then Company XYZ's debt ratio is:

Debt-to-Equity Ratio = $10,000,000/$10,000,000 = 1.0 times or 100%

This means that for every dollar of Company XYZ owned by the shareholders, Company XYZ owes 1 to creditors.

Du Pont Analysis

Dupont analysis is an extended examination of return on equity (ROE) of a company which analysis net profit margin, asset turnover and financial leverage. This analysis was developed by the Dupont Corporation in the year 1920.

In simple words, it breaks down the ROE to analyse how corporate can increase the return for their shareholders.

Return on Equity= Net Profit margin x Asset turnover ratio x Financial leverage

= (Net Income / Sales) x (Sales / Total Assets) x (Total Assets / Total Equity)The company can increase its return on equity if it-

1. Generates a high net profit margin. The company can increase its return on equity if it-
2. Effectively uses its assets so as to generate more sales
3. Has a high financial leverage

DuPont analysis examines the return on equity (ROE) analyzing profit margin, total asset turnover, and financial leverage. It was created by the DuPont Corporation in the 1920s.

How it works:

The DuPont analysis is also referred to as the DuPont identity. In a DuPont analysis, the formula for ROE is:

ROE = Profit Margin x Total Asset Turnover x Leverage Factor

The formula breaks down further to:

ROE = (Net Income/Revenues) x (Revenues/Total Assets) x (Total Assets/ Shareholders' Equity)

For example, let's consider the following information for Company XYZ: Using the formula above, we can calculate that Company XYZ's ROE is :

ROE = (2,000/10,000) x (10,000/25,000) x (25,000/5,000) = 0.20 x 0.40 x 5 = 0.40 or 40%.

The DuPont analysis analyzes the numbers shown in profit margin (2,000/10,000), total asset turnover (10,000/25,000) and leverage factor (25,000/5,000) to find Company XYZ's ROE.

The Dupont equation can be further decomposed to have an even deeper insight where the net profit margin is broken down into EBIT margin, tax burden and interest burden.

Return on Equity= EBIT margin x Interest burden x Tax burden x Asset turnover ratio x Financial leverage

ROE = (EBIT / Sales) x (EBT / EBIT) x (Net Income / EBT) x (Sales / Total Assets) x (Total Assets / Total Equity)

3.7 ADVANTAGES AND LIMITATIONS OF RATIO ANALYSIS ADVANTAGES

(i) **Forecasting and Planning:**

The trend in costs, sales, profits and other facts can be known by computing ratios of relevant accounting figures of last few years. This trend analysis with the help of ratios may be useful for forecasting and planning future business activities.

(ii) **Budgeting:**

Budget is an estimate of future activities on the basis of past experience.

Accounting ratios help to estimate budgeted figures. For example, sales budget may be prepared with the help of analysis of past sales.

(iii) **Measurement of Operating Efficiency:**

Ratio analysis indicates the degree of efficiency in the management and utilisation of its assets. Different activity ratios indicate the operational efficiency. In fact, solvency of a firm depends upon the sales revenues generated by utilizing its assets.

(iv) **Communication:**

Ratios are effective means of communication and play a vital role in informing the position of and progress made by the business concern to the owners or other parties.

(v) **Control of Performance and Cost:**

Ratios may also be used for control of performances of the different divisions or departments of an undertaking as well as control of costs.

(vi) **Inter-firm Comparison:**

Comparison of performance of two or more firms reveals efficient and inefficient firms, thereby enabling the inefficient firms to adopt suitable measures for improving their efficiency. The best way of inter-firm comparison is to compare the relevant ratios of the organisation with the average ratios of the industry.

(vii) **Indication of Liquidity Position:**

Ratio analysis helps to assess the liquidity position i.e., short-term debt paying ability of a firm. Liquidity ratios indicate the ability of the firm to pay and help in credit analysis by banks, creditors and other suppliers of short-term loans.

(viii) **Indication of Long-term Solvency Position:**

Ratio analysis is also used to assess the long-term debt-paying capacity of a firm.

Long-term solvency position of a borrower is a prime concern to the long-term creditors, security analysts and the present and potential owners of a business. It is measured by the leverage/capital structure and profitability ratios which indicate the earning power and operating efficiency. Ratio analysis shows the strength and weakness of a firm in this respect.

(i) **Indication of Overall Profitability:**

The management is always concerned with the overall profitability of the firm. They want to know whether the firm has the ability to meet its short-term as well as long- term obligations to its creditors, to ensure a reasonable return to its owners and secure optimum utilisation of the assets of the firm. This is possible if all the ratios are considered together.

(ii) **Signal of Corporate Sickness:**

A company is sick when it fails to generate profit on a continuous basis and suffers a severe liquidity crisis. Proper ratio analysis can give signal of corporate sickness in advance so that timely measures can be taken to prevent the occurrence of such sickness.

(iii) **Aid to Decision-making:**

Ratio analysis helps to take decisions like whether to supply goods on credit to a firm, whether bank loans will be made available etc.

(iv) **Simplification of Financial Statements:**

Ratio analysis makes it easy to grasp the relationship between various items and helps in understanding the financial statements.

LIMITATIONS

Limitations of Financial Statements:

Ratios are calculated from the information recorded in the financial statements. But financial statements suffer from a number of limitations and may, therefore, affect the quality of ratio analysis.

i. Historical Information:

Financial statements provide historical information. They do not reflect current conditions. Hence, it is not useful in predicting the future.

ii. Different Accounting Policies:

Different accounting policies regarding valuation of inventories, charging depreciation etc. make the accounting data and accounting ratios of two firms non- comparable.

iii. Lack of Standard of Comparison:

No fixed standards can be laid down for ideal ratios. For example, current ratio is said to be ideal if current assets are twice the current liabilities. But this

conclusion may not be justifiable in case of those concerns which have adequate arrangements with their bankers for providing funds when they require, it may be perfectly ideal if current assets are equal to or slightly more than current liabilities.

iv. Quantitative Analysis:

Ratios are tools of quantitative analysis only and qualitative factors are ignored while computing the ratios. For example, a high current ratio may not necessarily mean sound liquid position when current assets include a large inventory consisting of mostly obsolete items.

v. Window-Dressing:

The term 'window-dressing' means presenting the financial statements in such a way to show a better position than what it actually is. If, for instance, low rate of depreciation is charged, an item of revenue expense is treated as capital expenditure etc. the position of the concern may be made to appear in the balance sheet much better than what it is. Ratios computed from such balance sheet cannot be used for scanning the financial position of the business.

vi. Changes in Price Level:

Fixed assets show the position statement at cost only. Hence, it does not reflect the changes in price level. Thus, it makes comparison difficult.

vii. Causal Relationship Must:

Proper care should be taken to study only such figures as have a cause-and-effect relationship; otherwise ratios will only be misleading.

viii. Ratios Account for one Variable:

Since ratios account for only one variable, they cannot always give correct picture since several other variables such Government policy, economic conditions, availability of resources etc. should be kept in mind while interpreting ratios.

ix. Seasonal Factors Affect Financial Data:

Proper care must be taken when interpreting accounting ratios calculated for seasonal business. For example, an umbrella company maintains high inventory during rainy season and for the rest of year its inventory level becomes 25% of the seasonal inventory level. Hence, liquidity ratios and inventory turnover ratio

will give biased picture.

3.8 LET'S SUM UP

Ratios are a powerful tool in the interpretation of the accounts and can discover issues and problems not immediately evident from the accounts and financial information provided in the annual report. They can provide the basis for inter-firm comparisons allowing managers to benchmark the performance and efficiency of the firm against its competitors. Trends can then be examined and analysed. Stakeholders may use ratios to support their decision making. Employees, for example may use profit ratios to support pay claims and creditors can use liquidity ratios to evaluate whether debts will be repaid.

KEYWORDS

Financial Ratios, Meaning of Financial Ratios, Ratio Formula, Procedure for Computation of Ratios, Objective of Ratio Analysis, Importance of Ratio Analysis, Types of Financial Ratios, Financial Performance Evaluation, Decision-Making Tool, Trend Analysis, Benchmarking

3.9 REVIEW QUESTIONS

1. Which of the following is NOT a type of financial ratio?

   a) Liquidity Ratio

   b) Profitability Ratio

   c) Colour Ratio

   d) Solvency Ratio

2. The main objective of ratio analysis is to:

   a) Decorate financial reports

   b) Assess the financial performance and position of a business

   c) Increase the number of transactions

   d) Prepare legal contracts

3. What is meant by ratio analysis? Discuss its objectives and limitations.

4. "Ratio analysis is a tool to examine the health of business with a view tomake financial results more intelligible." Explain.

5. Briefly describe the steps of ratio analysis.

6. Write a detailed note on DuPont analysis.

7. _____ Ratios are used to measure a company's ability to meet its short-term obligations.
8. The primary objective of ratio analysis is to assess the _____ performance and financial position of a business.
9. Profitability ratios indicate a firm's ability to earn _____ in relation to its sales, assets, or equity.
10. One of the major limitations of ratio analysis is that it ignores the effects of _____ changes over time.

UNIT-4: COST REDUCTION AND COST CONTROL

STRUCTURE

4.1 Concept of Cost control

4.2 Steps in cost control

4.3 Concept of Cost Reduction

4.4 Difference between Cost Control and Cost Reduction

4.5 Scope of Cost Reduction

4.6 Tools and Techniques of Cost Reduction and Control

4.7 Let's Sum Up

4.8 Review Questions

LEARNING OUTCOME

After studying this module, you shall be able to

- Understand the concept of cost control and cost reduction
- Identify the steps involved in cost control
- Differentiate between cost control and cost reduction
- Know the tools and techniques of cost control and cost reduction

Bloom's Level of Learning

The topics listed under **Cost Control and Cost Reduction** — including concepts, steps, differences, scope, tools, and techniques — primarily fall under the **Understanding** and **Applying** levels of **Bloom's Taxonomy of Learning**. At the **Understanding level**, learners are expected to explain ideas, describe processes, and differentiate between concepts like cost control and cost reduction. They comprehend how various tools and techniques function in managing and minimizing costs. At the **Applying level**, learners should be able to use the steps, tools, and techniques of cost control and cost reduction in business scenarios, and suggest appropriate measures to improve operational efficiency. This set of topics builds foundational knowledge that encourages learners to both grasp the concepts and practice applying them in realistic financial and managerial decision-

making situations.

4.1 CONCEPT OF COST CONTROL

CIMA, London has defined cost control as “the regulation by executive action of the cost of operating an undertaking particularly where action is guided by cost accounting”

Cost Control is a process which focuses on controlling the total cost through competitive analysis. It is a practice which works to maintain the actual cost in accordance with the established norms. It ensures that the cost incurred on an operation should not go beyond the pre-determined cost.

Cost Control involves a chain of functions, which starts from preparation of the budget in relation to the operation, thereafter evaluating the actual performance, next is to compute the variances between the actual cost & the budgeted cost and further, to find out the reasons for the same, finally to implement the necessary actions for correcting discrepancies.

4.2 STEPS IN COST CONTROL

Establishing norms:

The first step in cost control is to set norms or standards which may serve as yardsticks for measuring performance. These standards are set on the basis of past performance adjusted for changes in future and on the basis of studies conducted.

Comparison with actual:

The actual cost incurred are compared with established standard costs to know the level of achievement. The variations are analysed so as to arrive at the causes which are controllable.

Corrective Action:

Remedial measures are taken to avoid the recurrence of variation in future and for revision of standards wherever necessary

4.3 CONCEPT OF COST REDUCTION

Cost Reduction is a process, aims at lowering the unit cost of a product manufactured or service rendered without affecting its quality by using new and improved methods and techniques. It ascertains substitute ways to reduce the cost of a unit. It ensures savings in

per unit cost and maximisation of profits of the organisation.

Cost Reduction aims at cutting off the unnecessary expenses which occur during the production, storing, selling and distribution of the product. To identify cost reduction, the following are the major elements:

- Savings in per unit cost.
- No compromise with the quality of the product.
- Savings are non-volatile in nature.

4.4 DIFFERENCE BETWEEN COST CONTROL AND COST REDUCTION

- The activity of maintaining cost as per the established norms is known as cost control. The activity of decreasing per unit cost by applying new methods of production in such a way that it does not affect the quality of the product is known as cost reduction.
- Cost Control focuses on decreasing the total cost while cost reduction focuses on decreasing per unit cost of a product.
- Cost Control is temporary in nature. Unlike Cost Reduction which is permanent.
- The process of cost control is completed when the specified target is achieved. Conversely, the process of cost reduction has no visible end as it is a continuous process that targets for eliminating wasteful expenses.
- Cost Control does not guarantee quality maintenance, however 100% quality maintenance is assured in case of cost reduction.
- Cost Control is a preventive function as it ascertains the cost before its occurrence. Cost Reduction is a corrective action.
- Cost control is the achievement of pre- determined targets of costs.
    - Cost reduction is the achievement of the real and permanent reduction in costs.
- Cost control tends to assume a static state of affairs and that standard once set are not challenged.
    - Cost reduction assumes the existence of concealed potential saving in the standards or pre- determined costs set for cost control and that these

standards are always subject to challenge.

- Cost control is concerned with predetermining costs, comparing it with actual costs, analysing the variances and taking corrective measures.
    - Cost reduction is not concerned with maintenance of performance according to predetermined targets. It is rather concerned with finding out new product design, methods, etc
- Cost control is a part of cost accounting function.
    - Cost reduction may be achieved even when no cost accounting system is in operation
- Cost control lacks dynamic approach to cost improvement.
    - Cost reduction is more dynamic approach to cost improvement.

Table 1: Comparison Chart

BASIS OF COMPARISON	**COST CONTROL**	**COST REDUCTION**
Meaning	A technique used for maintaining the costs as per the set standards is known as Cost Control	A technique used to economies the unit cost without lowering the quality of the product is known as Cost Reduction
Savings in	Total Cost	Cost Per Unit
Retention of Quality	Not Guaranteed	Guaranteed
Nature	Temporary	Permanent
Emphasis on	Past and Present Cost	Present and Future Cost
Ends when	The pre-determined target is	No end

	achieved	
Type of Function	Preventive	Corrective

4.5 SCOPE OF COST REDUCTION

(1) Product Design

- One area in which manufacturers are finding ways to meet this challenge is the often-overlooked area of product design. The design of the product provides the greatest scope for cost reduction.
- There are two basic points that need to be remembered while effecting cost reduction in product design

(a) The product should perform all the functions for which it was intended, and

(b) The product should retain its esteem or aesthetic value.

Improvement in product design may result in cost reduction as illustrated below:

(a) Material Cost: change in design of the product may result in saving the material cost. Economical substitution for existing material may also be considered. Eg: in manufacturing kitchen utensils, brass may be substituted with cheaper alloys.in curtain rings, metal may be substituted with plastic.

(b) labour cost: improvement in design may result in reduced operating time.

(c) factory overhead: reduce labour time will also reduce factory overhead.

(d) packing and transporting: compact design will reduce cost.

(e) cost of tools jigs and fix12tures can be reduced.

(2) Organisation

Cost reduction may also be achieved by improving factory organisation in the form of clear-cut lines of authority and responsibility, well-defined channels of communication, co- ordination and co-operation conductive to efficiency.

(3) Production

A cost reduction programme should make a study of sequence of operations to find

out the best one, to use the most suitable machines for the work, to use jigs and fixtures to reduce operating time, to reduce idle time, to reduce scrap by the use of better quality tools ,to provide better working conditions conductive to efficiency.

4) Administration

An organisation should make efforts to reduce the cost of administrative expenses, as there is ample scope to do so. A company may evaluate and reduce the cost of following expenses, but not the cost of efficiency:

a. Modifying range of discounts

b. Modifying internal and external communication Telephone expenses

c. Travelling expenses Salary by reducing staff

d. Reduction in cost of stationery Postage and Telegrams

e. Eliminate Subscriptions a15nd Memberships

(5) Marketing

In this function, costs can be reduced by revising the methods of remuneration of salesmen, re- arrange territorial responsibilities of sales representatives, modifying current methods of advertising, improving product design and production quality so as to reduce after sales service, economizing channels of distribution, improving packing etc.

(6) Finance

Finance is an important area where cost reduction is possible through the following measures

a. control over utilisation of finance meant for both working capital and fixed capital needs.

b. proper evaluation of investments in new projects.

c. appropriate control of capital expenditure.

d. profitable employment of capital with the objective of maximum return.

4.6 TOOLS AND TECHNIQUES OF COST REDUCTION AND CONTROL

(1) Standard Costing

Standard costing is one of the prominently used systems of cost control. It aims at establishing standards of performance and target costs which are to be achieved under a given set up working conditions. It is a pre-determined cost which determines what each product or service should cost under certain situation.

Standard costing is defined as the preparation and use of standard costs, their comparison with actual costs and the measurement and analysis of variances to their causes and points of incidence. Standard costs should be obtained under efficient operations.

It starts with an estimate of what a product should cost during a future period given reasonable efficiency Standard costs are established by bringing together information collected from various sources within the company.

The degree of success is measured by a comparison of actual performance and standard performance.

Advantages of Standard Costing

(i) It helps in establishing a yardstick with which the efficiency of performance is measured that helps to exercise control.

(ii) It provides how the clear goal is to be achieved by providing incentive and motivation to work.

(iii) It provides the management the basic information to fix selling price, transfer pricing, etc.

(iv) It facilitates delegation of authority and fixation of responsibility.

(v) It helps in achieving optimum utilisation of plant capacity.

(vi) It provides means for cost reduction.

(vii) Variance analysis and reporting is helpful for taking corrective measures.

Limitations of Standard Costing

(i) Application of standard costs is quite difficult in practice.

(ii) Frequently, standards become rigid over time and do not keep pace with changes in conditions.

(iii) If the standards are outdated, loose, inaccurate and unreliable, they are more harmful.

(iv) It standards set are higher than reasonable, they act as discouraging factor.

(v) When there are random factors, it is difficult to explain variance properly.

(vi) Standard costing may be found to be unsuitable and costly in the case of firms dealing in non-standard products.

(vii) It is difficult to distinguish between controllable and non-controllable variances.

(viii) Setting the standard costing are highly technical and mechanical

(2) Budgetary Control

Budgetary control is a system of controlling costs through preparation of budgets. Budgeting is thus only a part of the budgetary control.

Characteristics

(a) Establishment of budget for each function/department of the organisation

(b) Comparison of actual performance with the budgets on a continuous basis

(c) Analysis of variation of actual performance from that the budgeted performance to know the reasons

(d) Taking suitable remedial action, where necessary

(e) Revision of budgets in view of changes in condition

Objective of Budgetary Control

(i) Planning

(ii) Co-ordination

(iii) Communication

(iv) Motivation

(v) Control

(vi) Performance evaluation

Limitations of budgetary control

(i) The budget plan is based on estimates

(ii) Danger of rigidity

(iii) Budgeting is only a tool of management

(iv) Opposition from staff

(v) Expensive technique

(3) Inventory Control

ICAI defines inventory as tangible property held,

i) for sale in ordinary course of business.

ii) in the process of production for such sale.

iii) in the form of maintenance and supplies to be consumed in the process of production.

Meaning of inventory control.

The process whereby the investment in materials and parts carried in Stock is regulated within predetermined limits set in accordance with the inventory policy established by management.

It involves:

i) fixation of limits within which inventory is to be held.

ii) laying down of inventory policies.

iii) setting out the investment pattern.

iv) Examining the work of inventory policy and effecting changes when needed.

Techniques of inventory control

(i) min max plan

(ii) two bin system

(iii) order cycling system

(iv) ABC analysis.

(v) VED analysis

(vi) inventory turnover ratios

(4) Production Planning and Control

Production planning is a plan for the future production, in which the facilities needed are determined and arranged. A production planning is made periodically for a

specific time period, called the planning horizon. It can comprise the following activities:

1. Determination of the required product mix and factory load to satisfy customer's needs.
2. Matching the required level of production to the existing resources.
3. Scheduling and choosing the actual work to be started in the manufacturing facility"
4. Setting up and delivering production orders to production facilities.

In order to develop production plans, the production planner or production planning department needs to work closely together with the marketing department and sales department. They can provide sales forecasts, or a listing of customer orders.

Different types of production methods, such as single item manufacturing, batch production, mass production, continuous production etc. have their own type of production planning.

Production planning can be combined with production control into production planning and control, or it can be combined and or integrated into enterprise resource planning.

Production planning is used in companies in several different industries, including agriculture, industry, amusement industry, etc.

The planning process, then, provides for two types of control mechanisms: feed forward, which provides a basis for control at the point of action (the decision point); and feedback, which provides a basis for measuring the effectiveness of control after implementation. Management's role is to feed forward a futuristic vision of where the company is going and how it is to get there, and to make clear decisions coordinating and directing employee activities. Management also oversees the development of procedures to collect, record, and evaluate feedback.

(5) Standardisation and Simplification

Simplification is a process of product analysis through which unnecessary varieties and designs are eliminated. Only a limited number of grades, types and sizes of the product are retained.

Standardisation is the second step after simplification towards interchangeable manufacturing. Having selected the varieties and grades of the products to be retained as much of its manufacturing details are standardised as possible.

Since manufacturing involves a large number of decisions from selection of raw material to the process used for finishing, standardisation of some of these items reduce unnecessary repetition of work.

Use of standard components reduces inventory costs, ensures interchangeability and makes future maintenance easier.

It also reduces component cost since standard components are manufactures by mass production methods and are cheaper.

Selection of standard materials ensures physical performance and guarantees failure- free operation. Use of standard methods of production enables comparisons to be made of the standard and actual manufacturing time

The purpose of standardisation is, therefore, on one hand, efficiency and economy in the use of human effort and on the other economy of material varieties and stock quantities and therefore reduction in cost and increase in turnover.

(6) Operational research and statistical techniques

The Operation Research approach is particularly useful in balancing conflicting objectives (goals or interests) where there are many alternative courses of action available to the decision-makers. In a theoretical sense, the optimum decision must be one that is best for the organization as a whole it is often called the global optimum. A decision that is best for one or more sections of the organization is usually called suboptimum decision. Operation Research attempts to resolve the conflicts of interest among various sections of the organization and seeks the optimal solution which may not be acceptable to one department but is in the interest of the organization as a whole.

(7) Value analysis

Value analysis is an approach to cost saving that deals with product design. Here, before buying any equipment or materials, a study is made as to what purpose these things serve? Would other lower- cost design work as well? Is there a cheap material which can serve the same purpose? So value analysis is a procedure which specifies the function of products or components, establishes appropriate costs, determines the alternatives and evaluates them.

Thus the objective of value analysis is the identification of such costs in a product that do not in any manner contribute to its specification or functional value. Thus, it is the process of reducing the cost without sacrificing the predetermined standards of

performance. It is a supplementary device in addition to the conventional cost reduction methods.

Value analysis is closely related to Value Engineering. It is very helpful in industries where production is done on a large scale and in such cases even a fraction of savings in cost would help the firm significantly.

(8) Automation

Automation is certainly a means of reducing costs. It also reduces human interaction. It is the advance of automatic techniques which has changed the face of industry and commerce. The proportion of people working in manual and semi-skilled jobs has been drastically reduced.

Automation is the use of automatic control equipment to operate and control machines. Automation is being used at an increasing speed, spurred on by the development of the large scale integrated circuits printed on to silicon chips. The next stage in the development of automated controls was the use of analogue computers. It is a machine designed to process electronic signals.

There are three tasks for which automatic equipment can be used to replace the human being.

They are:

(a) Measurement,

(b) Control, and

(c) Data Processing.

There are a variety of reasons for introducing automation among which is the following:

(i) To reduce costs,

(ii) To increase quality, and

(iii) To meet shortages in skilled people.

(9) Job evaluation and Merit rating

Job evaluation is the process of analysis assessment of jobs to ascertain their comparative labour worth.

Each job factor is given a relative weightage and is allotted a number of points. Depending upon its nature and requirement of different factors in its performance, a

job secures a number of points.

Merit Rating is the comparative appraisal of individual merits of an employee. it is the qualitative or quantitative assessment of an employee's performance or his personality made by his supervisor or any other competent person.

(10) Work Study

Work Study is the systematic examination of the methods of carrying out activities such as to improve the effective use of resources and to set up standards of performance for the activities carried out.

A generic term for those techniques, particularly method study and work measurement, which are used in the examination of human work in all its contexts, and which lead systematically to the investigation of all the factors which affect the efficiency and economy of the situation being reviewed, in order to effect improvement.

(11) Method Study

Method-study concerned with "the way in which work is done (i.e., method)". It is used to simplify the way to accomplish a work and to improve the method of production.

Method-study results in a more effective use of material, plant, equipment and manpower.

Method study is essentially concerned with finding better ways of doing things.

It adds value and increase the efficiency by eliminating unnecessary operations, avoidable delays and other forms of waste

Work Measurement is the application of techniques designed to establish the time for a qualified worker to carry out specified jobs at a defined level of performance or at a defined rate of working.

Technique of work measurement: a) Time study: short cycle repetitive jobs b) Work sampling: Long cycle jobs c) Predetermined motion time standards: manual operations confined to one work centre.

Objective

- To analyse the present method of doing a job, systematically in order to develop a new and better method
- To measure the work content of a job by measuring the time required to do the

job for a qualified worker and hence to establish standard time

- To increase the productivity by ensuring the best possible use of human, machine and material resources and to achieve best quality product/ service at minimum possible cost

Organisation and Methods study

Organisation and Methods are defined as, "The systematic examination of activities in order to improve the effective use of human and other material resources". It is generally accepted to be concerned with improving the administrative work, the way it is organised and the way methods and procedures are used.

(12) Quality Control

Quality Control refers to all those functions or activities that must be performed to fill the company's Quality objectives. Quality Control aims at investigating the root cause for defects unidentified by inspection and take corrective action to overcome the defects for future production. Quality Control helps to minimize the cost of Inspection and Rejection. Quality Control is an approach to prevent the defects rather than detecting the defects. The ultimate aim is to provide products which are dependable, Satisfactory, Economical.

4.7 LET'S SUM UP

Cost control and cost reduction are essential financial strategies that help businesses manage their expenses and improve profitability, though they differ in purpose and approach. **Cost control** is the process of monitoring, regulating, and maintaining costs within pre-set budgets or standards. It is a preventive function that ensures expenses do not exceed the planned level. The process involves several steps: setting cost standards, measuring actual performance, comparing actual costs with the standards, identifying deviations, and taking corrective actions to address those differences. This helps organizations maintain operational efficiency and keep costs in check.

In contrast, **cost reduction** goes a step further by aiming for real and permanent savings in costs without compromising the quality or functionality of products and services. It is a corrective and continuous process that seeks innovative ways to lower the cost per unit of output. Unlike cost control, which focuses on adhering to existing budgets, cost reduction

involves redesigning products, improving processes, eliminating waste, and finding more economical methods of operation. The **key difference** between cost control and cost reduction lies in their objectives and scope—cost control is about maintaining expenses within limits, while cost reduction seeks to lower those limits sustainably over time.

The **scope of cost reduction** is broad and applies across various business functions such as product design, manufacturing methods, procurement strategies, inventory management, administrative tasks, and marketing. Businesses may redesign products using less costly materials, automate production to improve efficiency, or negotiate better deals with suppliers to reduce procurement costs. Streamlining administrative functions and using technology to automate tasks can also contribute to cost reduction.

To implement cost control and reduction effectively, organizations use various **tools and techniques**. These include standard costing and budgetary control to monitor and evaluate cost performance; value analysis to explore alternative, cost-effective methods; work study to improve labour and operational efficiency; and quality control to prevent defects and rework. Techniques like Just-in-Time (JIT) help reduce inventory costs by minimizing storage time and waste. Together, these methods support businesses in enhancing productivity, reducing unnecessary expenditures, and achieving long-term profitability.

KEYWORDS

Cost Control, Monitoring expenses, Standard costing, Variance analysis, Corrective action, Setting standards, Measuring performance, Cost comparison, Deviation analysis, Cost correction, Cost Reduction, Cost saving, Process improvement, Efficiency enhancement, Innovation, Product design, Production methods, Procurement, Inventory management, Administration, Marketing, Standard costing, Budgetary control, Value analysis, Work study, Quality control

4.8 REVIEW QUESTIONS

1. Which of the following best describes *cost control*?

   A. A strategy to reduce the price of raw materials

B. A process to maintain expenses within set limits

C. A technique for increasing product prices

D. A method of eliminating product features to cut costs

2. Which of the following is **not** a step in the cost control process?

A. Setting standards

B. Measuring performance

C. Increasing product price

D. Taking corrective action

3. The key difference between cost control and cost reduction is:

A. Cost control is temporary, cost reduction is permanent

B. Cost control is optional, cost reduction is compulsory

C. Cost control increases cost, cost reduction maintains cost

D. Cost control is a future strategy, cost reduction is a past review

4. Which of the following is a **technique used for cost reduction**?

A. Product bundling

B. Value analysis

C. Market segmentation

D. Customer feedback

5. _____ is a preventive function that ensures costs do not exceed the set budget.

6. _____ is a corrective and continuous process that aims at permanent cost

savings without reducing quality.

7. What do you mean by cost control and cost reduction?
8. Differentiate between cost control and cost reduction.
9. Enumerate the inventory control as a tool of cost control and reduction.
10. Write a brief note on budgetary control.

UNIT-5: INTRODUCTION TO TYPES OF COSTING

STRUCTURE

5.1 Introduction

5.2 Meaning of Cost, Costing and Cost Accounting

5.3 Evolution of Cost Accounting

5.4 Costing as an Aid to Management

5.5 Classification of Cost

5.6 Elements of Cost

5.7 Methods of Costing

5.8 Let Us Sum Up

5.9 Review Questions

LEARNING OUTCOME

After studying this unit, you will be able to:

- Learn about cost accounting.
- Describe the evolution of cost accounting.
- Know about the classification of cost

Bloom's Level of Learning

The topics outlined—ranging from the meaning and evolution of cost accounting to the classification, elements, and methods of costing—primarily align with the lower to middle levels of Bloom's Taxonomy of learning, specifically the **"Remembering," "Understanding," and "Applying"** stages. At the *remembering* level, learners are expected to recall definitions and basic concepts such as the meaning of cost, costing, and cost accounting. The *understanding* level is reflected in grasping the evolution of cost accounting and its role in aiding management decisions. Topics like classification and elements of cost, as well as methods of costing, also engage the *applying* level, where students must use their knowledge in practical scenarios such as analyzing cost structures or choosing suitable costing methods. Overall, these topics are foundational and aim to build a strong conceptual base for more advanced learning in the domain of cost and

management accounting.

5.1 INTRODUCTION

You are aware that a business firm is formed to earn profit. For this, it has to invest money and incur expenses on various items and then generates incomes by selling goods or generating services for reasonable consideration. In these activities, therefore, cost is an important subject matter to which the firm has to take care of. This is the reason why Cost Accounting is taught in commerce and management and practiced in business firms. In this unit we will discuss the preliminaries of cost accounting covering the meaning, objectives, nature and scope of cost accounting, costing as an aid to management and relationship between cost and financial accounting.

5.2 MEANING OF COST, COSTING AND COST ACCOUNTING

The term **'Cost'** is understood in a variety of ways. In ordinary language cost means expenses. But in Cost Accounting it has special meaning. The Institute of Cost and Management Accountants, London, has defined the term as, ''the amount of expenditure, actual or notional, incurred on or attributable to a given thing''. Therefore, when we use the term cost, it must relate to certain thing. Cost is always attached to a good, or service which is consumed or used. It can be regarded as the price paid for attaining the objective. When a manufacturer buys raw materials he pays price at a certain rate. This price paid by the manufacturer is called cost of materials. If he incurs additional expenses, e.g., freight, insurance etc., to bring the raw materials to his place of work, these are also added to the price to find out the cost of raw materials.

Thus cost represents a sacrifice of values, a foregoing or a release of something of value. It is the price of economic resources used as a result of producing or doing the thing costed. It is the amount of expenditure incurred on a given thing. Therefore, cost has been defined as ''the amount measured in money or cash expended or other property transferred, capital stock issued, services performed or a liability incurred in consideration of goods or services received or to be received''.

'Costing' is the process of ascertainment of cost. It is different from cost accounting. It is referred to as classifying, recording and appropriate allocation of expenditure for

determination of the costs of products or services. It consists of rules and principles of ascertainment of costs of a product or service.

Cost accounting is a broader term. The system of accounting which is concerned with determination of costs of doing something which can be manufacturing or producing an article or rendering service or even conducting any activity or function, and keeping records of costs is called 'Cost Accounting'.

To be very precise, **'Cost Accounting'** is the application of costing principles, methods and techniques in the ascertainment of costs and analysis of savings and/or expenses as compared with previous experience or with standards. Cost accounting is the technique and process of ascertainment of costs, which begins with recording of expenses or the bases on which they are calculated and ends with preparation of statistical data.

The costing terminology of I.C.M.A. London, defines cost accounting as, ''The process of accounting for costs from the point at which expenditure is incurred or committed to the establishment of its ultimate relationship with cost centres and cost units. In its widest sense, it embraces the preparation of statistical data, the application of cost control methods and the ascertainment of the profitability of activities carried out or planned.''

Wheldon defines cost accounting as, ''Classifying, recording and appropriate allocation of expenditure for determination of costs of products or services and for the presentation of suitably arranged data for purposes of control and guidance of management.''

5.3 EVOLUTION OF COST ACCOUNTING

The evolution of cost accounting as a specialised subject of study and research can be attributed to several factors. These are explained below: Earlier development of cost accounting can be credited to the mathematicians and industrial engineers, rather than accountants and managers. After industrial revolution the development of industry and system of different forms of organisation and capitalistic structure of organization gave new dimension to accounting. This had given rise to the need for more detailed analysis of costs.

Initially, attempt was made to cover transactions concerned with movement of material within the organisation and emphasis was laid on cost ascertainment.

Towards the end of 18th century, emphasis was shifted to present costing information to management so that it can assist in planning, controlling and decision making. The

need was felt for more elaborate study in Cost Accounting in U.S.A., England and later in India. After 1st World War, the Institute of Cost and Works Accountants was set up in U.K. and National Association of Cost Accountants was set up in U.S.A. The Institute of Cost and Works Accountants was set up in India in 1944 and statutory recognition was given to it in 1959. The Companies Act amendments in 1965 brought about provisions regarding maintenance of cost accounting records in cases, where Central Government considers it necessary.

5.4 COSTING AS AN AID TO MANAGEMENT

Cost accounting plays a crucial role in helping management make informed decisions by providing accurate and detailed cost-related information. Since management is the primary user of cost accounting data, it benefits in multiple ways, leading to improved efficiency, cost control, and profit optimization. Below are the key advantages of cost accounting for management:

1. Cost Control and Reduction

Cost accounting enables management to identify areas where costs can be controlled or reduced. By analyzing cost data, managers can detect inefficiencies, wastage, and unnecessary expenditures. This helps in implementing corrective measures such as process improvements, better resource utilization, and elimination of non-value-adding activities.

2. Scientific Pricing of Products

One of the critical functions of cost accounting is determining the actual cost of production. With accurate cost data, management can set and revise product prices in a structured and logical manner. This ensures that prices cover costs while remaining competitive in the market, thus safeguarding profitability.

3. Minimization of Material Losses

Proper inventory management techniques in cost accounting help reduce losses due to mishandling, theft, or obsolescence of materials. Techniques such as Just-in-Time (JIT) inventory, Economic Order Quantity (EOQ), and ABC analysis ensure optimal stock levels, reducing carrying costs and wastage.

4. Standard Costing and Variance Analysis

Standard costing sets predetermined costs for products, and variance analysis compares actual costs with these standards. Any deviations (variances) highlight inefficiencies in production, labour, or material usage. Management can take timely corrective actions to improve performance and maintain cost efficiency.

5. Optimal Production Level and Plant Utilization

Cost accounting helps determine the most efficient production level where costs are minimized, and output is maximized. By analyzing cost behaviour (fixed vs. variable costs), management can decide the best production volume to achieve economies of scale and fully utilize plant capacity.

6. Ideal Product Mix Decisions

If a company produces multiple products, cost accounting assists in determining the most profitable product mix. Techniques like marginal costing and contribution analysis help identify which products contribute the most to profits, allowing management to prioritize high-margin items.

7. Profit Planning and Maximization

Through budgeting, break-even analysis, and cost-volume-profit (CVP) analysis, cost accounting aids in profit planning. Management can forecast future profits, set financial targets, and implement strategies to maximize profitability by adjusting production, pricing, and cost structures.

8. Basis for Quotations and Tenders

When bidding for contracts or tenders, management relies on cost accounting data to prepare competitive yet profitable quotations. Accurate cost estimation ensures that the company does not underprice or overprice its offerings, securing business while maintaining profitability.

Cost accounting is an indispensable tool for management, providing essential data for decision-making, cost efficiency, and profit optimization. By leveraging cost accounting techniques, management can enhance operational efficiency, control expenses, and make strategic business decisions that drive long-term success. The benefits extend beyond the

organization, influencing stakeholders such as investors, employees, and consumers, making cost accounting a vital aspect of modern business management.

5.5 CLASSIFICATION OF COST

Very often, the management desires to know, the detail nature of costs incurred and costs to be incurred. The purpose to know the cost after it is incurred and before it is incurred, might be different. It may be to keep the cost within control or it may be used for profit planning. Many times, it is required to submit tenders, to give quotations, to prepare the price lists etc. In this unit we will focus on different classes of costs and elements of costs. We will also discuss the preparation cost sheet.

There are different types of costs. The list of all these costs which are incurred is quite long. Therefore, these costs are classified on some basis. Cost Classification is the process of grouping costs according to their common features. Costs are to be classified in such a manner that they are identified with cost centre or cost unit. Costs are generally classified on the following basis.

5.5.1 On the basis of Time: (i) Historical Cost (ii) Pre-determined Cost.

5.5.2 On the basis of Behaviour: (i) Variable Cost (ii) Fixed Cost.

5.5.3 On the basis of Controllability (i) Controllable Cost (ii) Non- Controllable Cost.

5.5.4 On the basis of Function: (i) Manufacturing Cost, (ii) Administration Cost, (iii) Selling and Distribution Cost, (iv) Research and Development Cost (v) Production Cost.

5.5.5 On the basis of Elements of Cost:

5.5.1.1 Direct cost (Direct Material, Direct Labour, Direct Expenses),

5.5.1.2 Indirect Cost, (Indirect Material, Indirect Labour, Indirect Expenses),

5.5.6 Other basis: (i) Conversion Cost, (ii) Normal Cost,

(iii) Avoidable Cost, (iv) Unavoidable Cost.

A. (i) Historical Cost :

Historical Costs are past costs which are already incurred. These are recorded

costs. These costs may be already paid or an obligation to pay in a future date as given. ***Example:*** *Salary Paid, Rent Paid, Furniture Purchased, Carriage, Goods Purchased etc.*

(ii) ***Pre-determined Cost :***

These are estimated costs. These costs may be budgeted cost or standard cost. For planning and decision making purpose these costs are considered.

Example: Budgeted Salary, Budgeted Rent, Standard cost of Material per unit etc.

B. *(i) Fixed Cost:*

Fixed cost is that portion of the total cost which remains constant irrespective of output up to the capacity limit. It is also called period cost as it depends upon the passage of time. The amount of this cost is unaffected by variations in output. These costs provide conditions for production rather than costs of production. They are created by contractual obligations and managerial decisions.

Examples of fixed cost: Rent of Building, insurance, salaries, cost of fixed assets etc.

(ii) *Variable Cost :*

Variable costs are those costs which vary in proportion to output. In other words, it is a cost which changes according to the changes in output. If the output increases, variable cost also will increase. These are generally direct costs. It is concerned with output or product. Therefore, it is called as a ‘‘product’’ cost. If the output is doubled, variable cost will also be doubled.

Example of Variable Cost: Direct material Cost, Direct labour Cost, Direct expenses, Variable overheads.

(iii) *Semi-variable Cost :*

Semi-variable Cost is also referred to as semi-fixed or partly variable cost. It remains constant up to a certain level and changes after point. These costs vary in some

degree with volume but not in direct or same proportion. Examples of Semi-variable Cost:

Repairs and maintenance of machinery, Depreciation, Supervision,

Telephone charges, Maintenance of building, etc.

C. (i) Controllable Cost:

Controllable Costs are those costs which can be influenced by the action of manager. In formulating cost reduction measures these are used. Examples of Controllable Cost: Stationery Cost, Indirect Wages, Electricity, Telephone Expenses, Traveling expenses, etc.

(ii) Non-Controllable Cost :

It is the cost which cannot be influenced by the action of manager. These costs are committed costs.

Examples of Controllable Cost: Salary, Rent, Bank Interest etc.

(D) *(i) Manufacturing cost :*

Manufacturing costs are related directly to production function. It is the cost of operating the manufacturing process.

Examples of Manufacturing cost : Direct materials, Direct labour, Direct expenses, and overheads relating to production.

(ii) *Administration cost :*

Establishment expenses are generally called Administration costs. These are the costs which are incurred for formulating the policy, directing the organisation and controlling the operations.

Examples of Administration cost: Office Rent, Stationery, Salary to office staff, Office lighting etc.

(iii) *Selling and distribution cost :*

These are the cost which relate to sale and delivery. Distribution cost is incurred for distribution of products.

Example of Selling & distribution cost: Commission to sales man, After sales expenses, Advertisements, Market research, warehousing, Packing and cartage etc.

(iv) *Research and development costs :*

These cost are incurred to discover new ideas, processes, products by experiment. It includes the cost of the process which begins with the implementation of the decision to produce a new or improved product.

Example of Research and development costs: Salary to Scientists and other staff engaged in research, Patent cost etc.

5.6 ELEMENTS OF COST

You have come through various types of costs. After discussing the classification of costs, let us now focus on Elements of Cost. Elements mean nature of items. A cost is composed of three elements: material cost, labour cost and expenses. All these three elements of cost may be direct or indirect. These are explained below.

Direct cost :

It is the cost which is directly chargeable to the product manufactured. It is easily identifiable. Direct cost consists of direct material, direct labour and direct expenses. These three elements are explained below.

Direct Material: It is the cost of basic raw material used in production of a product. No finished product can be produced without basic raw materials. It becomes a part of the product. This cost is easily identifiable and chargeable to the product. For example, leather in shoe, pulp in paper, steel in steel furniture, sugarcane for sugar etc. But it is also true that raw material for one manufacturer may be finished product for another.

Direct Labour: This cost is known as direct wages and it is the amount of wages paid to those workers who are engaged in the production process for conversion of raw materials into finished goods. The amount of wages can be easily identified and directly charged to the product.

Direct Expenses: Direct Expenses are the directly chargeable expenses to the product. It can be easily identified with the product. For example, rental charges paid nor payable for a special machine used for manufacturing a product, cost of designing the product or architects fees, surveyors fees, octroi duty, royalty on production, etc.

Indirect Cost :

Indirect costs are those in the total costs which cannot be identified and charged directly to the product. It has to be allocated, apportioned and absorbed over the units

manufactured on a suitable basis. It consists of the three elements:

a) **Indirect Material:** These are expenses on such materials which are not visible in the product but are required to produce it. Examples of indirect materials are lubricants, grease, cotton waste, stationery etc.
b) **Indirect Labour:** It is the amount of wages paid to those workers who are not engaged on the manufacturing process.
c) **Indirect Expenses:** It is the amount of expenses which is not chargeable to the product directly. It is the cost of giving service to the production department. It includes factory expenses, administrative expenses, selling and distribution expenses etc.

5.7 METHODS OF COSTING

Costing refers to the technique and processes of determining cost of goods manufactured or services rendered. The basic principles of ascertaining costs are the same in every system of cost accounting. But the method of analyzing and presenting the cost may vary from industry to industry. The method of costing to be used for collecting and presenting cost depends on the nature of production. There are actually two methods of costing, namely Job costing and Process Costing.

JOB COSTING: The terminology of ICMA defines job costing as "that form of specific order costing which is applies where work is undertaken to customers' special requirements" The job is usually carried out in the factory. Each job is treated as a distinct unit and all cost incurred for the job is recorded separately.

Merits and Demerits:

The following are the merits of Job Costing:

- It records costs in a more accurate and systematic manner.
- It facilitates comparison of cost of two jobs and thereby enables the management to ascertain the profitability of the jobs and accept only such jobs which are more profitable.
- It facilitates preparation of cost estimates of similar jobs.
- It facilitates better cost control through comparision of actual costs with that of the estimates.

- It facilitates easy identification and control of spoilage and defectives.
- It serves as a good basis for preparation of price quotations and tenders.

The demerits of the Job Costing are as follows:

- It proves to be an expensive method of costing because all costs related

  to the job are accumulated and ascertained separately which involves a great deal of time, energy and money.
- The possibilities of mistakes are high as the cost of one job may be wrongly posted to another job.
- Cost comparision may not always be possible particularly when there occurs a drastic change in the economy.

Applicability:

This type of costing is suitable to printers, machine tool manufacturers and furniture manufacturers, interior decorators, advertising firms etc.
The variants of job costing are:

BATCH COSTING: The ICMA defines it as "that form of specific order costing which applies where similar articles are manufactured in batches either for sale or for use within the undertaking. In most cases the costing is similar to job costing." In other words, it is a method of costing where the cost of a group of product is ascertained. Costs are collected according to batch order number and the total cost is divided by the numbers in a batch to find out the cost of each product.

Merits and Demerits:

The following are the merits of Batch Costing:

- The burden of accounting work is considerably reduced as the batch constitutes a group of homogeneous items.
- The variations in the costs arising under this method of costing are smoothened as a result of averaging the costs. It means the total costs are spread

  over the articles produced by a particular batch. As such, it gives a consistent cost of production for every article produced

by a particular batch.

- It can reap the benefit of reduced cost of production arising from the use of economic batch quantity.
- Under this method, the supervision can be made more effective and the supervision cost can also be minimized by spreading the same over all the units that constitute the batch. Thus, it can take care of the problem of idle time of the supervisor as well as of the workers.
- The loss of time occurring in job costing as a result of inter-transfer of materials and labour etc is minimized under this batch costing.

This method of costing also suffers from the following demerits:

- There arises some problems with regard to determination of an ideal quantity or size for forming the batch.

  When quantity of goods to be produced differs from customer to customer, then again it creates some problem to determine the batch size.
- Another problem associated with this method is that if the production of a batch is wrongly carried out either owing to use of sub-standard

  materials or defective operations then the whole batch of goods will have to be discarded which is a big loss to the producer.

Applicability:

It is usually applied in general engineering factories, biscuit factories and pharmaceutical industries, ready made garments, electronic items like TV and computers etc.

CONTRACT COSTING: The ICMA defines it as "that form of specific order costing which applies where work is undertaken to the customers' special requirements and each order is of a long duration. The work is usually of a contractual nature. Generally, the method is similar to job costing, although it has some distinctive features." For each individual contract, a separate account is kept to record the expenses incurred for executing the contract. The method is applicable in construction works, construction industries engaged in roads, building and bridges construction etc.

PROCESS COSTING:

Process Costing is used where production moves from one processor department to the

next until its final completion and there is a continuous mass production of identical units through a series of processing operations. According to Kohler's Dictionary for Accountants, process costing is "a method of cost accounting whereby costs are charged to processes or operations and averaged over the units produced." ICMA defines process costing as "that form of operation costing which applies where standardized goods are produced."

Characteristics or features of Process Costing:

The following are the characteristics of features of Process Costing:

- The factory is divided into various process or departments which perform a certain limited operation. Each process or department is known as cost centre.

  The finished products are uniform in all respects such as size, shape, quality, colour and chemical content etc.
- The output of one process goes to constitute input of subsequent processes until finished product emerges therefrom.
- It is not possible to distinguish finish products while they are in the stage of processing.
- The cost of production of one process transferred to subsequent processes.
- It is quite common to incur normal loss and wastages. There may

  occur some extra loss in addition to the normal loss owing to some abnormal reasons.
- Production of the main output is sometimes accompanied by some secondary output. It is called 'joint product' if the secondary output is

  also of equal value. It is known as by-product if it is of small economic value.
- The semi-finished products (work-in-progress) are expressed in terms of completed units. It is technically called 'equivalent units'.
- The cost of production is ascertained at each stage of production process and finally after completion of the final products.
- Production process is predetermined and a definite sequence of production is

followed. The raw materials and work-in-progress flow from one process to another according to the sequence of production.

- The process itself constitutes the cost unit for the purpose of costing.
- Production is undertaken on a continuous basis and on a large scale in anticipation of demands.

Merits and Demerits of Process Costing:

The following are some of the important merits of Process Costing:

- The cost of process as well as finished products can be computed conveniently at short intervals.
- Control over costs and production can be gainfully reaped as predetermined and actual data are available for comparision for each of the processes.
- It involves lesser amount of time, energy and money because of simplicity of this method of costing. The average cost of homogeneous products can be easily computed.
- Expenses can be allocated to different processes on some rational or equitable basis and therefore there is no difficulty in calculating the accurate cost.
- It enables correct valuation of the closing inventories.

This method of costing is not free from some demerits. The following are some of the demerits of Process costing:

- The cost ascertained at the end of the process is called historical costing which is of little use for managerial control. Since it is based

  on historical costs, it naturally has all the weaknesses of historical costing.
- The system of costing conceals weaknesses and inefficiencies in processing.
- It does not evaluate the efforts put in by the individual workers and supervisors.
- The valuation of work-in-progress on the basis of degree of completion is merely a guess work or estimation.
- If production is not homogeneous, then the average cost may give a misleading

picture of cost.

Applicability:

The Process costing is applied in the following industries:

- Manufacturing industries such as iron and steel, cement, paper, rubber, ceramics, automobile plants and paint industries.
- Chemical industries such as perfumery, soap, oil and medicines etc.
- Mining industries such as mineral oil, coal, gold, iron, zinc and gas etc.
- Public utility industries such as generation and distribution of electricity, gas supply and water supply etc.

5.8 LET US SUM UP

The concept of **cost** is always associated with specific goods or services, representing the monetary expression of the resources used to produce or obtain them. **Costing** refers to the body of rules, principles, and techniques used to determine the cost of a product or service in a systematic and standardized manner. It serves as the foundation for accurate cost measurement, supporting managerial decisions. **Cost accounting** is the technique and process of ascertaining costs, recording related expenses, and preparing relevant statistical data to aid internal control and decision-making. It broadly encompasses three major functions: **ascertainment of costs**, **cost control**, and **cost reporting or presentation** to management.

Cost accounting starts where **financial accounting** ends. While financial accounting focuses on recording and summarizing financial transactions for a specific period, cost accounting dives deeper into analyzing the cost of production and operations. One of the key functions of cost accounting is the **classification of costs**, which involves grouping costs based on their common characteristics. Costs can be classified according to **time**, **behavior** (fixed, variable, semi-variable), **controllability** (controllable or uncontrollable), **function** (production, administration, selling), and **elements** (material, labor, and expenses).

When classified by **elements**, costs are divided into **direct** and **indirect costs**. **Direct costs** are those that can be directly traced to a product, such as direct materials, direct labour, and direct expenses. On the other hand, **indirect costs** cannot be directly attributed to a single product and include indirect materials, indirect labour, and indirect expenses. Understanding these classifications is crucial for effective cost control, budgeting, and pricing strategies in any organization.

KEYWORDS

Monetary value, goods/services, resource usage, Rules and principles, cost ascertainment, standard method, Cost Accounting, Technique, expense recording, statistical data, managerial tool, Functions of Cost Accounting, Ascertainment, control, reporting, Relation with Financial Accounting, Starts where financial accounting ends, Financial Accounting, Records transactions, financial nature, specific period, Time, behaviour, controllability, function, elements, Behaviour of Cost, Elements of Cost, Direct Costs, Indirect Costs.

5.9 REVIEW QUESTIONS

1. Which of the following best describes *Cost Accounting*?

   A. It records income and expenses of a business.

   B. It is used for the preparation of financial statements.

   C. It provides detailed cost information for control and decision-making.

   D. It is concerned with tax calculations.

2. Which method of costing is most suitable for the construction industry?

   A. Process Costing

   B. Batch Costing

   C. Contract Costing

D. Job Costing

3. In cost classification, which of the following is considered a fixed cost?

   A. Direct material

   B. Factory rent

   C. Power used in machines

   D. Packing material

4. Which of the following is *not* a direct element of cost?

   A. Direct materials

   B. Direct wages

   C. Direct expenses

   D. Office rent

5. The three main elements of cost are:

   A. Material, Labour, and Overheads

   B. Assets, Liabilities, and Capital

   C. Revenue, Expense, and Profit

   D. Fixed, Variable, and Semi-variable

6. The process of determining the cost of a product or service is known as _____.
7. Costs that vary with the level of output are called _____ costs.
8. What is Cost Accounting?
9. ‘‘Cost Accounting has become an essential tool of management’’. Comment.
10. Discuss the various classification of cost.

UNIT-6: MARGINAL, ABSORPTION AND ACTIVITY BASED COSTING

STRUCTURE

6.1 Introduction

6.2 Important Terms

6.3 Characteristics of Marginal Costing

6.4 Determination of Cost and Profit under Marginal Costing

6.5 Absorption Costing

6.6 Characteristics of Absorption Costing

6.7 Activity-Based Costing

6.8 Benefits of Activity-Based Costing

6.9 Differences in Approach

6.10 Let Us Sum Up

6.11 Review Questions

LEARNING OUTCOME

After studying this unit, you will be able to:

- Describe the important terms of marginal costing.
- Discuss the determination of cost and profit under marginal costing.
- Learn about the concept of absorption costing

Bloom's Level of Learning

This chapter ranging from marginal costing and absorption costing to activity-based costing (ABC)- cover fundamental concepts in managerial accounting, each requiring different levels of cognitive engagement as per **Bloom's Taxonomy**. The initial sections, such as **important terms** and the **characteristics of marginal and absorption costing**, primarily align with **Level 2 (Understanding)**, where learners must grasp and explain key definitions and principles. Moving forward, sections like **determination of cost and profit under marginal costing** and **absorption costing** involve **Level 3 (Applying)**, as they require students to use formulas and techniques to compute costs and profits. The

discussion on **differences in approach** and **benefits of ABC** elevates the learning to **Level 4 (Analyzing)**, where learners compare methods and assess their advantages.

6.1 INTRODUCTION

Management accounting serves as a vital decision-making tool for organizations by providing relevant financial and operational information. As introduced in our first unit, this system equips managers with the necessary data to make informed choices that drive business performance. Among its various analytical techniques, marginal costing emerges as a particularly valuable method for examining the critical relationship between cost, production volume, and profitability. This approach focuses on how costs behave at different activity levels and how this behaviour impacts overall financial results.

Marginal costing, fundamentally, is a specialized costing technique that distinguishes between variable and fixed costs to better understand their effect on business decisions. At its core, it examines the incremental costs associated with producing one additional unit - what we term as 'marginal cost'. To fully grasp and effectively apply this concept, we must first establish clear definitions of key terms such as marginal cost itself, contribution margin, break-even point, and profit-volume ratio. These foundational elements form the building blocks for more advanced applications of marginal costing in real-world business scenarios.

The technique's true value lies in its ability to provide clarity in decision-making processes. By separating variable costs from fixed costs, marginal costing offers managers a transparent view of how changes in production levels affect overall profitability. This clarity becomes particularly crucial when making important business decisions regarding pricing strategies, product mix optimization, cost control measures, and determining the most profitable levels of operation. As we delve deeper into this unit, we will explore how marginal costing serves not just as a theoretical concept, but as a practical tool that directly contributes to sound financial management and strategic planning in organizations.

6.2 IMPORTANT TERMS

The important terms have been defined as follows:

1. **Marginal Cost:** Marginal cost as understood in economics is the incremental cost of production which arises due to one-unit increase in the production quantity. As we understood, variable costs have direct relationship with volume of output and fixed costs remains constant irrespective of volume of production.

   Hence, marginal cost is measured by the total variable cost attributable to one unit.
2. **Marginal Costing:** It is a costing system where products or services and inventories are valued at variable costs only. It does not take consideration of fixed costs. This system of costing is also known as direct costing as only direct costs forms the part of product and inventory cost. Costs are classified on the basis of behavior of cost (i.e. fixed and variable) rather functions as done in absorption costing method.
3. **Direct Costing:** Direct costing and Marginal Costing is used synonymously at various places and it is so also. But the relation of costs with respect to activity level must be understood. Some costs are variable at batch level but fixed for unit level and likewise variable at production line level but fixed for batches and units.
4. Differential and Incremental Cost: Differential cost is difference between the costs of two different production levels. It is a relative representation of costs for two different levels either increase or decrease in cost. Incremental cost, on the other hand, is the increase in the costs due change in the volume or process of production activities. Incremental costs are sometime compared with marginal cost but in reality there is a thin line difference between the two. Marginal cost is the change in the total cost due to production of one extra unit while incremental cost can be both for increase in one unit or in total volume. In the Example 2 above, `1,220 is the incremental cost of producing one extra unit but not marginal cost for producing one extra unit.

6.3 CHARACTERISTICS OF MARGINAL COSTING

The technique of marginal costing is based on the distinction between product costs and period costs. Only the variables costs are regarded as the costs of the products while the fixed costs are treated as period costs which will be incurred during the period regardless of

the volume of output. The main characteristics of marginal costing are as follows:

1. All elements of cost are classified into fixed and variable components. Semi-variable costs are also analyzed into fixed and variable elements.
2. The marginal or variable costs (as direct material, direct labour and variable factory overheads) are treated as the cost of product.
3. Under marginal costing, the value of finished goods and work–in–progress is also comprised only of marginal costs. Variable selling and distribution are excluded for valuing these inventories. Fixed costs are not considered for valuation of closing stock of finished goods and closing WIP.
4. Fixed costs are treated as period costs and are charged to profit and loss account for the period for which they are incurred.
5. Prices are determined with reference to marginal costs and contribution margin.
6. Profitability of departments and products is determined with reference to their contribution margin.

6.4 DETERMINATION OF COST AND PROFIT UNDER MARGINAL COSTING

For the determination of cost of a product or service under marginal costing, costs are classified into variable and fixed. All the variable costs are part of product and services while fixed costs are charged against contribution margin.

Cost and Profit Statement under Marginal Costing

Details	**Amount**	**Amount**
Revenue		xxx
(A)		
Product Cost:	xxx xxx	
- Direct Materials	xxx xxx	
- Direct employee (labour)	xxx	
- Direct expenses	xxx xxx	
- Variable manufacturing overheads		Xxx xxx

Product(Inventoriable)Costs:		
(B)	xxx xxx	xxx xxx
Product Contribution Margin {A – B}		
- Variable Administration overheads		
- Variable Selling & Distribution overheads		xxx xxx
Contribution Margin: (C)		
Period Cost:		
(D)		
Fixed Manufacturing expenses		
Fixed non-manufacturing expenses Profit/ (loss) {C – D}		

Product (Inventoriable) Costs: These are the costs which are associated with the purchase and sale of goods (in the case of merchandise inventory). In the production scenario, such costs are associated with the acquisition and conversion of materials and all other manufacturing inputs into finished product for sale. Hence, under marginal costing, variable manufacturing costs constitute inventoriable or product costs. Finished goods are measured at product cost. Work-in-process (WIP) inventories are also measured at product cost on the basis of percentage of completion.

1. **Contribution:** Contribution or contribution margin is the difference between sales revenue and total variable costs irrespective of manufacturing or nonmanufacturing.

   Contribution (C) = Sales Revenue (S) – Total Variable Cost (V)

   It is obtained by subtracting variable costs from sales revenue. It can also be defined as excess of sales revenue over the variable costs. The contribution concept is based on the theory that the profit and fixed expenses of a business is a ‘joint cost’ which cannot be equitably apportioned to different segments of the business. In view of this difficulty the contribution serves as a measure of efficiency of operations of various segments of the business. The contribution

forms a fund for fixed expenses and profit as illustrated below: Example: Variable Cost = `50,000,

Fixed Cost = ` 20,000, Selling Price = ` 80,000

Contribution = Selling Price – Variable Cost

= ` 80,000 – ` 50,000 = ` 30,000

Profit = Contribution – Fixed Cost

= ` 30,000 – ` 20,000

= `10,000

Since, contribution exceeds fixed cost; the profit is of the magnitude of ` 10,000. Suppose the fixed cost is ` 40,000 then the position shall be:

Contribution – Fixed cost = Profit or,

= ` 30,000 – ` 40,000

= - ` 10,000

The amount of `10,000 represent extent of loss since the fixed costs are more than the contribution. At the level of fixed cost of `30,000, there shall be no profit and no loss.

2. **Period Cost:** These are the costs, which are not assigned to the products but are charged as expenses against the revenue of the period in which they are incurred. All fixed costs either manufacturing or non-manufacturing are recognized as period costs in marginal costing.

6.5 ABSORPTION COSTING

Absorption costing represents a traditional costing methodology that accounts for all manufacturing expenses associated with product creation. This systematic approach enables businesses to fully absorb both direct and indirect production costs into product pricing. The direct costs component encompasses tangible production inputs such as raw materials and direct labour, while indirect costs incorporate manufacturing overheads including factory expenses, administrative costs, regulatory compliance expenditures, and insurance premiums.

The fundamental distinction between absorption costing and marginal costing lies in their treatment and classification of expenses. Absorption costing adopts a functional classification approach, categorizing costs based on their operational purpose within the manufacturing process. This method allocates fixed manufacturing overheads across production units using a predetermined absorption rate, typically based on estimated output levels. However, this approach presents a potential limitation - when actual production deviates from projected volumes, it results in either over-absorption or under-absorption of fixed costs, creating accounting discrepancies.

In contrast, marginal costing demonstrates greater flexibility in this regard by maintaining a clear separation between variable and fixed costs. The contribution margin (sales revenue minus variable costs) serves as a dynamic fund that absorbs fixed expenses, eliminating the estimation challenges inherent in absorption costing. This fundamental difference in cost treatment makes marginal costing particularly valuable for decision-making scenarios requiring precise cost-volume-profit analysis, while absorption costing remains essential for financial reporting and inventory valuation purposes under standard accounting principles. Both methods offer distinct advantages and serve complementary roles in comprehensive cost management systems.

6.6 CHARACTERISTICS OF ABSORPTION COSTING

1. The costs observed under **absorption costing** include variable costs, fixed costs, and semi-variable costs. Variable costs increase or decrease in the proportion of the goods produced. Fixed costs do not alter irrespective of the quantity of production. Semi-variable costs increase or decrease in batches.
2. Absorption costing is part of **accounting** methods and procedures. Absorption costing determines the cost of the **inventory** at the end of an accounting period. The closing inventory also consists of fixed costs, thus increasing the **value** of the inventory. This method of inventory valuation increases the profit of the company.
3. Absorption costing is also known as full costing since it includes all the costs associated with production. Variable costs are direct labour and material costs.

   Fixed costs include rent, security, and **insurance** expenses. Semi-variable costs include electricity charges for the factory. Thus, under full costing, all the **expense** are absorbed by the product irrespective of the product being sold.

4. Absorption costing enables **precise** accounting for the overall cost of production, unlike in variable costing, which considers only variable costs. The method of absorption costing enables reporting of high profit with a high value of closing inventory. This is because the cost of production is completely absorbed.
5.

The presentation of information to management under the two costing techniques is as under:

Income Statement (Absorption Costing)

Details	**Amount**
Sales	xxx
Production Costs:	
Direct material consumed Direct labour cost	xxx xxx
Variable manufacturing overhead Fixed manufacturing overhead	xxx xxx xxx xxx
Cost of Production	
Add: Opening stock of finished goods	xxx xxx
(Value at cost of previous period's production)	xxx
Less: Closing stock of finished goods (Value at production cost of current period) .	xxx
Cost of Goods Sold	
Add: (or less) Under (or over) absorption of fixed Manufacturing overhead	xxx xxx xxx
Add: Administration costs xxx	
Selling and distribution costs xxx	
Total Cost	
Profit (Sales – Total cost)	

Income Statement (Marginal Costing)

Details	**Amount**
Sales	xxx
Variable Manufacturing Costs: Direct material consumed Direct labour	xxx xxx
Variable manufacturing overhead Cost of Goods Produced	xxx xxx
Add: Opening stock of finished goods	xxx
(Value at cost of previous period's production)	xxx
Less: Closing stock of finished goods (Value at production cost of current period)	
Cost of Goods Sold	xxx xxx
Add: Variable administration, selling and dist. overhead Total Variable Cost	xxx xxx
Add: Selling and distribution costs Contribution (Sales-Total variable cost)	xxx xxx
Less: Fixed costs (Production, admin, selling and dist.)	xxx
Net Profit	

It is evident from the above that under marginal costing technique the contributions of various products are pooled together and the fixed overheads are met out of such total contribution. The total contribution is also known as gross margin. The contribution minus fixed expenses yields net profit. In absorption costing technique cost includes fixed overheads as well.

6.7 ACTIVITY-BASED COSTING

Activity-based costing, also known as ABC, is an accounting method that identifies a company's activities and assigns costs to units produced by the company based on the number of activities used by each unit. Activity-based costing first determines the

purpose and cost of each activity performed by a company and then assigns a proportionate cost to every individual unit produced based on its consumption of those activities.

Let's say a company spends Rs. 2, 00,000 per year on equipment setup. Under activity-based costing, it would then attempt to assign a proportion of that 2, 00,000 to each unit it produces. However, doing so is not just a simple matter of taking that 2,00,000 and dividing it by the number of units produced. Instead, the company would need to figure out which units or products utilize which equipment the most, and then assign each unit a cost based on its individual consumption of that usage.

Companies rely on activity-based costing to better understand the true costs of manufacturing or producing products. The downside of activity-based costing is that it can be a time-consuming system to follow. Furthermore, some indirect costs can be difficult to assign to an individual unit or product produced. For example, if a company pays 1,00,00,000 in administrative staff salaries and manufactures a number of different products, it can be tricky to assign that 1,00,00,000, or portions thereof, to individual products or units.

The ABC calculation is as follows:

1. Identify all the activities required to create the product.
2. Divide the activities into cost pools, which includes all the individual costs related to an activity—such as manufacturing. Calculate the total overhead of each cost pool.
3. Assign each cost pool activity cost drivers, such as hours or units.
4. Calculate the cost driver rate by dividing the total overhead in each cost pool by the total cost drivers.
5. Divide the total overhead of each cost pool by the total cost drivers to get the cost driver rate.
6. Multiply the cost driver rate by the number of cost drivers.

6.8 BENEFITS OF ACTIVITY-BASED COSTING

Activity-based costing (ABC) enhances the costing process in three ways. First, it expands the number of cost pools that can be used to assemble overhead costs. Instead of accumulating all costs in one company-wide pool, it pools costs by activity.

Second, it creates new bases for assigning overhead costs to items such that costs are allocated based on the activities that generate costs instead of on volume measures, such as machine hours or direct labor costs.

Finally, ABC alters the nature of several indirect costs, making costs previously considered indirect—such as depreciation, utilities, or salaries—traceable to certain activities. Alternatively, ABC transfers overhead costs from high-volume products to low-volume products, raising the unit cost of low-volume products.

6.9 DIFFRENCES IN APPROACH

Absorption costing and activity-based costing differ in approach. Absorption costing assigns costs to individual units, whereas activity-based costing focuses on company activities as a central cost and then attempts to assign indirect costs to units.

One major advantage of activity-based costing is that it allows companies to understand the true cost and profitability of individual units produced or services rendered. This increased accuracy is achieved by essentially converting indirect costs to direct costs. In fact, activity-based costing can be applied to all business costs, not just production-related overhead. For instance, a company can assign its marketing costs directly to the individual units it produces. Because of this, activity-based costing can paint a more precise picture than absorption costing. On the other hand, activity-based costing can be an expensive system to implement, and it may not be as useful to companies whose overhead costs are primarily volume-related, or to companies whose overhead represents a small proportion of their overall costs.

Absorption costing, meanwhile, is easier to implement yet recognized as perfectly compliant with generally accepted accounting principles and IRS reporting requirements. The downside, however, is that it may offer less insight to those charged with making strategic decisions regarding production practices and costs.

6.10 LET US SUM UP

Modern managerial accounting relies on several sophisticated costing techniques to enhance decision-making, optimize resource allocation, and improve profitability. **Marginal costing** serves as a dynamic tool that distinguishes between

variable and fixed costs, focusing on contribution margin analysis to evaluate cost-volume-profit relationships. This approach provides clarity in short-term decision-making, particularly for pricing strategies, product mix optimization, and break-even analysis. In contrast, **absorption costing**, compliant with traditional financial reporting standards, allocates both variable and fixed manufacturing overheads to products, offering a comprehensive view of total production costs but potentially distorting profitability analysis due to inventory valuation effects.

The emergence of **activity-based costing (ABC)** addresses limitations in conventional methods by tracing overhead costs to specific activities and cost drivers, enabling more accurate product costing in complex, multi-product environments. ABC proves particularly valuable for identifying process inefficiencies, eliminating non-value-added activities, and enhancing cost transparency. While marginal costing excels in tactical decisions and absorption costing meets external reporting requirements, ABC provides strategic insights for long-term cost management. Together, these approaches form a complementary toolkit, allowing managers to select the most appropriate method based on specific organizational needs—whether for operational control, financial compliance, or strategic cost reduction initiatives. The judicious application of these techniques empowers businesses to achieve both financial precision and competitive advantage in increasingly complex market environments.

KEYWORDS

Managerial Accounting, Costing Techniques, Decision-Making, Resource Allocation, Marginal Costing, Variable Costs, Fixed Costs, Contribution Margin, Cost-Volume-Profit (CVP) Analysis, Pricing Strategies, Product Mix Optimization, Break-Even Analysis, Absorption Costing, Financial Reporting Standards, Manufacturing Overheads, Inventory Valuation, Activity-Based Costing (ABC), Cost Transparency, Strategic Cost Management, Operational Control

6.11 REVIEW QUESTIONS

1. In marginal costing, fixed costs are treated as:

   a) Product costs

b) Period costs

c) Variable costs

d) Semi-variable costs

2. Absorption costing differs from marginal costing in:

   a) Including only variable costs in product cost

   b) Including both fixed and variable costs in product cost

   c) Ignoring fixed costs completely

   d) Considering only administrative costs

3. Activity-Based Costing (ABC) is particularly useful for:

   a) Allocating fixed costs equally to all products

   b) Tracing overhead costs based on cost drivers

   c) Ignoring indirect costs in decision-making

   d) Using only direct labour hours for cost allocation

4. The break-even point is the level of sales where:

   a) Total revenue exceeds total costs

   b) Total revenue equals total variable costs

   c) Total revenue equals total costs (no profit or loss)

   d) Fixed costs are minimized

5. In marginal costing, the excess of sales over variable costs is known as _____.

6. Absorption costing follows _____ accounting standards, whereas marginal costing is used for internal decision-making.

7. Activity-Based Costing allocates overhead costs based on _____ rather than traditional volume-based methods.

8. Under marginal costing, inventory is valued at _____ cost only.

9. What is ABC?

10. Difference between marginal and absorption costing.

UNIT-7: COST VOLUME PROFIT ANALYSIS

STRUCTURE

7.1 Introduction

7.2 Basic Assumptions

Objectives of C-V-P analysis

7.3 Let Us Sum Up

7.4 Review Questions

LEARNING OUTCOME

After studying this unit, you will be able to:

- Know the meaning objectives and assumptions of cost-volume profit analysis.
- Know the importance of cost-volume profit analysis
- Know what contribution margin is and how it is calculated.

Bloom's Level of Learning

The content presented corresponds primarily to the lower and intermediate levels of Bloom's Taxonomy, specifically **Remembering (Level 1)** and **Understanding (Level 2)**. The **Introduction** serves to establish foundational knowledge by defining Cost-Volume-Profit (C-V-P) analysis and outlining its general purpose, which aligns with the **Remembering** stage, as it involves the recall of fundamental concepts. The **Basic Assumptions** section advances to the **Understanding** level, as it requires learners to comprehend and interpret the underlying principles that govern C-V-P analysis. Meanwhile, the **Objectives of C-V-P Analysis** remains at the **Understanding** level as it simply enumerates the goals of the analysis.

7.1 INTRODUCTION

In simple words, cost-volume-profit (hereafter C-V-P) analysis is the most fundamental tool because it provides straightforward ways to study the effects of changes in costs and volume on a company's profits. C-VP analysis is important in profit planning. In C-V-P analysis executives and accountants recognize that there are many interacting variables

that affect an organization's profits-such things as the sales price of a product, the variable costs per unit, and the volume of production and sales. C-V-P analysis evaluates the relationships among these interacting variables and the effect that changes in these variables have on an organization's profits.

As the term itself suggests, C-V-P analysis is an analytical technique which examines costs and revenue behavioral patterns and their relationships with profit. The analysis separates costs into fixed and variable components and determines the levels of activity where costs and revenues are in equilibrium. We define C-V-P analysis as a mature model in such management decisions as setting selling prices, determining cost, determining the best product mix, and making maximum use of production facilities. The usual starting point in such an analysis is the determination of the company's break – even point. Thus, break – even analysis forms often a key component of the total system of C-V-P analysis which gives the executives and accountants many insights in profit planning.

7.2 BASIC ASSUMPTIONS

The technique of C-V-P analysis rests on a set of assumptions. These assumptions may be identified as the fundamental base of such analysis. The importance of identifying and criticising the underlying assumptions of C-V-P analysis rests on the practical application of C-V-P analysis. The assumptions underlying C-V-P analysis are mentioned below:

1. Total costs are separated into fixed and variable costs.
2. A firm's total revenue changes in direct proportion to changes in its unit sales volume. That is, the average sales price per unit of product is constant.
3. Fixed costs remain fixed over a relevant range of activity.
4. Variable cost per unit is also constant. Therefore, total variable costs are directly proportional to volume. There is either no inflation or, if it can be forecasted, it is incorporated into the C-V-P analysis. This eliminates the possibility of cost changes.
5. Selling prices are constant per unit.
6. Prices of factors of production e.g. material price, wage rate etc. are constant.
7. There will be no changes in firm's efficiency or productivity.
8. Changes in activity are the only factors that affect costs.

9. All units produced are sold. Inventories are significants.

10. In a multi-product firm , the sales mix will remain constant. If this assumption is not made, no weighted average contribution margin could be computed for the company.

11. There will not be any significant change in the inventory level at the beginning and at the end of the year.

12. The firm is assumed to make analysis under short run .

13. The analysis will be effective for a limited range of operation over which the firm was operating in the past and expected to operate in future. It is known as relevant range. Relevant range is the levels of activities within which a particular cost behavior does not change. [C-V-P analysis under uncertainty is treated separately.]

14. Uncertainty and risks do not exist.

It is frequently found that students are quite happy to apply C-V-P analysis principles in theoretical setting but may be unaware of these assumptions and how restrictive they really are. When these assumptions are not valid, the results of C-V-P analysis may be inaccurate. These assumptions are also termed as limitations of C-V-P analysis.

7.3 OBJECTIVES OF C-V-P ANALYSIS

Accountants and executives are uncertain about some of the variables used in C-V-P analysis, though this analysis can be used to answer several types of questions and can be helpful in decision making. The basic objective of C-V-P analysis is to establish what will happen to the financial position if the output level fluctuates. This analysis will help the management to:

1. Make reasonably accurate forecast of future profits;

2. Assess the degree of risk involved in output fluctuation. If the present activity level of the organization is very near to no profit no loss situation or the proportion of fixed cost in the cost structure is very high, the degree of risk will be high in as much as a slight fall in output will lead to a significant fall in profit. This is also known as operating risk.

3. Take different decisions that are important for the operations of business. It includes pricing decisions, make or buy decision, shut down decision and like.

4. Prepare budget for future activities.

Contribution Margin (CM)

One of the key relationships in C-V-P analysis is the contribution margin (CM) . Contribution margin is the excess of revenue over the variable costs of generating that revenue.

Contribution Margin = Sales - Variable Costs of Sales

This amount indicates the rupee figure available to 'contribute' to the coverage of all fixed expenses, both manufacturing and nonmanufacturing. The sequence here is that, contribution margin is used first to cover the fixed expenses, and then whatever remains goes toward profit. If the contribution margin is not sufficient to cover the fixed, then a loss occurs for the period.

We will use following relevant data of the Appollo Company to understand various important issues relating to C-V-P analysis.

Unit selling price : 50 Unit variable costs : 30

Total monthly fixed costs : 20,000 Units sold : 1000 units

Therefore, Contribution Margin is = **(Sales – Variable Costs of Sales)**

= (1000 units □ 50) – (1000 units □ 30)

= 20,000

This contribution margin is then available to cover fixed costs and to contribute income for the Appollo Company. In other words, this indicates how much revenue is available to cover all period expenses and potentially to provide net income.

Viewers differ as to the best way to express contribution margin (CM). Some executives and accountants favor a per unit basis. We can determine the amount that each unit contributes – the contribution margin per unit (CMPU) in two different ways. In the first way we merely subtract the variable cost per unit from the sales price per unit:

Contribution Margin Per Unit (CMPU): Sales price per unit – Variable costs per unit

At the Appollo Company, the CMPU is 20, computed as follows: CMPU= 50 – 30 = 20

The contribution margin per unit indicates that for every unit sold, Appollo company will have 20 to cover fixed costs and contribute to income. In other words, this means that each time a unit is sold, it provides, or contributes, the difference between the sales price and the unit variable costs toward covering the fixed costs needed to produce and sell all the units.

Since fixed costs are 20,000, Appollo must sell 1000 units (20,000 / 20) before there is any income. Above the sales volume of 1000 units, every unit sell will contribute 20 to

income. The second way to calculate CMPU is to divide the total contribution margin by the number of units sold.**:**

Contribution Margin Per Unit (CMPU) =Total Contribution margin

Units sold

Contribution Margin Ratio (CM Ratio) =In addition to being expressed on a per unit basis, sales revenue, variable costs, and contribution margin, some executives and accountants prefer to use another relationship in C-V-P analysis , which is known as contribution margin ratio. The contribution margin ratio (or contribution margin percentage) represents the portion of total sales that remains after the variable costs have been subtracted. This ratio is computed as follows:

Contribution Margin Ratio = Total contribution margin □ 100 %

Total sales in rupees

= 20,000 □ 100

(1000 units □ 50)

= 40%

The CM ratio of 40% means that 40 paisa of each sales rupee(1 □ 40%) is available to apply to cover fixed costs and to contribute to income. The CM ratio is extremely useful since it shows how the contribution margin will be affected by a change in total sales. This expression of contribution margin is important in determining the effect of changes in sales on income.

A second way to determine the CM ratio involves the contribution margin per unit and sales price per unit:

CMPU Contribution Margin Ratio = CMPU □ 100 %

Sales price per unit

Finally, we can calculate the contribution margin ratio by subtracting the variable cost percentage from 100%, or the variable cost ratio from 1.

Contribution Margin Ratio = 100%- Variable Cost % (VC%) or

1 – Variable cost % (VC%)

If the contribution margin ratio = 1 – VC%, then conversely the VC% =1 – CM ratio. Variable cost percentage (VC%) can be calculated in total sales form or per unit basis in the following ways:

Variable cost percentage (VC%) = Total variable costs of Sales □ 100%

Total sales Or

Variable cost per unit Sales price per unit

7.4 LET US SUM UP

Cost-Volume-Profit (C-V-P) analysis serves as a critical managerial accounting tool that examines the relationship between costs, sales volume, and pricing to determine their collective impact on profitability. At the core of this analysis lies the concept of contribution margin, which represents the residual revenue after accounting for variable costs and plays a pivotal role in break-even analysis, product profitability assessment, and operational decision-making. Contribution margin can be calculated through three distinct yet interrelated approaches: contribution margin per unit (selling price minus variable cost per unit), which is particularly valuable for unit-level decisions like pricing strategies; contribution margin ratio (contribution margin divided by sales revenue), which facilitates profitability comparisons across different products or business segments; and total contribution margin (total revenue minus total variable costs), which provides a comprehensive view for broader financial planning. The selection of the most appropriate calculation method depends on both the nature of the available data and the specific analytical objective at hand, as each approach offers unique insights tailored to different decision-making contexts. While contribution margin per unit aids in granular, product-specific evaluations, the ratio method enables relative performance assessment, and the total contribution margin supports high-level financial forecasting. A thorough command of all three methodologies equips managers with the flexibility to adapt their analytical approach based on situational requirements, thereby enhancing the accuracy and relevance of their financial insights. This versatility underscores the practical utility of C-V-P analysis in dynamic business environments, where informed decision-making hinges on the ability to select and apply the most suitable financial metrics to diverse operational scenarios. Ultimately, the strategic application of these contribution margin concepts enables organizations to optimize pricing structures, control costs effectively, and maximize overall profitability.

KEYWORDS

Cost-Volume-Profit (C-V-P) Analysis, Contribution Margin, Variable Costs, Break-even

Analysis, Profitability Assessment, Contribution Margin per Unit, Contribution Margin Ratio, Total Contribution Margin, Pricing Strategies, Financial Planning, Operational Efficiency, Revenue Optimization, Managerial Accounting

7.5 REVIEW QUESTIONS

1. Which of the following is ***not*** a basic assumption of Cost-Volume-Profit analysis?

   A. Total costs can be separated into fixed and variable components.

   B. The selling price per unit remains constant.

   C. Variable cost per unit increases with output.

   D. The volume of activity is the only relevant factor affecting cost behavior.

2. The main objective of CVP analysis is to:

   A. Calculate depreciation of fixed assets

   B. Determine the optimum production mix

   C. Evaluate the impact of cost and volume on profits

   D. Estimate the payback period of a project

3. In CVP analysis, the break-even point is the level of sales at which:

   A. Total cost exceeds total revenue

   B. Fixed costs equal variable costs

   C. Total revenue equals total costs

   D. Profit margin is maximized

4. CVP analysis is most useful for:

A. Long-term strategic planning

B. Budget preparation

C. Short-term profit planning

D. Evaluating intangible assets

5. In CVP analysis, the _____ cost remains constant regardless of the level of production.
6. The break-even point occurs where total revenue equals _____.
7. CVP analysis assumes that all units produced are _____.
8. Explain the key assumptions of Cost-Volume-Profit analysis and discuss why they are important for managerial decision-making.
9. Discuss how a manager can use CVP analysis to make decisions regarding pricing, product mix, and profit planning. Provide examples.
10. Define the objectives of CVP analysis. How does understanding the break-even point assist businesses in planning their operations more effectively?

UNIT-8: BREAK-EVEN ANALYSIS

STRUCTURE

8.1 Introduction

8.2 Algebraic Method of Break Even Analysis (BEA)

8.3 How to draw Break Even Chart

8.4 Computation of Break Even Point

8.5 Break Even Point of Merged Plant

8.6 Margin of Safety

8.7 Angle of Incidence

8.8 Let Us Sum Up

8.9 Review Questions

LEARNING OUTCOME

After studying this unit, you will be able to:

- Learn use of algebraic method to do break even analysis.
- Know how to calculate break-even point (BEP) in units and in value with relevant information.
- Draw a break even chart with given information and data.
- Evaluate cash, financial and composite break-even point.
- Calculate break even capacity of merged plants, margin of safety and angle of incidence

Bloom's Level of Learning

The topics outlined in sections covering Introduction to Break-Even Analysis, Algebraic Method of BEA, Break-Even Chart Drawing, Computation of Break-Even Point, Merged Plant BEP, Margin of Safety, and Angle of Incidence—primarily correspond to the **"Application" and "Analysis" levels** of Bloom's Taxonomy of Learning. These levels involve not just remembering and understanding theoretical concepts but also applying

mathematical and graphical techniques to solve practical business problems.

At the **Application level**, learners are expected to use learned methods such as the algebraic formula for break-even point and graphical representation through charts to compute and interpret results. This includes calculating the break-even point, margin of safety, and evaluating the impact of sales and cost structures on profit outcomes.

At the **Analysis level**, students are required to go deeper by comparing different scenarios (e.g., individual vs. merged plant break-even), interpreting the **angle of incidence** to assess profitability trends, and drawing logical conclusions from data. This level of learning enhances a learner's ability to break down complex cost-volume relationships and use them in decision-making processes, which is essential for managerial roles in business planning and control.

8.1 INTRODUCTION

Break Even Analysis (BEA) is a significant technique of **CVP** analysis. BEA can be called as a mechanism wherein correlation among variables namely, cost of production, volume of production, sales, and profit is determined and a detailed analysis is done as to determine the effect on profits due to changes in these variables. The BEA helps to compute the predictable profits at any stated level of activity or output. The BEP is the sales level at which the contribution margin is just equal to the fixed cost, and the firm has no profit no loss. At the BE sales level, the total costs (fixed + variables) are just covered. Any sales level below the BEP level will therefore results in loss and sales level above the BEP level will bring profit to the company. Once the BE sales level has been achieved, the net profit will increase by the unit contribution.

8.2 ALGEBRAIC METHOD OF BREAK EVEN ANALYSIS (BEA)

The BE equation may be derived from the basic relationship between sales and profit as follows: Net Profit (NP) = Sales (S) – Variable Cost (VC) – Fixed Cost (F)

Or, S = VC + F + NP

No of units sold (N) * SP = (No. of units sold * VC per unit) + F + NPN* SP = N * VC+ F + NP

The BE equation may be derived from the basic relationship between sales and profit as follows:
Net Profit (NP) = Sales (S) – Variable Cost (VC) – Fixed Cost
(F)Or, S = VC + F + NP
No of units sold (N) * SP = (No. of units sold * VC per unit) + F
+ NPN* SP = N * VC + F + NP

Where Contribution/Sales ratio (P/V Ratio) = Contribution margin / sales * 100

Thus the BE analysis can be done by determining the value of one or more of these variables. If the selling price is known, we can find out the number of units to be sold to break even. If the number of units expected to be sold are known, we can use this equation to find out the minimum selling price to break even.

Computation of break-even point in units and in value

XYZ Ltd. manufactures files for office use. The breakup of its cost and sales is as follows:

Variable cost per file	= Rs. 40
Fixed cost	= Rs. 60,000
Selling price	= Rs. 100 per file.

Compute the break-even point in units as well as in value (in terms of rupees).

Solution

Selling price per file	Rs. 100
Less: Variable cost per file	Rs. 40
Contribution per file	Rs. 60

Break-even point (Units) = total fixed cost/contribution per unit

= Rs. 60,000 / Rs.60 = 1000 files Now, Break-even point (value) = total fixed cost/P/V ratio

Where P/V ratio = (contribution per unit/selling price)*100 = (60/100)*100 = 60%

Break-even point (Value) = 60,000/60% = Rs.1,00,000.

8.3 HOW TO DRAW BREAK EVEN CHART

How to draw break even chart with given information and data.

For example, Fixed Cost = Rs. 6,00,000

Variable cost per unit = Rs. 40

Selling Price per unit = Rs. 70

Prepare Break even chart from the above information.

Solution:

No of units	Fixed Cost	Variable cost	Total cost	Sales	Profits/Loss
0	6,00,000	0	6,00,000	0	0
5000	6,00,000	2,00,000	8,00,000	3,50,000	-4,50,000
10000	6,00,000	4,00,000	10,00,000	7,00,000	-3,00,000
15000	6,00,000	6,00,000	12,00,000	10,50,000	-1,50,000
20000 profit,	6,00,000 no loss)	8,00,000	14,00,000	14,00,000	BEP
25000	6,00,000	10,00,000	16,00,000	17,50,000	1,50,000
30000	6,00,000	12,00,000	18,00,000	21,00,000	3,00,000

8.4 COMPUTATION OF BREAK-EVEN POINT

8.4.1 Computation of cash break-even point:

Assume that the total fixed cost of a firm for the year 2013-14 is Rs. 18,60,000 which includes a depreciation of machinery amounting to Rs. 2,50,000, notional rent of the building Rs. 1,50,000 and a deferred revenue expenditure of Rs. 60,000. If the selling priceper unit is Rs. 250 and the variable cost per unit is Rs. 150, compute the cash break even point.

Solution:

Cash contribution per unit = Rs. 250 – Rs. 150 = Rs. 100.\

Cash fixed cost = total fixed cost – non-cash fixed cost

= 18,60,000 – (2,50,000+1,50,000+60,000) = 14,00,000.

Cash break-even point (units) = cash fixed cost/cash contribution per unit

= 14,00,000/100 = 14,000 units.

Cash BEP (Rs.) = 14,000units *Rs.250 = Rs.35,00,000

Cash break-even point (value) = cash fixed cost/P/V ratio

= 14, 00,000/((250-150)/250*100

= Rs.35, 00,000

8.4.2 Computation of financial break-even point:

Assume that the information given below is provided about the firm: Share capital: Equity shares (Rs. 10) Rs.7,00,000

14% preference shares (Rs. 100) Rs.3,00,000

16% Debentures Rs.2,00,000

Tax Rate 40%

EBIT Rs.102000

Calculate EPS:

8.4.2.1 At the existing level of EBIT.

8.4.2.2 If EBIT increases to Rs. 1,50,000

SOLUTION:

Particulars	Existing EBIT (Rs. 1,02,000)	Increased EBIT (Rs. 1,50,000)
EBIT	Rs. 1,02,000	Rs. 1,50,000
Less: Interest (16% on Debentures)	Rs. 32,000	Rs. 32,000
EBT (Earnings Before Tax)	Rs. 70,000	Rs. 1,18,000
Less: Tax (40%)	Rs. 28,000	Rs. 47,200
EAT (Earnings After Tax)	Rs. 42,000	Rs. 70,800
Less: Preference Dividend (14%)	Rs. 42,000	Rs. 42,000
Earnings Available for Equity Shareholders	Rs. 0	Rs. 28,800
Number of Equity Shares	70,000	70,000
EPS (Earnings Per Share)	Rs. 0	Rs. 0.4114

NOTE:

1. At the existing EBIT (Rs. 1,02,000):
   - After paying interest, taxes, and preference dividends, no earnings remain for equity shareholders.
   - Hence, EPS = Rs. 0.
2. At the increased EBIT (Rs. 1,50,000):
   - Earnings available for equity shareholders = Rs. 28,800.
   - EPS = Rs. 28,800 / 70,000 shares = Rs. 0.4114 per share.

8.4.3 Computation of composite break-even point

ABC manufactures three types of products P1, P2 and P3. The relevant data is as under:

Product	Output	Selling price/unit	Variable cost/unit
P1	10,000 units	4	3
P2	20,000 units	5	3
P3	5,000 units	6	3

The fixed costs are Rs. 15,000. Calculate composite P/V ratio and composite break-even point.

SOLUTION

Calculate Contribution per Unit

Contribution per Unit=Selling Price−Variable CostContribution per Unit=Selling Price−Variable Cost

Product	Contribution/Unit (Rs.)
P1	4−3=14−3=1
P2	5−3=25−3=2
P3	6−3=36−3=3

Compute Total Contribution for Each Product

Total Contribution=Contribution/Unit×OutputTotal Contribution=Contribution/Unit×Output

Product	Total Contribution (Rs.)
P1	1×10,000=10,0001×10,000=10,000

Product	Total Contribution (Rs.)
P2	2×20,000=40,0002×20,000=40,000
P3	3×5,000=15,0003×5,000=15,000
Total	65,000

Calculate Total Sales Revenue

Sales Revenue=Selling Price×OutputSales Revenue=Selling Price×Output

Product	Sales Revenue (Rs.)
P1	4×10,000=40,0004×10,000=40,000
P2	5×20,000=1,00,0005×20,000=1,00,000
P3	6×5,000=30,0006×5,000=30,000
Total	1,70,000

Compute Composite P/V Ratio

Composite P/V Ratio=(Total Contribution/Total Sales Revenue)×100

Composite P/V Ratio = (65,000/1,70,000)×100= 38.24%

Calculate Composite Break-Even Point (in Rs.)

Composite BEP (in Rs.)=Total Fixed Costs/Composite P/V Ratio

=15,000/0.3824=Rs. 39,230(approx.)

Answer in Table Format:

Particulars	Value (Rs.)
Total Contribution	65,000
Total Sales Revenue	1,70,000
Composite P/V Ratio	38.24%
Composite Break-Even Point (BEP)	39,230

- The Composite P/V Ratio (38.24%) indicates that 38.24% of total sales revenue contributes to covering fixed costs and profit.
- The Composite BEP (Rs. 39,230) is the minimum sales revenue required to cover all fixed costs when all three products are sold in the given ratio.

8.5 BREAK EVEN POINT OF MERGED PLANT

Suppose two manufacturing units A and B are operating under the same management. The management desires to merge these two plants. The given below particulars are available (amount in Rs.Lakhs):

	Plant A	Plant B
Level of capacity utilization	100 %	60%
Sales	400	120
Variable costs	320	150
Fixed costs	40	20

Compute:

a) What would be the capacity of the merged plant to be operated for the purpose of break even, and

b) What would be the profitability on working at 75% of the merged capacity

Solution:

Compute Merged Plant Data

Particulars	Merged Plant
Total Sales (A + B)	400 + 120 = **520**
Total Variable Costs	320 + 150 = **470**
Total Contribution	520 - 470 = **50**
Total Fixed Costs	40 + 20 = **60**
Current Profit (Loss)	50 - 60 = **-10**

Calculate Break-Even (BEP) for Merged Plant

BEP (in Rs.)=Total Fixed /CostsP/V Ratio

P/V Ratio=(Contribution/Sales)×100=(50/520)×100=9.62%

BEP=60/0.0962=623.70 *Lakhs*(*Rs.*)

Capacity Utilization at BEP:

BEP Sales=623.70 Lakhs

Capacity Utilization for BEP=(623.70/520)×100=120

(a) The merged plant must operate at **120% capacity** to break even.

Profitability at 75% Merged Capacity

Contribution=390−352.5=37.5Lakhs

Profit=Contribution−Fixed Costs=37.5−60=−22.5Lakhs

(b) At **75% capacity**, the merged plant incurs a **loss of Rs. 22.5 Lakhs**.

Answer Table:

Particulars	Value (Rs. Lakhs)
Total Sales (Merged)	520
Total Variable Costs	470
Total Contribution	50
Total Fixed Costs	60
Current Profit (Loss)	-10
P/V Ratio	9.62%
Break-Even Sales (BEP)	623.70
Capacity Utilization for BEP	120%
Sales at 75% Capacity	390
Profit at 75% Capacity	-22.5 (Loss)

NOTE:

1. Break-Even Requirement:
    - The merged plant must operate at 120% capacity to cover all costs.
    - This suggests inefficiencies in Plant B, which is dragging profitability down.
2. Profitability at 75% Capacity:
    - The merged plant cannot sustain profitability at 75% capacity and incurs a loss of Rs. 22.5 Lakhs.
    - Suggestion is to Improve cost efficiency (especially in Plant B) or increase sales volume to achieve profitability.

8.6 MARGIN OF SAFETY

The margin of safety is the difference between total sales (actual or projected) and the break- even sales. Low BEP may leave a higher margin of safety. It may be expressed in the monetary terms or as a percentage of actual or projected sales.

Margin of Safety (M/S) = Actual or Sales Projected – Break even sales(Monetary terms) Margin of Safety (Units) = Profit /Contribution per unit

Margin of Safety (%) = (Actual sales – Break even sales) / Actual Sales * 100 Margin of Safety can also be calculated with the help of P/V ratio as under: Margin of Safety (M/S) = Profit / (P/V ratio)

The size of margin of safety is an extremely valuable guide to the strength of a business. It indicates by how much sales may decrease before a loss occurs. If large, this means that there can be substantial falling off sales and yet a profit can still be made. It is a sign of soundness of business since even with a substantial reduction in sales, profit is earned by the business. On the other hand, if the margin is small, any loss may be a serious matter. Low margin of safety implies that reduction in sales even by a small amount may the profitability of the firm very adversely. Margin of safety serves as an indicator of strength of the business firm.

If the margin of safety is unsatisfactory, possible steps to rectify the matter are listed below:

- Increase in the selling price – for this to be possible, the company must be in a very strong and favorable position. It should be able to influence the price charged. The demand must be inelastic; otherwise, the same quantity will not be sold.
- Reduce fixed costs.
- Reduce variable costs.
- Substitution of existing product(s) by more profitable lines.
- Increase the volume of output.

8.7 ANGLE OF INCIDENCE

This is the angle at which the sales line cuts the total costs line. A large incidence angle indicates that profits are being made at high rates. Together with Margin of Safety, it indicates an extremely favorable position, and sometimes may indicate monopolistic conditions. The smaller angle of incidence depicts that while profits are being made they are being made under less favorable conditions.

8.8 LET US SUM UP

The break-even point (BEP) is a crucial concept in cost and managerial accounting that indicates the level of production or sales at which a business covers all its costs, resulting in neither profit nor loss. At this point, the total revenue earned from selling goods or services is exactly equal to the total costs incurred, which include both fixed and variable costs. The BEP can be calculated using two main methods: by dividing the total fixed costs by the contribution per unit (selling price minus variable cost per unit), or by multiplying the ratio of total fixed costs to total contribution with the total sales. This helps businesses determine the minimum output or sales volume required to avoid losses.

To better understand and analyze the break-even point, businesses often use a break-even chart. This is a graphical representation that shows the relationship among various factors like total costs, total revenue, volume of output, and the resulting profit or loss at different levels of activity. It visually displays how changes in output affect profitability, making it a useful tool for decision-making and financial planning.

Another important concept related to BEP is the margin of safety, which measures how much actual or projected sales exceed the break-even sales. It serves as a cushion, indicating the extent to which sales can drop before the business starts incurring losses. A higher margin of safety signifies greater financial stability and lower business risk.

Lastly, the angle of incidence in a break-even chart refers to the angle formed at the point where the total cost line intersects the total revenue line. This angle provides insight into the profitability of the business. A larger angle indicates a higher rate of profit beyond the break-even point, while a smaller angle suggests slower profit growth. All these concepts

collectively help businesses evaluate their cost structure, set sales targets, and make informed financial decisions.

8.9 REVIEW QUESTIONS

1. Which of the following is correct regarding the Break Even Point (BEP)?
   A. It occurs when fixed costs equal variable costs
   B. It represents the output level at which total profit is maximized
   C. It is the level of output where total costs equal total revenue
   D. It occurs when contribution per unit is zero
2. In a break-even chart, the point at which the total cost and total revenue curves intersect is known as:
   A. Margin of Safety
   B. Angle of Incidence
   C. Contribution Point
   D. Break Even Point
3. Which of the following formulas is used to compute the Break Even Point in sales value?
   A. Fixed Costs × Selling Price
   B. (Fixed Costs / Contribution) × Sales
   C. Variable Costs / Fixed Costs
   D. Fixed Costs / Profit
4. A larger angle of incidence in a break-even chart indicates:
   A. Low contribution per unit
   B. Higher losses beyond the break-even point
   C. Higher rate of profit after BEP
   D. Lower sales volume
5. The formula for calculating Break Even Point in units is _____.
6. The _____ represents the excess of actual sales over the break-even sales.
7. In a merged plant analysis, the overall BEP is affected by the combined _____ and _____ of both plants.
8. A company produces and sells a product with the following details:
   Selling Price per unit: ₹50
   Variable Cost per unit: ₹30

Fixed Costs: ₹2,00,000

Actual Sales: ₹6,00,000

Tasks:

a) Calculate the Contribution per unit

b) Compute the Break Even Point in units and in sales value

c) Determine the Margin of Safety in sales value and percentage

d) Explain the significance of the Margin of Safety and Angle of Incidence based on the values obtained.

9. What do you mean by angle of incidence?
10. Explain the methods of Break-even analysis.

UNIT-9: MEANING, NATURE AND SIGNIFICANCE OF STANDARD COSTING

STRUCTURE

9.1 Introduction

9.2 Standard Cost

9.3 Standard Costing

9.4 Need of Standard Costing

9.5 Merits of Standard Costing

9.6 Standard costing as a management tool

9.7 Limitations of Standard Costing

9.8 Let's sum up

9.9 Review Questions

LEARNING OUTCOME

After studying this module, you shall be able to:

- Understand the meaning and concept of standard costing
- Know the need for standard costing
- Learn the merits of standard costing
- Understand standard costing as a tool of management

Bloom's Level of Learning

The content outlined in sections 9.1 to 9.7, which encompasses the concepts of standard cost, standard costing, its necessity, advantages, managerial relevance, and limitations, corresponds primarily to the **"Understanding"** and **"Analyzing"** levels of **Bloom's Taxonomy of Educational Objectives**.

Initially, the foundational sections engage learners in the **understanding** of key concepts—such as the definition of standard cost and the methodology of standard costing—by enabling them to comprehend, explain, and differentiate between related cost accounting terms. As the learner progresses to the later sections, the focus shifts towards **application and analysis**. Here, students explore the practical need for standard

costing in managerial contexts, assess its merits, and examine how it functions as a strategic tool for cost control, budgeting, and performance evaluation.

Finally, the discussion of limitations invites a deeper level of cognitive engagement, requiring learners to **analyze** the drawbacks of standard costing in modern, dynamic business environments and, where applicable, **evaluate** its effectiveness compared to alternative costing methods.

Overall, this module fosters a progression from conceptual understanding to critical analysis, and as such, it aligns with the **mid to upper levels** of Bloom's taxonomy—specifically the levels of **Understanding, Applying, Analyzing**, and potentially **Evaluating**.

9.1 INTRODUCTION

If you were to scheme a cost accounting system with no accounting education other than financial accounting courses, you would probably devise an accounting system that collects, summarizes, and reports actual costs. This approach would be consistent with the implicit assumption throughout all financial accounting courses that when financial statements report historical cost data, such as is done for cost-of-goods-sold and ending inventory, that the information reported represents actual costs. Therefore, it comes as a surprise to most students that the initial journal entries to record the production and movement of inventory in the costing systems of most manufacturing firms are not based on actual costs at all, but rather are based on budgeted per-unit costs. In most manufacturing entities, the initial journal entries to debit work-in-process, finished goods and cost-of-goods-sold are based upon the actual quantity of output produced, multiplied by budgeted data about the input which are necessary to produce those outputs, and the budgeted costs of such inputs. Then, at the end of the month (or possibly quarterly), an "adjusting" or "closing" entry is made to record in the inventory accounts the difference between actual costs incurred, and the budgeted information that has formed the foundation for the journal entries during the month. The nature of this adjusting entry depends on the materiality of the amounts concerned. If the differences between actual costs and budgeted costs are small, this adjusting entry might be made in a speedy manner, involving only cost-of-goods-sold, but if the differences are huge, the adjusting entry might also involve work-in-process and finished goods inventory accounts. The

accounting system described above is called a standard costing system, and it is extensively used by companies in the manufacturing sector of the economy. The present chapter describes standard costing systems, and explains why companies use them. But before that we discuss a related concept, standard costs, which constitutes an important component of standard costing systems.

9.2 STANDARD COST

The word "**Standard**" means a "**Yardstick**" or "**Bench Mark**." The term "Standard Costs" refers to Pre-determined costs. Standard Cost can be defined as a Pre-determined Cost which determines what each product or service should cost under given circumstances. According to it standard costs represent planned cost of a product. Standard cost as defined by the Institute of Cost and Management Accountants, London "is a pre-determined cost based on technical estimates of material, labour and overhead for selected period of time and for prescribed working conditions". A standard, that is a term usually used in management accounting, is a budgeted amount for a single unit of output. A standard cost for one unit of output is the budgeted production cost for that unit. Standard costs are calculated using engineering estimates of standard quantities of inputs, and budgeted prices of those inputs. For example, for an apparel manufacturer, standard quantities of inputs are requisite yards of fabric per jean and requisite hours of sewing operator labor per jean.

Budgeted prices for those inputs are the budgeted cost per yard of fabric and the budgeted labor wage rate.

Standard quantities of inputs can be established on the basis of ideal performance, or on expected performance, but are usually based on efficient and attainable performance.

Research in psychology has determined that most people will exercise the greatest effort when goals are somewhat difficult to achieve, but not extremely difficult. If goals are easily achieved, managers and employees might not work as hard as they would if goals are challenging. But also, if goals appear not viable, managers and employees might quit themselves to falling short of the goal, and might not work as hard as they otherwise would. For such reason, standards are often established on the basis of efficient and achievable performance.

9.3 STANDARD COSTING

Standard Costing is a concept of accounting for determination of standard for each element of costs. Such pre-determined costs are compared with actual costs to find out the deviations known as "Variances." Identification and analysis of causes for such variances and remedial measures should be taken in order to overcome the reasons for Variances.

Chartered Institute of Management Accountants England defines Standard Costing as "the Preparation and use of standard costs, their comparison with actual costs and the analysis of variances to their causes and points of incidence."

From the above definition, the technique of Standard Costing may be summarized as follows:

(1) Determination of appropriate standards for every element of cost.

(2) Ascertainment of information regarding actuals and use of Standard Costs.

(3) Comparing actual costs with Standard Costs, their differences defined as Variances.

(4) Analysing Variances to find out the causes of Variances.

(5) Reporting to the responsible authority for taking remedial measures. Historical costs are the actual costs that are ascertained after these are incurred. During the initial stages of development of cost accounting, historical costing systems like process costing, contract costing, service costing etc. were available for ascertaining the costs. The historical costing methods are used to find out the cost incurred for the production of particular products or completion of a particular job.

The historical costing systems suffer several limitations; some of which are as follows:

1. There is no basis for cost control

2. It does not have any yardstick for measuring efficiency

3. There is always a delay in availability of information

4. It is expensive

Although standard costing attempts to overcome the limitations of historical costing system, it is not a substitute to the existing historical costing. Standard costing is the

most extensively used technique of controlling costs. Standard costing is a technique that establishes predetermined estimates of the cost of products and services and compares these costs with the actual costs as they incur. Standard costing can be considered to be a yardstick to measure the efficiency with the actual cost incurred. Hence, standard costing is system of costing which makes a comparison between the standard cost of each product or service with its actual cost to determine the efficiency of the operation, with a view to take corrective actions at the earliest possible time.

Process of standard costing

1. Setting of standard costs

2. Ascertainment of actual costs

3. Comparison between the above two and measurement of variances

4. Analysis of variances

5. Reporting to responsibility centers to take appropriate and necessary remedial actions

9.4 NEED OF STANDARD COSTING

1. To facilitate the setting of budgets and evaluating managerial performance.
2. To act as a control device by stressing those activities which do not match to plan and thus alerting decision-makers to those situations that may be out of control and in need of remedial action.
3. To provide a forecast of future costs that can be used for decision making purposes.
4. To simplify the task of tracing costs to products for inventory evaluation purposes.
5. To provide a challenging target which individuals are motivated to achieve.
6. To improve cost /cost control, simplify stock valuation and to also improve costing and pricing of products.
7. To assess the performance of each segment.
8. To examine the impact of cost if sale volume increase/decrease by certain percentage.
9. To design the performance measurement system to encourage the employees to

participate for the betterment of the organisation.

10. To the efficiency of production.

9.5 MERITS OF STANDARD COSTING

1. **Measurement of Efficiency**: - Standard cost provides a benchmark to measure efficiency or inefficiency. This makes possible application of the principle "Management by Exception". The operations, which meet standard performance or reaches closely, are ignored. The attention of the management would be drawn to those distinct areas, where action is essentially needed.

2. **Fixing Prices & Formulating Policies**: - Prices may be fixed, by adding the standardised margin of profits to standard cost. It enables the management to quote prices for potential orders to push sale of products. This serves as a valuable aid to management in formulation of pricing policies.

3. **Eliminates Wastages**: -By fixing standards, certain wastes such as material wastages, idle time and machine hours etc. can be reduced.

4. **Inventory Valuation**: -Standard Costing simplifies valuation of inventory. This ensures uniform valuation of stock-raw materials, work-in-progress and finished goods over a period. The difference between standard and actual cost is transferred to a variance account.

5. **Creates Cost Consciousness**: -Targets are laid down for the employees to achieve them. Additionally, Standard Costing creates cost consciousness amongst the employees, irrespective of their level, in the whole organization.

6. **Economical and simple**: -Standard costing is simple and economical in implementing the costing system. It results in reduction of paper work in accounting and needs lesser number of forms and records. This results in reduction of costs.

9.6 STANDARD COSTING AS A MANAGEMNET TOOL

A valuable management tool, standard costing is part of cost accounting. Instead of using actual costs for direct material, labor and manufacturing overhead, standard costs are used to easily track variances and estimate profit. Although actual costs are still paid, standard costing is frequently used for inventories and cost of goods sold. The difference between standard and actual costs is called variances. These variances are the ones that make

standard costing such a valuable practice for management. Management can speedily become aware of changes in budgeted costs by tracking the variances. When standard costing is used, you will frequently take notice of the terms unfavorable or favorable variance. This refers to changes in actual costs relative to planned or standard costs. A favorable variance takes place when actual costs dip below standard costs. On the contrary, if actual costs are greater than standards, the variance is unfavorable. In regards to manufacturing companies, standard costs would first be seen as individual parts or pieces of the finished product. This means that the final standard cost will be the sum of the standard costs of each of the individual pieces of the product. When variances are recurrently evaluated, standard costing operates as an effective management tool. Increased production, for example, should enable a company to benefit from economies of scale. That increased production quantity, although, may lead to unfavorable variances that emphasize inefficiencies not acknowledged previously. Conversely, a favorable variance may indicate that efficiencies attained were greater than anticipated. Given the financial importance of inventory, regular evaluations of standard costing may also illustrate instances where profitability is not possible with current actual inventory costs. Management can then evaluate the appropriateness of raising prices or seeking other ways to lower costs or improve production efficiency. When variances are frequently evaluated, standard costing functions as an effective management tool. Increased production, for example, should enable a company to benefit from economies of scale. That increased production quantity, although may bring about unfavorable variances that emphasize inefficiencies not acknowledged earlier.

With regular analysis of variances, standard costing enables managers to compare expectations to actual costs and profit margins. Basic causes of those variances can then be addressed. By then recalibrating costs and prices, manufacturers can maintain themselves in the middle of uncertainties.

9.7 LIMITATIONS OF STANDARD COSTING

1. **Difficult to Establish**: -It is difficult to establish standard costs. Inaccurate standards do more harm than good. Inaccurate and unreliable standards may give misleading results. Standards set may not enjoy the confidence of the users.

2. **Conditions Change**: - A business may not be able to keep standards up-to-date. Conditions change in business and are never static. Standards too have to change with the

times. Revision of standards is costly. If firms ignore to revise standards, suitably, damage would occur. If business does not revise the standard in manufacturing conditions, in tune with changes, even the current level of progress may not be sustained. Firms may not revise standards as it is a difficult and costly affair.

3. **Morale and Motivation may be affected**: - If standards are set too high, it may have an adverse impact on the morale and motivation of the employees. Staff might not be capable of operating the system.

4. **Expensive**: -Operation of standard costing system is a costly affair for small concerns. It is unsuitable for the small organizations.

5. **Unsuitable for Job Orders**: - Standard costing is expensive and unsuitable in job order industries and units engaged in manufacturing non-standardized products. For example, in a repair workshop, for each repair job undertaken, standards need to be set. Setting standards for each individual job is quite expensive.

9.8 LET'S SUM UP

Standard costing serves as a vital management tool aimed at both cost control and product costing by providing a consistent yardstick for evaluating actual performance against predefined standards. It enables organizations to understand expected outcomes, measure actual results, and take corrective actions when deviations occur. This process may involve modifying resources—such as labor, materials, or machinery—or adjusting overly ambitious standards. As an ongoing activity, standard costing supports the optimal utilization of resources and minimizes wastage by identifying variances and recommending corrective actions. The system promotes accountability by establishing cost centers and assigning responsibility to specific departments or individuals, thereby enhancing the delegation of authority. When implemented with active participation across management levels, it fosters a positive and cost-conscious organizational culture. Additionally, standard costing encourages the periodic review of methods, materials, and techniques to reduce unfavorable variances. It directs managerial attention to areas that deviate from planned performance, making it a strategic tool not only for cost control but also for broader functions such as budgeting, planning, marginal costing, and inventory valuation.

KEYWORDS

Cost control, Performance measurement, Variance analysis, Resource optimization,

Predetermined cost, Cost estimation, Target cost, Efficiency measurement, Cost comparison, Variance analysis, Control mechanism, Accounting technique, Budgeting aid, Decision-making, Planning and control, Performance evaluation, Cost control, Delegation of responsibility, Minimizes waste, Supports budgeting, Strategic planning, Cost monitoring, Inventory valuation, Marginal costing, Operational efficiency, Rigidity

9.6 REVIEW QUESTIONS

1. Which of the following best describes a "Standard Cost"?

   A. The actual cost incurred for production

   B. The historical cost recorded in books

   C. The predetermined cost established for performance evaluation

   D. The cost that includes only fixed overheads

2. One of the major needs for standard costing in an organization is:

   A. To prepare the balance sheet

   B. To eliminate the need for budgeting

   C. To provide a base for cost control and performance analysis

   D. To increase selling price

3. Which of the following is not a merit of the standard costing system?

   A. Encourages cost-consciousness

   B. Facilitates variance analysis

   C. Guarantees profitability

   D. Helps in performance evaluation

4. The limitation of standard costing includes:

   A. It is cost-effective in all environments

   B. Suitable for fast-changing industries

   C. It can be time-consuming and expensive to implement

   D. It eliminates the need for management intervention

5. _____ is the difference between the standard cost and the actual cost.
6. Standard costing acts as a _____ tool for management in areas like budgeting and inventory valuation.
7. A key limitation of standard costing is that it may not be effective in _____ production environments.
8. What do you mean by Standard costing? Is there a difference between standard cost and Standard costing? If yes, state.
9. State the merits and limitations of standard costing.
10. Explain standard costing as a management tool.

UNIT-10: SETTING UP OF STANDARDS

STRUCTURE

10.1 Introduction

10.2 Various elements and steps involved in setting standards

10.3 Circumstances under which standards are revised

10.4 Meaning of standard cost card

10.5 Let's Sum up

10.6 Review Questions

LEARNING OUTCOME

After studying this module, you shall be able to**:**

- Know the need for setting standards
- Identify the different types of standards
- Learn the steps involved in setting standards
- Evaluate circumstances under which Standards are revised
- Understand the meaning of Cost Card

Bloom's Level of Learning

The topics related to standard costing primarily engage **Bloom's Taxonomy levels of "Remembering," "Understanding," and "Applying."** At the **"Remembering"** level, learners recall basic concepts such as the definition of standard costs and the components of a standard cost card. The **"Understanding"** level is demonstrated when students explain the various elements and steps involved in setting standards, as well as the circumstances that necessitate their revision. Finally, the **"Applying"** level comes into play when learners use this knowledge to prepare or interpret a standard cost card in practical scenarios. While these topics establish a strong foundation in standard costing principles, they primarily focus on lower-to-mid cognitive skills, with potential to progress to **"Analyzing"** and **"Evaluating"** in more advanced applications.

10.1 INTRODUCTION

In the previous module, we have learnt about meaning, nature & significance of Standard Costing. Let us now learn about the concept of setting up of standards in this module. Let us first define a standard cost, in order to know its meaning.

The term 'Standard' has been called by different names in Accounting, for example, 'a norm', 'a model', a yardstick', a benchmark', 'a criterion of excellence' etc. therefore, a standard maybe a norm or a measure of comparison in terms of specific items such kilograms of material, labour hours or hours of plant capacity etc.

The Chartered Institute of Management Accountants [UK] defines standard costs as ''a predetermined cost which is calculated from management's standards of efficient operations and the relevant necessary expenditure. It may be used as a basis for price fixing and for cost control through variance analysis''.

The term standard cost should not be confused with the term ''estimated cost''. They are both different in following respects.

[i]- Estimated costs are, very often, less accurately determined. Standard costs are determined more realistically.

[ii]- Estimated costs are developed from projections using average of past data whereas standards are set more scientifically through the use of time & motion studies.

[iii]- Estimated costs are not helpful to management but standard costs provide basis for comparison to management.

[iv]-the emphasis of Estimated cost is on actual cost with which it is compared at the end of the period. Under standard costing, actual costs are ascertained only to facilitate their comparison with standard costs.

Setting Standards

The success of standard costing system will depend upon the setting up of proper standards. For the purpose of setting standards, a person or a committee should be given this job. In a big concern, a standard costing committee is formed for this purpose. The committee includes production manager, purchase manager, sales manager, personnel manager, chief engineer and cost accountant. The cost accountant acts as a co-coordinator of this committee.

Normally, setting up standards is based on the past experience. The total standard cost

includes direct materials, direct labour and overheads. Normally, all these are fixed to some extent. The standards should be set up in a systematic way so that they are used as a tool for cost control.

Types of standards

On the basis of attainability of standards and frequency with which the standards are revised, the standards are classified as ideal, normal, basic, current or expected actual standards; each one is being discussed, in brief, as under:

[i]-**Ideal/maximum efficiency/theoretic/perfect standards**- These standards can be attained under most favourable conditions. The level of performance under ideal can be achieved through the best combination of most favourable price for material, labour and maximum efficiency and capacity utilization. They represent the ultimate goal for which any company strives for & the attainment is not possible.

[ii]- **Normal standards**- these are average standards that can be attained during a period of time, which may be one complete business cycle. These are set on a normal capacity basis which represents a volume that averages out the peaks and lows.

[iii]- **Basic standards**- basic standards remain unaltered for a long period of time and they are seldom revised or updated. They are usually revised when a whole business cycle is completed.

[iv]-**Attainable or Expected Actual Standards**- current standards are those which are established for use over a short period of time & are related to current conditions. They represent current costs to be expected from efficient operations.

10.2 VARIOUS ELEMENTS AND STEPS INVOLVED IN SETTING STANDARDS

Various Elements which Influence the Setting of Standards

Standards are set around various cost centres. According to J. Betty, “A cost centre is a department or part of a department or an item of equipment or machinery or a person or a group of persons in respect of which costs are accumulated, and one where control can be exercised.” Cost centres are necessary for determining the costs. If the whole factory is engaged in manufacturing a product, the factory will be a cost centre. In fact,a cost centre describes the product while cost is accumulated. Cost centres enable the determination of costs and fixation of responsibility. A cost centre relating to a person is called personnel cost centre, and a cost centre relating to products and equipments is called impersonal

cost centre.

Different elements or variables which influence the setting up of standards are Material, Labour and Overheads. These are being discussed one by one:

[i]-Setting Standards for Direct Materials

There are several basic principles which ought to be appreciated in setting standards for direct materials. If material is used for a product, it is known as direct material. On the other hand, if the material cost cannot be assigned to the manufacturing of the product, it will be called indirect material. Therefore, it involves two things:

[a]-Quality of material [b]-Price of the material

When the organization wishes to purchase material, the quality and size should be determined. The standard quality to be maintained should be decided. The quantity is determined by the production department. This department makes use of historical records, and an allowance for changing conditions will also be given for setting standards. A number of test runs may be undertaken on different days and under different situations, and an average of these results should be used for setting material quantity standards.

The second step in determining direct material cost will be a decision about the standard price. Cost of Material will be decided in consultation with the purchase department. The cost of purchasing and store keeping of materials should also be taken into consideration. The procedure for purchase of materials, minimum and maximum levels for various materials, discount policy and means of transport are the other factors which have bearing on the materials cost price. It includes the following:

[a]-Cost of materials [b]-Ordering cost [c]-Carrying cost

The purpose should be to increase efficiency in procuring and store keeping of materials. The type of standard used, for example ideal standard or expected standard, would also affect the choice of standard price.

[ii]-Setting Direct Labour Cost

If the organization wishes to engage a labour force for manufacturing a product or a service for which some amount is to be paid, this is called wages. If the labour is engaged directly to produce the product, this is known as direct labour. The second largest amount of cost is the labour cost. The benefit derived from the workers can be assigned to a particular product or a process. If the wages paid to workers cannot be

directly assigned to a particular product, these will be known as indirect wages. The time required for producing a product would be ascertained and labour should be properly graded. Different grades of workers will be paid different rates of wages. The times spent by different grades of workers for manufacturing a product should also be studied for deciding upon direct labour cost. The setting of standard for direct labour will be done basically on the following parameters:

[a]-Standard labour time for producing a unit [b]-Labour rate per hour

Standard labour time indicates the time taken by different categories of labour force which are as under:

[a]-Skilled labour [b]-Semi-skilled labour [c]-Unskilled labour

For setting a standard time for labour force, previous experience, past performance records, test run result, work-study etc are taken into consideration. The labour rate standard refers to the expected wage rates to be paid for different categories of workers. Past wage rates and demand and supply principle may not be a safe guide for determining standard labour rates. The anticipation of expected changes in labour rates will be an essential factor. In case there is an agreement with workers for payment of wages in the coming period, these rates should be used. If a premium or bonus scheme is in operation, then anticipated extra payments should also be included. Where a piece rate system is used, standard cost will be fixed per piece. The object of fixed standard labour time and labour rate is to device maximum efficiency in the use of labour.

[iii]-Setting Standards of Overheads

The next important element comes under the head overheads. The very purpose of setting standard for overheads is to minimize the total cost. Standard overhead rates are computed by dividing overhead expenses by direct labour hours or units produced. The standard overhead cost is obtained by multiplying standard overhead rate by the labour hours spent or number of units produced. The determination of overhead rate involves three things:

[a]-Determination of overheads [b]-Determination of labour hours or units manufactured [c] Calculating overheads rate by dividing [a] by [b]

The overheads are classified into fixed overheads, variable overheads and semi-variable overheads. The fixed overheads remain the same irrespective of level of

production, while variable overheads change in the proportion of production. The expenses increase or decrease with the increase or decrease in output. Semi-variable overheads are neither fixed nor variable. These overheads increase with the increase in production but the rate of increase will be less than the rate of increase in production. The division of overheads into fixed, variable and semi-variable categories will help in determining overheads.

Method of setting standards

It is pertinent to note that a single standard set may not be suitable for all purposes and all situations. Realistic and rational standards only help to take good decisions. These standards help in taking pricing and capital expenditure decisions which are critical to very survival of the organization.

[i]- **High Standards**- a high standard is helpful in cost reduction and motivating employees to try to reach the targets. As high standards are not realistic they are not helpful in product costing, inventory valuation, capital expenditure decisions etc.

[ii**]-Low standards-**a standard which is low or loose can be attained by poor performance. It defeats the very purpose of standard costing & fails to disclose inefficiencies. These standards are not helpful in cost control as they are not accurate measures to compare actual results.

[iii]-**Current Standards**- A current standard is a standard which is established for use over a short period of time and is related to current condition. It reflects the performance that should be attained during the current period. The period for current standard is normally one year. It is presumed that conditions of production will remain unchanged. In case there is any change in price or manufacturing condition, the standards are also revised.

10.3CIRCUMSTNACES UNDER WHICH STANDARDS ARE REVISED

Revision of standards

It is a difficult situation and decision for any organization to contemplate revision of standards. Some experts argue that frequent revisions in standards would defeat the purpose of setting standards and destroy the effectiveness. Conversely, other experts are of the opinion that if the standards are not revised they destroy the utility in inventory

valuation & cost control.

Therefore, the standards require a continuous review and frequent change, at times. In this volatile world of fierce competition, changing technology, advent of multinational companies, new machineries, better quality labour, increased awareness of labour in terms of their rights etc, all these factors render the set standards obsolete and unfit for use. Under these circumstances, if old standards are used then it would result in poor cost control, unrealistic budgets and unreasonable unit cost for inventory.

Circumstances under which standards are revised

A company must make a plan or programme for revision of budgets so that standards can be set at a current attainable level. Hence the following must be done for revision of standards:

[i]- Labour rate standards should be revised for any change in labour rates

[ii]- Material standards should be revised for any change in the types of material, quality of material or method of production

[iii]- If a new machine is purchased to replace the old one, then labour time standards material quantity standards must be updated

[iv]-in every company, there should be a system in force to revise standards, at least, every year

Therefore, a periodic review or revision is not only desirable but also necessary, in order to accomplish the objectives of Standard Costing. For effective use of this technique, sometimes there is a need to revise the standards which follow for better control. Even standards are also subjected to change like the production method, environment, raw material, and technology. Standards may need to be changed to accommodate changes in the organization or its environment. When there is a sudden change in economic circumstances, technology or production methods, the standard cost will no longer be accurate. Standards that are out of date will not act as effective feed forward or feedback control tools. They will not help us to predict the inputs required nor help us to evaluate the efficiency of a particular department. If standards are continually not being achieved and large deviations or variances from the standard are reported, they should be carefully reviewed. Also, changes in the physical productive capacity of the organization or in material prices and wage rates may indicate that standards need to be revised. In practice, changing standards frequently is an expensive operation and can cause confusion. For this reason, standard cost revisions are usually made only once a year.

Revision variance

When a budget is revised, but the change discussed as above, is not incorporated in the standard cost rate, a variance would arise, which is known as the Revision Variance. Revision variance is the difference between basic standard cost and the revised standard cost.

10.4 MEANING OF STANDARD COST CARD

After standards have been determined and set, they are tabulated in one statement to show the total standard cost per unit of output. As soon as standards are evolved, a master standard cost card is prepared and kept. The actual cost may be recorded in the job card and subsequently, the job card can be compared with standard cost card so that the variance could be worked out.

A standard cost card contains an itemization of the standard amounts of materials, labor, and overhead required to create one unit of a product. The card also multiplies the standard cost of each of these line items to arrive at the total standard cost of a product. The card has two purposes:

[i]-To derive the standard cost of a product

[ii]-To serve as the basis for variance analysis when actual costs for the product are compiled

The number of units and their standard costs that are listed on the card should be reviewed and possibly revised regularly, due to all of the following factors. Otherwise, the standard cost card will gradually diverge from the actual results experienced when manufacturing a product.

The standard cost card is comprised of the expected quantities of materials to be used during the production process, which can vary somewhat from the actual amounts used. For example, it may contain a certain amount of scrap that may be higher or lower than the actual amount experienced. Also, the amount of spoilage encountered during the setup of a production run may also vary from the standard amount listed in the standard cost card.

The standard costs listed on the card may also vary from actual results. For example, there may be an expectation to purchase a component for rupees 1000/-, but because it

was actually purchased in a smaller unit quantity than anticipated when the standard was created, the supplier charges a higher price per unit.

The standard amount of labour stated on the card may be incorrect, because of changes in worker efficiency, altered equipment configurations, changes in the mix of experience levels used in a production team, and so forth. Similarly, the standard cost of the labour stated on the card may be incorrect, because of changes in the wages paid to employees, or in the amount of overtime paid, or in the mix of employees used in the manufacturing process.

The standard amount of overhead allocated to the product may also vary from actual results, since it is based on a combination of an expected pool of overhead costs and an expected volume of production to be generated during the period. If either estimate varies from actual results, then there will be a difference between the standard overhead cost and the actual overhead cost.

A physical card is rarely used to store standard costs. Instead, this information is stored in the computer system and printed as needed.

10.5 LET'S SUM UP

The term 'Standard' has been called by different names in Accounting, for example, 'a norm', 'a model', a yardstick', a benchmark', 'a criterion of excellence' etc. therefore, a standard maybe a norm or a measure of comparison in terms of specific items such kilograms of material, labour hours or hours of plant capacity etc. The success of standard costing system will depend upon the setting up of proper standards. For the purpose of setting standards, a person or a committee should be given this job.

On the basis of attainability of standards and frequency with which the standards are revised, the standards are classified as ideal, normal, basic, current or expected actual standards. Different elements or variables which influence the setting up of standards are Material, Labour and Overheads. It is pertinent to note that a single standard set may not be suitable for all purposes and all situations. Realistic and rational standards only help to take good decisions. These standards help in taking pricing and capital expenditure decisions which are critical to very survival of the organization. A company must make a plan or programme for revision of budgets so that standards can be set at a current attainable level. Revision variance is the difference between basic standard cost and the revised standard cost. A standard cost card contains an itemization of the standard

amounts of materials, labor, and overhead required to create one unit of a product. The card also multiplies the standard cost of each of these line items to arrive at the total standard cost of a product.

KEYWORDS

Standard, Norm, Model, Yardstick, Benchmark, Criterion, Comparison, Material, Labour hours, Machine hours, Proper Standards, Critical to success, Expert committee involvement, Ideal, Normal, Basic, Current, Expected Actual, Material, Labour, Overheads, Realistic standards, Decision-making, Strategic relevance, Revision of Standards, Budget updates, Attainable levels, Basic standard cost, Revised standard cost, Standard Cost Card, Itemization, Materials, Labour, Overhead, Unit cost

10.6 REVIEW QUESTIONS

1. Which of the following is not an element considered while setting up standards in standard costing?

   A. Material usage

   B. Labour efficiency

   C. Advertising expenditure

   D. Overhead absorption

2. Which type of standard is most likely to be revised regularly to reflect current conditions?

   A. Basic standard

   B. Ideal standard

   C. Expected actual standard

   D. Historical standard

3. The standard cost card is primarily used for:

A. Recording actual costs incurred

B. Preparing cash flow statements

C. Summarizing the standard quantities and costs for producing one unit

D. Assessing tax liabilities

4. Standards are usually revised under which of the following circumstances?

   A. When actual costs remain unchanged

   B. When there is no change in methods or materials

   C. When there is a change in production technology or input prices

   D. Only at the end of a financial year

5. The process of setting cost benchmarks for materials, labor, and overheads is called _____.
6. A _____ contains detailed information about standard input quantities and their costs per unit of output.
7. Revision of standards becomes necessary when there are changes in _____ or production methods.
8. What are the different types of Standards?
9. Explain the various elements of standards.
10. Why do you think revision of standards are required?

UNIT-11: KINDS OF VARIANCE

STRUCTURE

11.1 Introduction

11.2 Direct Material Variances

11.3 Direct Labour Variances

11.4 Overhead Variances

11.5 Let us Sum up

11.6 Review Questions

LEARNING OUTCOME

After studying this module, you shall be able to:

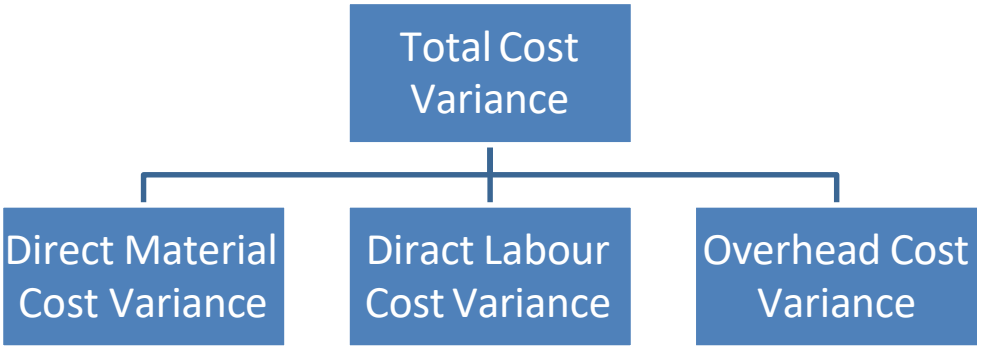
- Know the different kinds of material variances, labour variances and overhead variances,
- Identify the subcomponents of material variances,
- Categorize the labour variances,
- Understand the fact that overhead cost variance includes fixed cost variance and variable cost variance

Bloom's Level of Learning

This Chapter align with **Bloom's Understanding and Application levels** of learning. In Bloom's Taxonomy, the **Understanding level** involves grasping the meaning of concepts such as cost variances and their causes, while the **Application level** requires using this understanding to solve problems or calculate variances in given scenarios. The review questions assess both these levels by asking learners to explain concepts and perform calculations based on real or hypothetical data. In some cases, depending on the complexity of the questions, **Analysis** may also be involved, especially when students are asked to interpret variance results or diagnose cost control issues. Overall, these topics help learners move from theoretical comprehension to practical problem-solving in cost accounting.

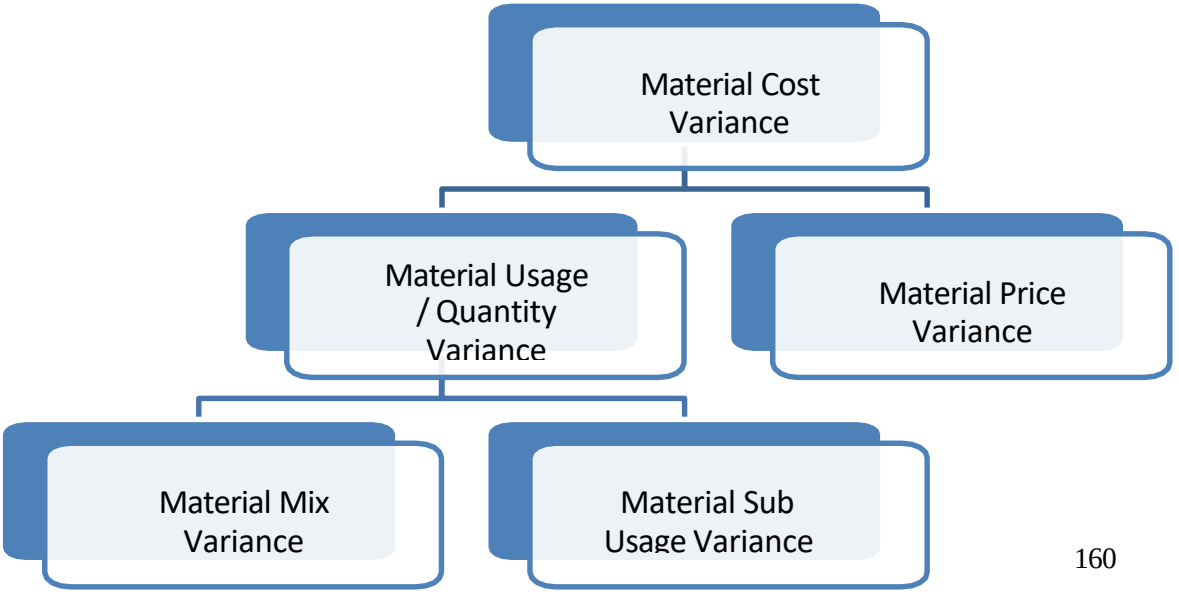
11.1 INTRODUCTION

The variance results when there is a difference between the standard cost allowed and the actual cost incurred. We should understand the kinds of variances before doing the computation work related to it. The total Cost variance (standard cost – actual cost) is divided into variance for each element of cost, that is, material cost variance, labour cost variance and overhead cost variance which are further bifurcated between various sub-components.



11.2 DIRECT MATERIAL VARIANCES

The variance arising with respect to the component 'material' constitutes material variance. The variance may arise as a result of the difference between the actual price paid and standard price expected, actual units produced from material and the standard expected, actual mix of material and the standard mix etc. The different types of material variances are discussed below:



11.2.1 Material cost variance is the difference between the standard cost of material allowed for the actual output achieved and the actual cost of material used. It is favourable when the actual cost of material is less than the standard cost of material for actual output and adverse when actual cost is higher than the standard cost. Material Cost Variance can be further sub divided into Material Price Variance and Material Usage / Quantity Variance

11.2.2 Material price variance arises when the firm purchases the raw material at a price different than the standard price.

11.2.3 Material Usage variance arises when there is a difference between the actual quantities of raw material and standard quantities of raw material to produce the output achieved. The Material Usage Variance is sub divided into Material Mix Variance and Material Revised Usage Variance or Material sub- usage Variance.

11.2.4 Material Mix Variance may arise in a situation where the firm uses different proportion (mix) of raw material to produce the finished product. The firm pre determines the standard mix to be used for production but its actual mix may differ than expected because of many reasons like, price rise in the cost of material, delay in purchasing the material etc.

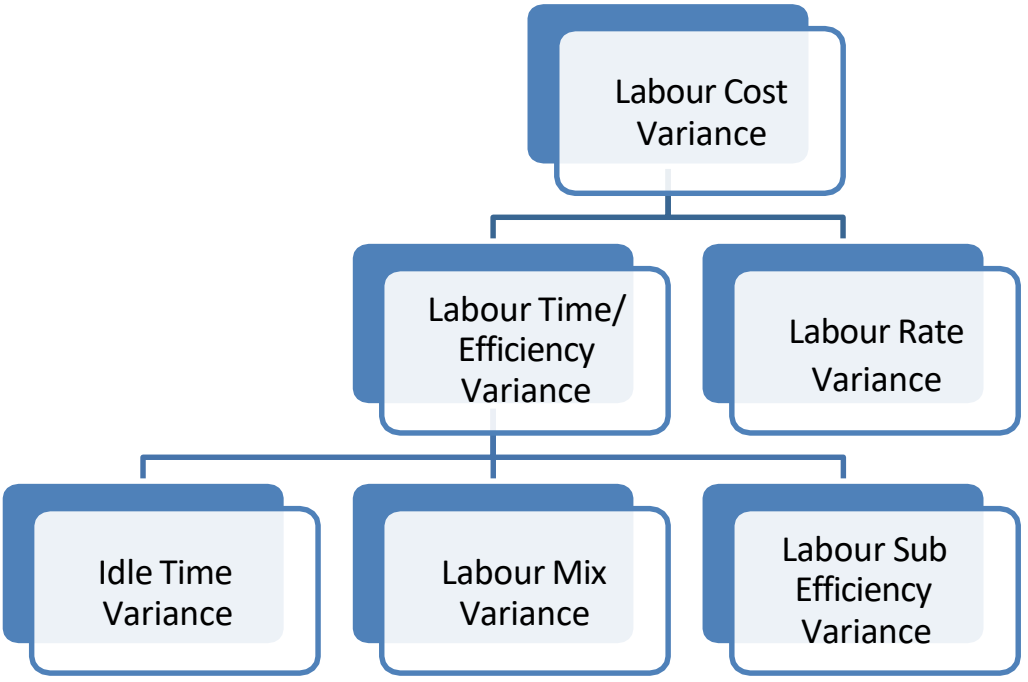
11.2.5 Material Revised Usage Variance takes into account those reasons other than those of material mix variance that attributes to material usage Variance. It undertakes the difference between the Standard Quantity and Revised Standard Quantity. When there is no difference between the total actual quantities of all the material and total standard quantities of all the material then the Revised Standard quantity would be same as Standard quantity resulting into no Material Revised Usage Variance.

11.2.6 Material Yield Variance measures the difference in actual output/ yield for a given set of inputs and the standard output/ yield for a given set of inputs.

11.3 DIRECT LABOUR VARIANCES

The term Labour Variance denotes variance with respect to the labour and it arises when there is a difference between the actual labour cost and standard labour cost. The

different kinds of labour variances are discussed in this section.



11.3.1 Labour cost variance is the difference between the standard cost of labour allowed for the actual output achieved and the actual cost of labour used. It can be further sub divided into Labour Rate Variance and Labour Time / Efficiency Variance.

11.3.2 Labour rate variance arises when the firm pays the wages at a rate different than the standard wage rate.

11.3.3 Labour Efficiency variance is that part of the labour cost variance which is due to the difference between the actual labour hours and standard labour hours specified to produce the output achieve.

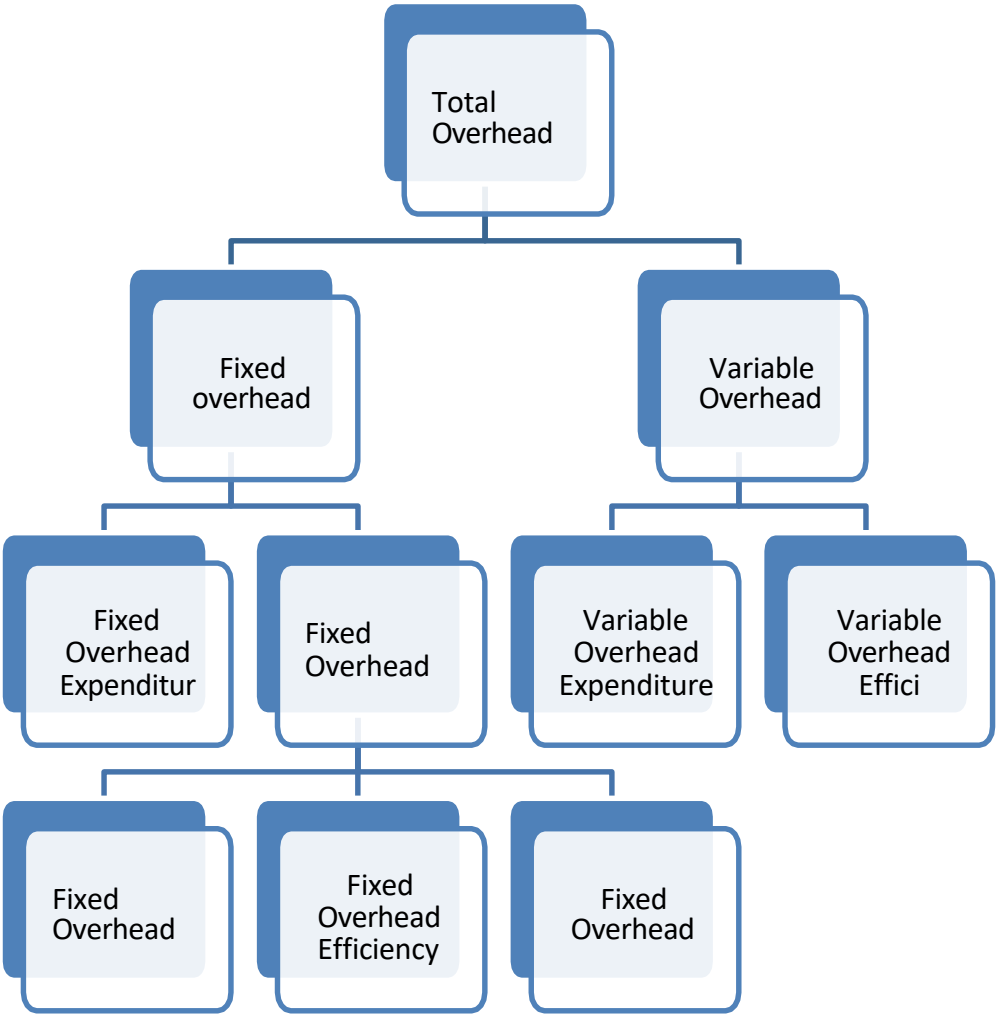
11.3.4 Idle time means the workers are not able to utilize the total time for which they are paid for and sitting idle for some time.

11.3.5 Labour Mix Variance may arise in a situation where the firm uses different proportion (mix) of labour to produce the finished product.

11.4 OVERHEAD VARIANCES

The total Overhead Cost Variance results whenever there is a difference between the

actual overhead cost incurred and the standard overhead cost for the actual output achieved. Overhead Cost variance is divided into Fixed Overhead Variance and Variable Overhead Variance. This section discusses Variable overhead variances and Fixed overhead variances.



Variable Overhead Variances

11.4.1 Variable overhead Cost - Variance measures the difference between Absorbed Variable overhead and Actual Variable Overhead. The Variable Overhead Cost Variance is sub divided into Variable Overhead Expenditure / Spending/ Budget Variance and Variable Overhead Efficiency Variance

11.4.2 Variable Overhead Expenditure / Spending/ Budget Variance arise when there is a

difference between standard variable overhead and actual variable overhead incurred.

11.4.3 Variable Overhead Efficiency Variance arises when there is a difference between adsorbed variable overhead and standard variable overhead.

Fixed Overhead Variances

a. Fixed Overhead Cost Variance (FOCV) is that portion of total overhead cost variance that arises due to difference between standard fixed overhead cost for actual output and actual fixed overhead. It is divided into two variances that is Fixed Overhead Expenditure Variance and Fixed Overhead Volume Variance.
b. Fixed Overhead Expenditure / Spending / Budget Variance is that portion of fixed overhead cost variance which arises due to the difference between budgeted fixed overhead and actual fixed overhead.
c. Fixed Overhead Volume Variance is that portion of fixed overhead cost variance which arises due to the difference between standard cost of overhead absorbed by actual production and the budgeted overhead. Fixed Overhead Volume Variance is sub- divided into Efficiency Variance, Capacity Variance and Calendar Variance.
d. Fixed Overhead Efficiency Variance is that portion of Fixed overhead volume variance that arises due to the difference between the actual output and the standard output expected, arising from the efficiency of workers or plant.
e. Fixed Overhead Capacity Variance is that portion of fixed overhead volume variance that arises due to difference in standard and actual output because of under and over capacity utilisation of plant and equipment attributing to factors like idle time, lockouts and strikes, material shortage, machinery break down, labour turnover.
f. Fixed Overhead Calendar Variance arises when there is a difference between the number of working days in the budgeted period and actual number of working days to which the budget is applicable.

11.5 LET US SUM UP

The variance results when there is a difference between the standard cost allowed and the actual cost incurred. The total Cost variance (standard cost – actual cost) is divided into variance for each element of cost, that is, material cost variance, labour cost variance and overhead cost variance which are further bifurcated between various sub- components.

The variance arising with respect to the component 'material' constitutes material variance. Material cost variance is the difference between the standard cost of material allowed for the actual output achieved and the actual cost of material used. Material Cost Variance can be further sub divided into Material Price Variance and Material Usage / Quantity Variance. The Material Usage Variance is sub divided into Material Mix Variance and Material Revised Usage Variance or Material sub- usage Variance.

The term Labour Variance denotes variance with respect to the labour and it arises when there is a difference between the actual labour cost and standard labour cost. Labour cost variance is the difference between the standard cost of labour allowed for the actual output achieved and the actual cost of labour used. Labour Cost Variance can be further sub divided into Labour Rate Variance and Labour Time / Efficiency Variance. The Labour Efficiency / Time Variance is sub divided into Idle Time Variance, Labour Mix Variance and Labour Sub – Efficiency Variance.

The total Overhead Cost Variance results whenever there is a difference between the actual overhead cost incurred and the standard overhead cost for the actual output achieved. Overhead Cost variance is divided into Fixed Overhead Variance and Variable Overhead Variance.

KEYWORDS

Variance, Standard Cost, Actual Cost, Total Cost Variance, Material Cost Variance (MCV), Labour Cost Variance (LCV), Overhead Cost Variance (OCV), Material Price Variance, Material Usage Variance, Material Mix Variance, Material Sub-Usage Variance, Labour Rate Variance, Labour Efficiency Variance, Idle Time Variance, Labour Mix Variance, Labour Sub-Efficiency Variance, Fixed Overhead Variance, Variable Overhead Variance, Actual Output Achieved, Cost Elements.

11.6 REVIEW QUESTIONS

1. Which of the following is **not** a component of direct material cost variance?
   a) Material Price Variance
   b) Material Usage Variance
   c) Material Mix Variance
   d) Material Efficiency Variance
2. If the actual labour hours worked are less than the standard hours for the same output, the variance will be:
   a) Favourable Labor Efficiency Variance
   b) Unfavourable Labor Rate Variance
   c) Unfavourable Labor Efficiency Variance
   d) Favourable Labor Rate Variance
3. The difference between actual variable overhead incurred and standard variable overhead for actual output is called:
   a) Fixed Overhead Volume Variance
   b) Variable Overhead Expenditure Variance
   c) Variable Overhead Efficiency Variance
   d) Total Overhead Variance
4. Standard costing is most useful for:
   a) Financial reporting only
   b) Budgeting and cost control
   c) Determining tax liabilities
   d) Calculating break-even point
5. The difference between the standard cost of materials and the actual cost incurred is called _____.
6. _____ arises when actual labour hours differ from standard labour hours allowed for actual production.
7. In overhead variance analysis, _____ is the difference between budgeted fixed overhead and applied fixed overhead.
8. What do you mean by variances? Discuss various types of material cost variances.
9. Explain various Labour variances.
10. Give a brief note on Overhead variances.

UNIT-12: VARIANCE ANALYSIS

STRUCTURE

12.1 Introduction

12.2 Meaning of Variance analysis

12.3 Concept of Variance analysis

12.4 Significance of Variance analysis

12.5 Let us Sum Up

12.6 Review Questions

LEARNING OUTCOME

After studying this module, you shall be able to:

- Know the meaning and concept of Variance analysis
- Understand the favorable variance and unfavorable variance
- Identify the variance that are controllable and uncontrollable
- Analyze that all adverse variances are not bad for the firm
- Appreciate how the variance analysis is significant to the management.

Bloom's Level of Learning

The given content, which includes sections like the introduction, meaning, concept, and significance of variance analysis, corresponds to the **"Understanding"** and **"Analyzing"** levels of Bloom's Taxonomy. These sections aim to help learners grasp the basic definition and purpose of variance analysis (Understanding), and also explore its significance and implications in decision-making (Analyzing). By breaking down the concept into smaller parts and examining its value in business operations, learners develop analytical thinking and are better prepared to apply the knowledge in practical scenarios.

12.1 INTRODUCTION

The term analysis of variance has two key words, analysis and variance. Let us understand the meaning of these words first. The difference between the actual and standard is the variance. This variance can be measured in terms of cost, profit or sales. The second key word is analysis, which means examining the total part by breaking it into the sub parts to look at their interrelationships. When we take both the terms together, that is, analysis of variance, it means analyzing the total variance, examining its sub parts , like the cost, profit or sales and their interrelationships in terms of good/ adverse, favorable/ unfavorable.

12.2 MEANING OF VARIANCE ANALYSIS

According to C.I.M.A., London Terminology,

‘‘Variance Analysis is the resolution into constituent parts and the explanation of variances.’’

This definition says that the total variance should be broken into its constituents and the Explanations of all the constituents are to look for. But the process does not end here.

The identification of explanations which causes the variance would demand the corrective action and a lesson for the future operations so that such causes won’t get repeated.

Therefore, the variance analysis involves three phases:-

Phase 1: Dividing the total variance into sub parts to analyse and assign the responsibility of each individual variance;

Phase 2: Explanation of variance from each individual responsibility centers; **Phase 3:** Variations demanding actions to be corrected and provides a base for conducting next period operations.

12.3 CONCEPT OF VARIANCE ANALYSIS

The concentration of variance analysis is in looking at the difference between the standard and the actual, whether it is favorable or adverse, controllable or uncontrollable and what does it indicate. These concepts of favorability, controllability raises queries in the minds of students which should be dealt beforehand. Does the unfavorable variance mean that the firm is not efficient? Are all adverse variances are bad? These are the basic questions which should be answered before getting into the technical details of this topic.

Favorable Variance vs. Unfavorable Variance

The simplest thing is to judge the favorability of the variance. We can say that the variance is favorable or good when:

1. Actual cost is less than standard cost:
2. Actual profit/ sale are more than standard profit/

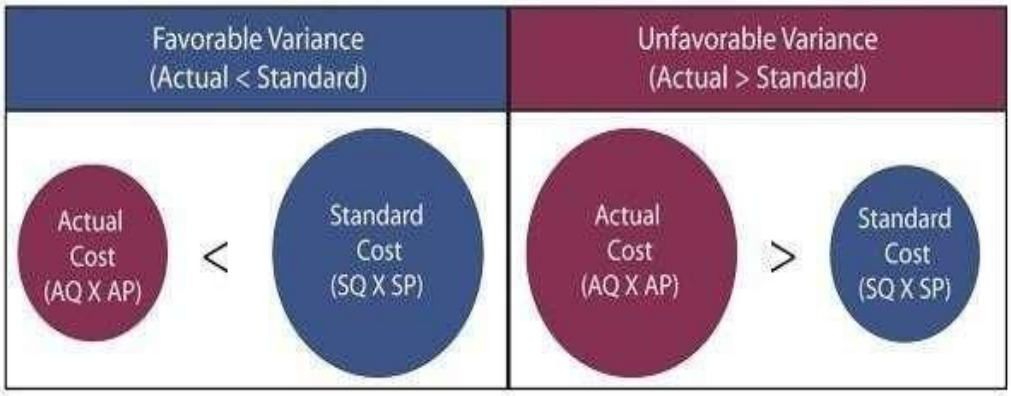
sale. On the other hand, variance is unfavorable when:

1. Actual cost is more than standard cost;
2. Actual profit/ sale are less than standard profit/ sales:

We have assigned (+) or (F) sign to indicate favorable variance (better than expected) and (-) or (A) sign for unfavorable or adverse variance(worse than expected).

Controllable Variances vs. Uncontrollable Variances

There can be many factors responsible for a favorable variance besides the straightforward relationship between cost, profit and sales. It sometimes, reflects the



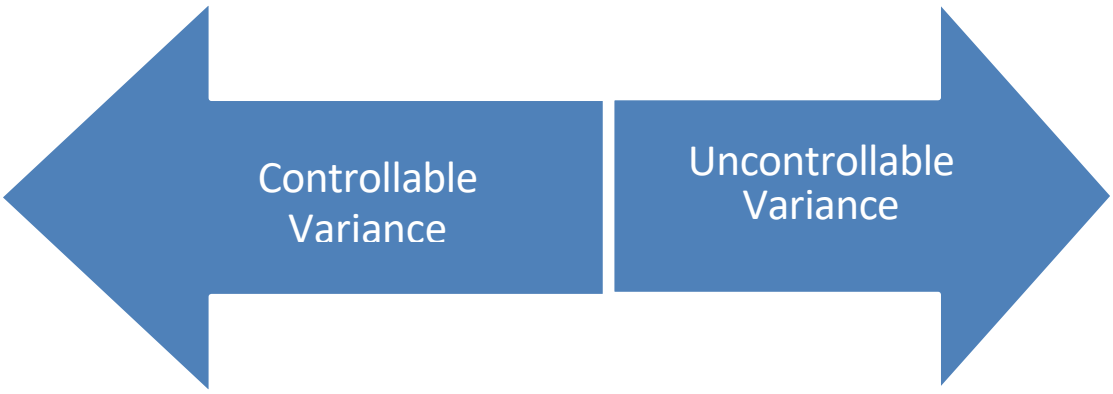
efficiency of an organization in managing all the given factors. The unfavorable variance generally is taken as an indicator of inefficiency of organization. We, as an analyst, have to see the underlying factors the favorable or unfavorable variance before declaring it as a contribution by efficient or inefficient management of an organization. The underlying factors can be internal to an organization or may relate to external environment. The examples of internal factor that may cause inefficiency are wrong standard setting, defect in the machine attributing to defective use of raw material etc. These factors can be corrected and controlled. That is why; these are known as controllable variance.

On the other hand, the external environment may also contribute to inefficiency of firms. For example, price increase, time lag in procurement of material etc. which are beyond the control of management affects the firm adversely. The variances arising because of these factors are uncontrollable variances as these are not controlled by any division or department of the organization.

The division of controllable and uncontrollable variance has to be seen in context of management by exception. The uncontrollable variance, over which the organization has no control, may be ignored. The important point here is to emphasize and investigate over controllable unfavorable variance that demands possible corrective action to convert them from adverse to favorable. This is in sync with the principle management by exception that is concentrates on matters which requires specialized attention as management cannot devote their precious time to all the deviation.

Can we say that all adverse variances are bad for a firm?

This is a common question that arises in our mind when we relate the word 'adverse' with anything bad happened. But let us tell you that it is not always. It depends on cause and quantum of the variance that decide its impact on a firm. The variance with small quantum generally requires no action and prompts the analyst to rather concentrate on those showing large deviations.



The other factor that we have to look is the cause of the adverse variance. The cause underlying may be anything, the long term external trend in the economy or internal problem of the company. These cause, although raises the cost expenditure of the firm results in increasing the sales in the given period. The result is adverse cost variance associated with increase in sales or profit. For example, the statement of budget shows higher actual cost of production than budgeted because of higher actual sales than budgeted.

Budget Statement (In Rs. 'crores)

	Budgeted	**Actual**	**Variance**	**F/A**
Sales: Product A	500	800	+ 200	F
Sales: Product B	600	650	+ 50	F
Sales: Product C	400	200	- 200	A
Total Sales	1,500	1,650		
Cost of Production (COP):				
Wages	150	200	- 50	A

Material	200	220	-20	A
Rent	130	100	+30	F
Marketing Expenses	225	250	-25	A
Other Overheads	230	240	-10	A
Total Cost	935	1,010	-75	A
Profit (Total Sales- Total Cost)	565	640	+75	F

In the above budget, sales and cost of production budgeted statement, we can see that sales and profit variance is favorable while cost variance is adverse. It establishes the relationship between sales and cost, that is, when sales increases, cost of production also increases resulting in adverse cost variance. This adverse cost variance is good for the company as it associated with the favorable sales and profit variance.

12.4 SIGNIFICANCE OF VARIANCE ANALYSIS

The variance analysis is significant for the management for following reasons:-

Cost control and measuring performance

Variance Analysis facilitates cost control by comparing the actual cost with the standard cost. It facilitates the managers to measure the performance of the respective divisions and departments.

Assigning responsibility and accountability

The identification of area deviating from the standard helps the management to see the responsible department and divisions and held them accountable for such deviations. It helps the management to extract the causes related to such deviations and once it is known, it is easier to implement changes in the structure.

Management by exception

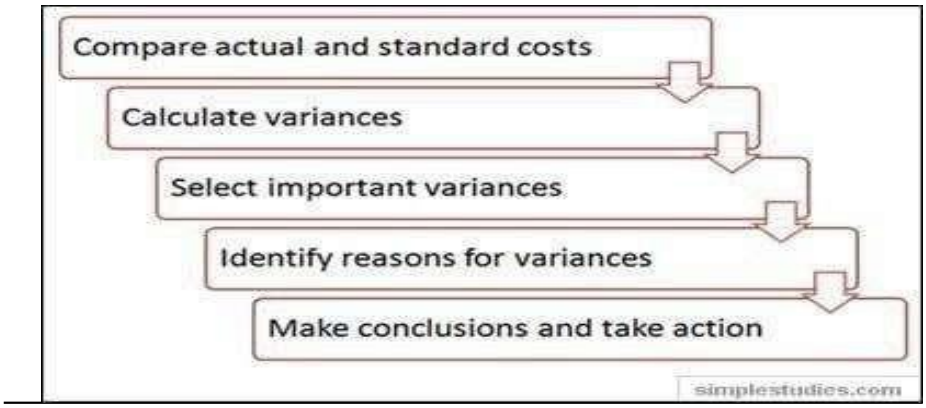
This principle says that the management should concentrate on core competencies and the areas where utmost attention is needed and ignoring the less significant areas. Variance analysis helps in identifying the areas where utmost attention is needed and thereby, in sync with the principle.

Helps in Decision making

Decision making involves taking right kind of decisions with respect to the area demanding action, after evaluating all possible aspects given at a point of time. Variance analysis identifies the area demanding action and prompts the management to draw their attention towards an urgent investigation of it. Here comes the role of management, that is, all the causes pertaining to the definite area identified should be extracted before making any decisions.

Warning for corrective action

The evaluation of performance by measuring the difference between the standard and the actual is not an end in itself. There is a need for action if there is any significant deviation. If the management takes no action then there is no point in discussing them.



12.5 LET US SUM UP

Variance analysis is a critical tool in managerial accounting that involves breaking down variances into sub-components to identify and understand their underlying causes. A variance is considered favourable when actual performance exceeds the standard or expected results, while an unfavourable variance occurs when actual results fall short of expectations. Some variances arise due to factors beyond management's control, known as uncontrollable variances, which may require adjustments rather than corrective actions. Importantly, not all adverse variances are inherently bad; their impact depends on the causes and magnitude, necessitating a thorough evaluation before labelling them as good or bad. Variance analysis holds significant importance for top management as it aids in

taking corrective measures, controlling costs, assessing performance, and making informed decisions. By analyzing variances, management can pinpoint inefficiencies, optimize resource allocation, and enhance operational effectiveness, ultimately contributing to better financial health and strategic planning for the organization.

KEYWORDS

Variance Analysis, Favourable Variance, Unfavourable Variance, Controllable Variance, Uncontrollable Variance, Standard Costing, Actual vs. Budget, Cost Control, Performance Measurement, Corrective Actions, Material Variance, Labour Variance, Overhead Variance, Budget Variance, Management Accounting, Operational Efficiency, Decision-Making, Financial Performance, Revenue Variance, Expense Variance

12.6 REVIEW QUESTIONS

1. Variance analysis helps in identifying the _____ between actual and budgeted performance.
2. A variance is considered _____ when actual costs are lower than standard costs.
3. Uncontrollable variances arise due to factors beyond the _____ of management.
4. What is the primary purpose of variance analysis?
   a) To calculate total sales revenue
   b) To compare actual performance with standard/budgeted performance
   c) To determine market trends
   d) To assess employee satisfaction
5. A favourable variance occurs when:
   a) Actual costs are higher than standard costs
   b) Actual revenue is lower than budgeted revenue
   c) Actual performance is better than expected
   d) There is no difference between actual and standard costs
6. Which of the following is an example of an uncontrollable variance?
   a) Increase in material costs due to poor inventory management
   b) Higher labour costs due to overtime mismanagement
   c) Rise in raw material prices due to inflation

d) Excessive power consumption due to machine inefficiency

7. Variance analysis is most useful for:

   a) Preparing financial statements

   b) Evaluating management performance and cost control

   c) Calculating tax liabilities

   d) Determining customer preferences

8. Discuss the concept of Variance analysis.
9. Differentiate between favourable and unfavourable variance analysis.
10. Differentiate between controllable and uncontrollable variance analysis.

UNIT-13: BUDGETING- CONCEPT AND CLASSIFICATION

STRUCTURE

13.1 Introduction

13.2 Meaning and Definition of Budget

13.3 Budgeting

13.4 Classification

13.4.1 Operating budgets

13.4.2 Capital Expenditure Budget

13.4.3 Research & Development Budget

13.4.4 Cash Budget

13.4.5 Budgeted Balance Sheet

13.4.6 Budgeted Statement of Changes in Financial Position

13.4.7 Fixed Budget

13.4.8 Flexible Budgeting

13.5 Let us sum up

13.6 Review Questions

LEARNING OUTCOME

After studying this module, you shall be able to:

- Know the meaning, features and basic ingredients of a budget
- Understand the concept, advantages and objectives of budgeting
- Know the various types of Operating and Financial Budgets Elaborate Sales Budget, Production Budgets
- Know the meaning, concept and features of Flexible Budget

Blooms Level of Learning

This chapter primarily align with **Bloom's Taxonomy levels of "Remembering," "Understanding," and "Applying."**

At the **"Remembering"** level, learners recall basic definitions and types of budgets. The **"Understanding"** level involves explaining concepts like fixed vs. flexible budgeting or the purpose of a budgeted balance sheet. Finally, **"Applying"** comes into play when learners classify budgets or prepare basic budget formats. This content is foundational, targeting **lower to mid-level cognitive skills** in Bloom's Taxonomy.

13.1 INTRODUCTION

WE ALL MAKE PLANS OR BUDGETS

Financial planning plays an important role in all the spheres of activities. Whether we talk about household or business or government, planning is the first basic exercise we undertake before venturing out for any activity. Even though we are not aware, but seeds of financial planning are sowed in us right in our childhood. We have all witnessed our mothers making estimates of household expenses in the beginning of the month, how she would stop us from making unnecessary purchases to keep expenditures within limits, how she would tell us to manage the whole month in that little pocket money! We all have also seen our fathers making estimates of how finances would be arranged to meet the EMIs of the housing loan, car loan or scooter loan etc. When we were students in college and had to organize college freshers party, or farewell party or cultural festivals, we used to spend hours with our committee members planning about how the funds shall be arranged through sponsorships, entry tickets, subscriptions, voluntary donations, cultural funds and how this collected fund shall be spent on catering, tents, decoration, DJs, food coupons to volunteers, souvenirs to be distributed, prizes and so on. At the time of marriages, we see how the families of the bridegroom and the bride list out items of expenditure and allocate budget for each head, compare the actual expenditure with the planned one and express shock and worry at the deviations. Government also indulges in planning. Each one of us eagerly waits for the Finance Budget, Railway Budget etc to get the feel of what the

government has in store for the common man. Thus if finance is the blood of business, financial planning is in the blood of all us! It is all pervasive and everyone be it home makers, traders, managers, government, business enterprises, salaried employees indulge in it.

13.2 MEANING AND DEFINITION OF BUDGET

3.1.1 Meaning

The word 'budget' is derived from a French term "bougette" denotinga leather pouch in which money is put in order to meet expected expenses. Similarly in business terms, a budget denotes a numerical statement showing how resources are to be required and used over a specified time period in future.

Definition

"Budget is an estimate o future needs arranged according to an orderly basis covering some or all the activities of an enterprise for definite period of time."
George R. Terry

"A budget is a comprehensive and coordinated plan expressed in financial terms for the operation and resources of an enterprise for some specific period in the future."
James M. Fremgen

"A budget is a pre-determined statement of management policy during a given period which provides a standard for comparison with the results actually achieved."
Brown and Howard

"A plan quantified in monetary terms prepared and approved prior to a defined period of time usually showing planned income to be generated and/or expenditure to be incurred during that period and the capital to be employed to obtain a given objective."
The Chartered Institute of Management Accountants, London

The definitions given above suggest that **budget is a formal expression of an action plan and provides a systematic and disciplined approach to implement and realize**

whatever was planned.

Essential Elements of Budget

Some of the important features or ingredients of a budget are:

A budget is a plan. A budget is a blueprint for the future action expressed in monetary terms. It is a roadmap prepared to guide the subordinates continuously while they engage in operations.

A budget is prepared in advance. Planning precedes an action. Similarly, a budget is prepared in advance so that one gets an idea about the future state of affairs and could therefore; well equip itself to avoid any crises later on.

A budget is concerned with operations as well as resources. A budget quantifies the revenues to be realized from products/services and the expenses to be incurred on goods/services. Further it is concerned with the planning of the various assets and sources of capital to finance these assets.

Budgets are prepared in financial terms. In an enterprise the various activities and operations are expressed in different units. If they have to be integrated in a plan, they must be expressed in comparable units of measurement say, in rupees, dollars, pounds etc.

A budget relates to a specified period of time. If a budget does not specify a time horizon, it will be meaningless. For example if a budget prescribes a sales revenue of 10 crores it makes no sense unless it qualifies that such a sales estimate is quarter wise estimate of the forthcoming year.

A budget is comprehensive. A budget includes all the activities and operations of an organization in it. It covers the organization as whole and not just some segments.

A budget is a coordinated plan. Budgets are prepared for different components/segments/divisions/activities of an enterprise so as to take care of the situations and problems facing each division. The budget for each of the components is prepared in harmony with each other resulting in a co - ordinate plan.

3.1.2 Forecast Vs Budget

Sometimes, the word 'budget' and 'forecast' are used interchangeably. In some ways, they do look similar like both relate to future events and both use predictions. However, forecast is concerned with **probable events** while budget is concerned with **planned events.** Forecast may be done for **longer time** but budget is prepared for **shorter time.** Forecast refers to the events over which there is **no control** while a budget is an **attempt to control** the events.

13.3 BUDGETING

Meaning & Definition of Budgeting

In the preceding section we learnt about budgets and their role in any organization. In this section we will learn about another related term, budgeting.

Budgeting is the formal process of financial planning using estimated financial and accounting data. It is the entire process of designing, implementing & operating budgets. The main emphasis of this process is on provisioning of resources to support plans which are being implemented.

Some important definitions are:

"Budgeting is the process of planning all flows of financial resources into, within and from an entity during some specified future period."- The National Association of Accountants (USA)

"The entire process of preparing the budget is known as budgeting."- Batty "Budgeting may be said to be the act of building budget." -Rowland & Hary Thus budgeting involves studying the business situations, understanding the management objectives and also the capacity of the enterprise. Budgeting is a planning function while its implementation is a control function.

Objectives/Functions of Budgeting

A proper budgetary exercise is indispensible for the success, survival and sustainability of a business firm. Budgeting has the following objectives/functions:

Budgeting compels planning

The planning process in any organization starts with defining a company's broad goals and objectives, developing strategies to achieve the desired goals and then setting up of the tentative schedules. A budget is one such schedule of proposed combination of the various factors of the production. Budgeting forces an organization to change its informal reactive style to a formal pro active style of management.

Budgeting improves coordination

Co ordination is achieved when all the factors of production and all departmental activities are synchronized and integrated to pursue the organizational goals. Budgeting helps in removing inconsistencies among the goals and actions of individual departments and reconciles their differences so that they work harmoniously.

Budgeting improves communication

Effective budgeting facilitates two way communications within the organization. The top management communicates the goals and objectives of the organization in its budgetary directives. Similarly, budgeting enables employees at middle and lower level to contribute their own ideas and provide feedback on goals and objectives.

Budgeting provides a basis for control and performance evaluation

After receipt and approval of the budgets of different departments, these budgets become their targets that set desirable limits on their spending. When the budget period ends, performances are judged and responsibilities are fixed on the basis of comparisons made between actual expenditure and budgeted expenditure.

Advantages of Budgeting

- Budgeting enables effective and efficient use of resources. It benefits the organization in the following ways:
- Budgeting forces management to make an early and timely study of its problems and generates a sense of cautions and care.
- Budgeting provides a valuable means of controlling income and expenditure of a business as it is a plan for spending.
- Budgeting instills cost consciousness within the organization and thereby

stimulates scarce resources to be allocated in the optimal way.

- Budgeting provides a mechanism through which managerial policies and goals are periodically evaluated, tested, improved and established as guidelines for future reference.
- Budgeting enables management to decentralize responsibility without losing control over the organization.
- Budgeting provides a yardstick for measuring performance of departments and individuals working in the organization. This enables them to self evaluate their actions and take suitable measures to improve themselves.
- Budgeting breeds healthy competition, provides incentives to good performers and gives a purpose to each individual in the organization thereby increasing their motivation, morale and productivity.
- Through budgeting, weakness, inefficiencies and deviations are identified. Variance analyses will prompt remedial actions wherever necessary.
- Budgeting draws attention of the management to information which has not gone according to the plan. This way, it goes with the principle of 'Management by Exception' and management can focus their time and energies on these issues.

Elements of a successful budgeting plan

In order to have a successful budgeting plan, the following elements should be present-

1. **Accurate Forecasting**- Budgets are made on the basis of forecasts and so their accuracy depends on the accuracy of the forecasts.
2. **Proper Coordination** - Since each budget has implications for other budgets, proper coordination among different individual budgets and different departments is necessary.
3. **Proper Communication**- The individual budgets should be communicated clearly and timely to all the units in the organization and their doubts and concerns should be clarified.
4. **Organization wide Cooperation-** A budgeting plan will be implemented successfully if people are ready to accept it and cooperate with it.
5. **Reasonable Flexibility-** Budgets should neither be too rigid nor too flexible. They

should set targets which are achievable and realistic. If the situation warrants, then modifications in the budgets should be carried out.

6. **Effective basis of Evaluation**- A properly developed budged can be used as a basis for the assessment of performances of operations, departments and individuals.

7. **Planning and control process in budgeting**

   Budgeting is a part of overall planning process. Planning involves deciding in advance what actions should be taken to help an organization to achieve its goals. Budgets are expression of these plans only showing how the resources are to be acquired and used over a certain period. This way budget is a means of translating the goals and strategies into operational terms. Planning identifies a desired output while budgeting identifies inputs needed to achieve that output. Similarly, controlling is a process which involves using feedback on actual performances and results, comparing the actual results with plans, measuring the deviations and finally taking corrective actions to bring future activities in accordance with plans. In budgetary control, this is what is exactly done. Actuals are compared with budgets, deviations are measured, variances are analyzed and corrective actions are taken. Thus budgeting is an integral part of both planning and control.

8. **Behavioral Dimensions of Budgeting**

   Since planning, forecasting and budgeting are not exact science, predictions made by them are not cent percent accurate. All these processes are based on assumptions, approximations, judgments and estimates which may not always turn out to be true. Further the budgeting process involves human beings at all the stages of formulation. Budgets are developed by people, budgets are executed by people, and budgets are used to evaluate the performance of the people. Hence management should take care of the behavioral and psychological effects of budgeting .It is a sensitive issue and management should try tackle it very carefully. Stated below are some of the behavioral implications of budgeting

- People associate the term ‘budget’ with ‘limit spending’. Just as the terms ‘restriction’ or ‘rationing’ evoke negative emotions; similarly people are

apprehensive of the word budget also. They perceive budgeting as an exercise to squeeze out every ounce of productivity out of the employees or to identify poor performance.

- The success and utility of budgeting depends upon the cooperation and participation of all the members of the organization. But in reality, budgets are imposed from above. Generally the top management retains total control of budgeting process and seeks only superficial participation from lower level managers. This kind of pseudo participation, which pretends to be participation but is actually only a lip service, is a form of deceit.
- Sometimes, departmental conflicts arise because of the competition for the resources allocation. Departments start blaming each other and try to pass the buck, when the targets are not achieved.
- The process of budgeting is lengthy and time consuming. Lot of energies and time are spent in getting the budget prepared and approved. But dynamism in the environment causes change in the projections and budgets lose their appeal creating frustration in employees.
- The budget process creates incentives among the managers to bias information that goes into the budget in order to get an increased allocation of resources for their department. More resource allocation means larger funds at their disposal, higher status in the peer group, higher responsibilities and higher prospects of promotions. Only a larger budget can justify larger allocations and so they tend to bias the information.
- Since budgeting is used for performance evaluation, many a times managers take sub optimal decisions to show their good performance e.g. in order to keep their cost within the limits, a manager may cut expenditure on R & D which may be detrimental to the overall interest of the organization.
- Managers compromise on long term goals of the organization in order to enhance their short term performance. For example, delaying expenditure on urgently needed repairs or offering huge discounts to push up sales of the current period.
- Budgeting may sometimes result in wasteful expenditure. This is because in order to fully utilize their allowances and to prevent future cutbacks, managers indulge in

careless and unnecessary spending.

- Sometimes, managers tend to overestimate costs so that they are not held responsible for overspending. The difference between the minimum possible expenditure and the expenditure shown in the budget is called **slack or padding** .Such a practice is widely adopted by the budgeter manager to create a buffer for the 'expected cuts' by the superior in the budgeted expenses proposed by him.
- Budgeting tends to bring rigidity in the system. It does not provide enough elbow space to dynamic managers and restricts their freedom of action. Further sticking to budgets often leads to poor decision making e.g. to keep costs within limits one may compromise on quality of inputs and go for the cheaper option.

Thus we can see how budgeting causes dysfunctional behavior amongst the people. Improper implementation of the budget, flaws in budget preparation process, the unrealistic management expectations, dominance of selfish interest over organizational goals etc. come in the way of reaping actual benefits from budgeting. The following principles may prove helpful in tackling the behavioral problems associated with budgeting –

1. Participative Budgeting

Participation by subordinate managers in preparing the budget increases their initiative in achieving the results, creates a sense of responsibility and also raises motivation, morale, and creativity. If subordinates managers are properly involved, the budgetary goals become their personal goals and these are readily accepted by them.

2. Top Management support

Top management should create a positive environment towards budget by giving impression to the subordinate managers about their commitment and support to the budgetary goals. Top management must review and approve budgets, take constant feedback, invite suggestions, and follow up the variances seriously.

3. Setting of realistic goals

To eliminate undesirable pressure, the budget goals should be reasonable. If targets are too high, they will be unachievable, causing frustration and resentment. Similarly if targets are too low, they will not remain exciting. The targets must be high enough to

motivate but realistic.

4. Transforming dysfunctional behavior into constructive behavior

Rigidity towards budgets should be avoided and top management should always be open to amend the budgetary targets if earlier targets are found impractical and unfeasible due to some changed circumstances or other reasons. Further efforts should be taken to bring about goal congruence i.e. align their personal goals with the organizational goals by devising proper rewarding system.

5. **Sound Communication system** .Expectations and goals of the top management should be communicated throughout the organization in clear cut terms. Further the communication channels should be kept open so that subordinates can convey their problems, concerns and apprehensions freely and fearlessly. Improvements and suggestions proposed by them, if found feasible, should be incorporated in final approved budget.

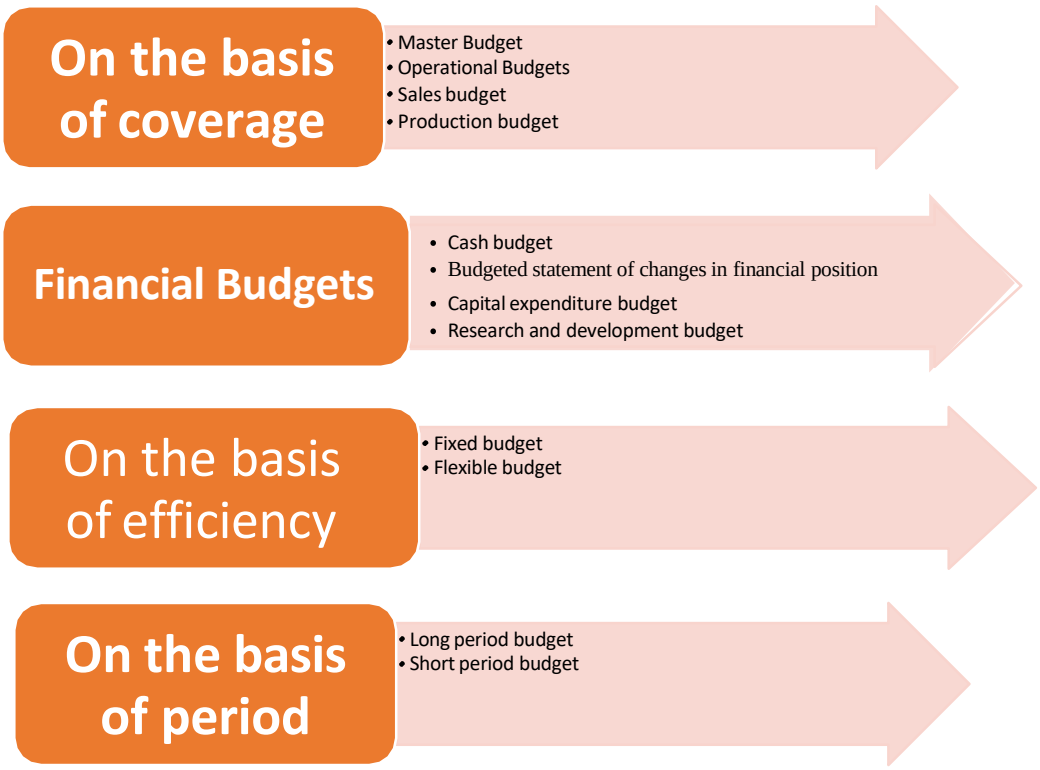
6. **Sound motivational structure** Budgeting will be a fruitful exercise only if it brings motivation. This can be brought about by following the policy of "fault fixing" rather than "fault finding". Instead of reprimanding a sub ordinate manager for not meeting the target, the effort should be directed to look into the real cause of deviation and take measures to plug them. Subordinates should be rewarded for achievements and if there is a deviation, investigation into the reasons be done rather than interrogation.

7. **Avoiding strict adherence to budgets.** To minimize the negative effects of budgeting, enlightened management should not demand strict adherence to a budget. They should emphasize on careful spending. They should seek justification for overspending and if justification is convincing, they should accept it.

If the above principles are followed it may be possible to minimize, if not eliminate, the undesirable repercussions of budgeting.

13.4 CLASSIFICATION

The budgeting process culminates in the preparation of the budgets. There are several kinds of budgets. We can classify them on the basis of coverage, capacity or periods as follows-



13.4.1 Operating budget

Operating budgets- The budgets related to the physical activities or operations of a business are called operating budgets. The various operating budgets are:

1. Sales budget
2. Product budget
3. Production cost budget
4. Ending inventory budget
5. Cost of goods sold budget
6. Selling expenses budget
7. Administrative expense budget
8. Budgeted income statement budget

1. SALES BUDGET

The sales budget is a forecast of total sales to achieve in a budget period expressed in

monetary or quantitative terms. It is prepared by the sales manager. This budget acts as a foundation for all the other functional budgets like production budget, cash budget, cost of goods sold budget etc., which depend on budgeted sales volume. It is the most challenging budget because estimating future sales is difficult due to continuous changes in the conditions of the economy and the company itself. Therefore, both internal and external factors are taken into account before making a sales forecast. The important factors that should be considered while making sales forecasts are as follows –

1. Past trend in sales- This is the most authentic and easily accessible source of information which companies use in predicting future sales. Analyzing past sales product wise, area wise and customer wise can prove very useful in devising appropriate marketing and pricing policies.
2. Estimates by sales force-Since salesmen are in direct contact with the customers and know their behavior better, they are in a position to estimate the potential demand more accurately.
3. Company specific factors- The plant capacity, the availability of raw materials and supplies, the size of workforce and procurement of labor with required skills, the arrangement of finance are some of the critical factors to be considered in making projections about sales.
4. Changes in firm's price policies-The firm should assess its pricing policies to see how the customer demand would react to the change in the prices of its commodities. Generally sales can be increased by reducing the prices and vice versa.
5. Changes in the product mix-The firm should also study the effect of changes in the product mix on the sales level as well as contribution margin. Then efforts should be made to increase the sales of the most promising products.
6. Marketing research- Surveys and market research should be conducted to collect information about customer tastes, preferences, purchasing patterns, disposable incomes etc. Such information leads to better product mix decisions and sales forecast.
7. Advertising and sales promotion plans-The impact of advertising campaign and sales promotion techniques should be assessed and incorporated in the sales forecast. If the company is proposing to undertake some major expansion or discontinuation

plan, it will affect the sales volume.

8. Competitors actions- The strategies and actions of competitors such as changes in their prices, product quality, promotion policies or new features in their products can greatly affect the sales of the organization and therefore must be considered.

9. General economic conditions- The sales forecast should be made in the light of general economic conditions prevailing in the economy like composition and purchasing power of population, growth in the national income and disposable income, seasonal and cyclical factors, international developments like petroleum prices and gold prices, labor market conditions, presence of substitutes etc.

10. Government policies- The government policies on exports -imports , taxes, interest, inflation, FDI, foreign exchange etc. have bearing on the operations of the business and should also be analyzed before making sales forecast. The forecasters should take note of government published predictions/forecasts for many economic indicators such as GDP, national income, production indexes, inflation etc. while making sales forecasts.

Having made sales forecast in the light of above factors, the sales budget is prepared according to products, territories, customers and period in the manner described below:

Product wise Sales Budget- Incase a company manufactures several products or variations of products, sales budget is prepared product wise. This will apprise management about the popularity of its various products

Territory wise Sales Budget-When a company sells its products in different geographical territories, then sales budget should be prepared for each territory. This will help management in fixing separate sales targets for each territory, distribution of costs among them, devising incentive schemes based on their performance etc. **Customer wise Sales Budget-** If a company sells its products to different customers such as wholesalers, retailers, public, government agencies, and foreign businesses and like, then it should prepare the budget customer wise. This will help in revealing the contribution of each type of customer to the total sales of the business firm and help in drawing proper marketing policies.

Period wise Sales Budget- The sales budget in terms of quantity and value should be broken down further into quarterly sales figures or monthly sales figures. This would

ensure coordination between production, purchasing and sales department and would also help in fixing quotas for salesmen. The specimen of a Sales Budget is given below:

ALPHA COMPANY LIMITED

SALES BUDGET FOR THE PERIODENDING ----------

Division	Product	Budgeted Sales Units	Budgeted Sales Price	Budgeted Sales Value
North Division Total	A B			
South Division Total	A B			
North + South	A			
North+ South	B			
Total				

2. PRODUCTION BUDGET

A production budget is a plan that shows the quantity of units of each product to be manufactured to cater to the sales forecast and to keep the desired level of inventories. It is prepared by the production manager. It is generally expressed in physical units. The production budget is based on sales budget but in those cases where sales cannot be precisely forecasted due to frequent change in fashion and style, it is based on past experience.

Units to be produced

=budgeted sales + desired level of closing inventory of finished goods–opening inventory of finished goods

The objectives of production budget are:

1. To plan what to produce, how much to produce, when to produce and where to

produce.

2. To lay down the production program that helps in maintaining optimum balance between sales, inventories and production at the minimal cost.
3. To serve as the foundation for the preparation of production cost budget, cash budget, material requirement budget, direct labor budget and factory cost budget.
4. To maintain the desired level of inventory so that there is neither shortage of goods during the peak season nor surplus stock during the slack season.
5. To make optimum utilization of plant capacity.
6. To make proper arrangements for material, labor, power etc. so that production can take place smoothly.
7. To highlight the weaknesses and source of potential trouble, which could be averted by taking timely actions?

In the production budget, the annual budget figures could be further broken down into monthly or quarterly estimates. The key factors i.e. factors which can limit the volume of production, should be identified. The following measures should be taken to insulate the production from the adverse effects of key factors:

Key factor	Preventive measures
Plant Capacity	Adoption of latest modern techniques Timely repair and maintenance Improving the plant lay out. Multiple shifts, night shifts
Raw Material	Scouting for alternative raw materials Re designing the product. Product modification To reduce intake of scarce raw material

Labor	Overtime working Proper incentive schemes Training of labor to improve efficiency Replacing labor intensive technology with capital intensive technology.
Power	Installing captive power unit Diesel generator for power supply Low power consumption devices.

Factors to be considered while preparing Production Budget are as follows:

1. Inventory policy of the firm – The inventory standard should be laid down so that there is neither surplus nor shortage of goods.
2. Sales requirements – The quantity of goods to be produced depends largely on the quantity of goods to be sold. Hence, preparation of sales budget precedes production budget.
3. Production stability – In order to have stability in employment, better use of plant facility and reduction of costs, the production should be evenly spread throughout the budget period.
4. Key factors – Key factors should be identified and proper preventive measures should be taken to minimize their adverse impact.
5. Time taken in production process – The production should commence depending upon the time it takes the factory to translate raw material into finished goods.

3. PRODUCTION COST BUDGET

Production budget is the key to the Production Cost Budget. After determining the volume of production, it is desired to determine the cost of producing this output. Just as cost of production comprises of material, labor and overhead, similarly production cost budget comprises of materials cost budget, direct labor cost budget and factory overhead budget.

Material Budget - A material budget specifies the expected quantities of each raw material required for the budgeted production. Materials used in the production process could be direct or indirect. The material budget includes only direct material and the indirect materials are considered in the factory overhead budget.

The material budget can be divided into two parts:

Material Usage Budget – which tells us the total raw material required to be used in the production process in terms of quantity/value.

Material Purchase Budget – which tells us the raw material required to be purchased for producing the required output in terms of quantity/value.

The material budget could be expressed in physical terms but when prices of materials are also considered, then we can ascertain the cost also.

The material budget helps purchasing department to prepare purchase schedule to enable uninterrupted delivery of materials for production. It helps finance managers to determine finances required to meet the production targets.

Factors to be considered while preparing materials budget are as follows:

1. Production Budget – The budgeted production will determine how much materials will be required for producing required output.

2. Prices of raw material – To determine the cost of raw material, quantity of raw material is multiplied by its price.

3. Standard usage – In case, raw material requirements can be standardized, the budget can be prepared precisely. Standard usage refers to input- output ratio i.e. how many units of material is used in producing one unit of output. If it is not possible to work out standardized raw material requirements, then percentage of raw material to total cost of product should be calculated on the basis of historical data and this ratio can be used as a basis for ascertaining material cost.

4. Normal loss/wastage in the production process –The material budget should also consider what percentage of raw material normally gets wasted in the production process. Efforts should be taken to keep normal loss to minimum.

5. Inventory Policy - The re order level, Economic Order Quantity, the desired level of

minimum stock and maximum inventory, operating inventory, closing inventory – all have bearing on material cost budget.

Direct Labour Budget – This budget gives the estimates of direct labor requirements essential for carrying out the budgeted output. This budget is devised to ensure that the required number of employees of relevant grades and suitable skills are available at the right time. This budget considers only the direct labor while the indirect labor is included in the factory overhead budget.

Direct labor budget may be expressed in physical terms (labor hours) but when wage rates are also available, then we can ascertain the cost also.

The Direct Labor Budget may be divided into two parts:

1. Direct Labor Requirement Budget – which specifies the total direct labor required for carrying the estimated production in terms of quantity or value.
2. Direct Labor Procurement Budget – which specifies the additional workers required to be recruited for carrying out the required production.

Direct Labor Budget enables the Personnel Department to recruit the required number of workers of suitable skills and provide them necessary training. It is also used for preparing the Cost of Sales Budget and Cash Budget.

Factors to be considered while preparing Labor Budget are as follows:

1. Production Budget – The budgeted production will determine how much of labor hours are required.

2. Wage rate – To determine the cost of labor, the labor hours are multiplied by the average wage rate per labor hour. Also overtime wage rate should be considered if workers work more than normal hours.

Standard labor usage – When labor usage can be standardized, the budget can be prepared precisely. Standard usage refers to input- output ratio i.e. how many labor hours are required to produce one unit of output. If it is not possible to standardize labor requirements, then percentage of labor cost to total cost of production should be calculated on the basis of historical data and this ratio can be used as a basis for ascertaining labor cost.

Labor cost = Labor cost per unit X No. of budgeted production.

Labor cost per unit = Number of labor hours required per unit of output X wage rate per labor hour.

FACTORY OVERHEAD BUDGET

This Budget sets out the estimated cost of indirect material, indirect labor and indirect factory expenses during the budget period.

Steps to be taken to prepare Factory Overhead Budget are as follows:

1. Classification of overheads into fixed, variable and semi variable expenses – Fixed overheads are those which remain constant irrespective of level of activity

e.g. property tax, supervisory salary, depreciation etc. Variable overheads are those which vary with the level of activity.

E.g. supplies, repairs etc. Semi variable overheads are those which are partly variable and partly fixed. Fixed overheads can be estimated with a fair degree of accuracy on the basis of past experience. Variable overheads are estimated on the basis of scheduled production and operating conditions in the budget period.

2. Departmentalization of expenses –The expenses should be classified department wise. This would help in determining how much expenses are incurred by various departments and the departmental heads can be held accountable.

3. Determining the level of activity – The level of activity will be determined on the basis of production schedule. Once the level of activity is determined, per unit overhead rate will be computed and applied on goods to find their cost of production.

4. ENDING INVENTORY BUDGET

Ending Inventory Budget shows the quantity as well as value of the closing inventory of the direct materials as well as finished goods of the company. These figures are used in the preparation of budgeted balance sheet of the organization. If varieties of direct materials are used, then category wise break up should be shown. Similarly, if numbers of products are manufactured then product wise break up should be shown. The quantities must be multiplied by the applicable rate to determine the cost of material in the inventory and cost of finished goods in inventory. The specimen of Ending Inventory Budget is given below

ALPHA COMPANY LIMITED

ENDING INVENTORY BUDGET FOR THE PERIOD _____

	Applicable Rate	Cost (Rest.)
Direct Material Inventories Material X Material Y		
Finished Goods Inventory Product A Product B		
Total cost of ending inventory		

Fig. 8.2 Specimen Cost of Ending Inventory Budget

5. COST OF GOODS SOLD BUDGET

Cost of Goods Sold Budget shows the budgeted manufacturing cost (direct material, direct labor and factory overhead) adjusted for beginning and ending inventory of finished goods. This budget can be prepared only after material cost budget, labor cost budget, and factory overhead budget and ending inventory budget have been prepared. This budget uses the estimates produced by the above said budgets to determine the budgeted cost of goods sold.

Specimen of cost of goods sold budget is given below:

ALPHA COMPANY LIMITED

Particulars	Amount (Rs.)
a) Cost of direct material used (from material budget)	
b) Cost of direct labor used (from labor budget)	

c) Cost of factory overheads (from factory overheads budget)	
Budgeted total manufacturing costs (a) + (b) + (c)	
+ Beginning finished goods (from production budget)	
- Ending finished goods (from ending inventory budget)	
Budgeted cost of goods sold	

Fig.8.3 Specimen Cost of Goods Sold Budget

6. SELLING EXPENSE BUDGET

This budget includes all expenses relating to selling, advertising, distribution and delivery of goods to the customers. Selling expenses includes salesmen salaries, sales office establishment expenses, travelling expenses, commissions etc. Advertising expenses include expenses relating to publicity, samples, advertisement expenses in journals, press, radio and television etc.

Distribution expenses include transportation, freight charges, packaging charges, warehousing charges, insurance etc. This budget is based on sales budget and is prepared on the basis of past experience, advertising policies, research programs, market analysis etc. It is generally prepared territory wise by the sales manager of each territory. The costs are segregated into fixed and variable cost.

Specimen of selling expenses budget is as under:

ALPHA COMPANY LIMITED

SELLING EXPENES BUDGET FOR THE PERIOD ENDING

Items	Cost	Total Cost
Variable Selling Expenses Sales Commission Salary & Wages Advertising Travelling Package Charges		

Fixed Selling Expenses Ware housing charges Advertising Marketing managers salary Depreciation		
Total Selling Expenses		

Fig.8.4: Specimen Selling Expenses Budget

7. ADMINISTRATIVE EXPENSE BUDGET

This budget covers the expenditure related to management, legal, financial, accounting and administrative activities of the organization. Formulation of this budget poses problems such as

1) Some expenses such as engineering cost, purchasing cost are such as they can be classified under both production as well as administrative costs. So, difficulty arises in deciding their correct classification.

2) It is difficult to identify the persons responsible for the incurrence and control of these costs.

Administrative expenses could be directors' remuneration, legal charges, audit fees, office expenses, rent, interest, taxes, postage etc. Generally these expenses are fixed and are shown in the budget under different catagories.

Specimen of Administrative Expenses Budget is given as under:

ALPHA COMPANY LIMITED

ADMINISTRATIVE EXPENES BUDGET FOR THE PERIOD ENDING ____

Items	Cost	Total Cost
Variable Administrative Expenses 1) Supplies		

Fixed Administrative Expenses Directors' fees Legal charges Salaries Rent Insurance		
Total Administrative Expenses		

Figure 8.5: Specimen Administrative Expenses Budget

8. BUDGETED INCOME STATEMENT BUDGET

A Budgeted Income Statement is prepared by using estimates provided by individuals operating budgets such as sales budget, cost of goods sold budget, selling expenses budget and administrative expenses budget. This budget shows the expected income that would be generated in the organization in the budgeted period. If tax rates are known, income before and after taxes can also be ascertained.

Specimen of Budgeted Income Statement is given below:

ALPHA COMPANY LIMITED

BUDGETED INCOME STATEMENT FOR THE PERIOD ENDING _____

Items	Amount
Sales	
Less: Cost of Goods Sold	
Gross Margin	
Less: Selling Expenses Admn. Expenses	
Income before taxes	

Less: Taxes on income	
Income after taxes	

Fig. 8.6 Specimen Budgeted Income Statement

13.4.2 CAPITAL EXPENDITURE BUDGET

The capital expenditure budget shows the estimated expenditure on acquisition of fixed assets such as plant, machinery, land, building etc. necessitated on account of replacement of old machines, increased demand for products, expansion of industry, modernization of the plant and adoption of new technology etc. Capital Budgeting is one of the most important areas of managerial decisions as such decisions involves huge outlay, have long term implications for the organization and are crucial for the achievement of its goals. Therefore, any error in this budgeting decision could prove fatal for the organization. The business firm should device proper procedural methods for evaluating the merits of the project before committing funds.

The alternative capital expenditure projects should be compared and evaluated in terms of expected sales, cost, and returns and then the most profitable alternative should be selected.

Capital expenditure budgets can be prepared for short range projects as well as long range projects. While short range projects are implemented during the current accounting period, long range projects cover several periods and so they are expressed only in general terms. When the time for their implementation approaches, then only they become budget commitments.

The capital expenditure budget should show the amount of original appropriations, the expenditure to date, new appropriations made, the amount of unutilized appropriations and the balance amount carried forward to next year.

13.4.3 RESEARCH AND DEVELOPMENT BUDGET (R&D)

Research & development refers to the investigative activities carried out by a business in order to experience future growth by developing new products or processes or improving existing products or procedures. The preparation of R & D budget facilitates the planning and controlling of R & D costs. This budget helps the business in planning

the R & D programmes it wishes to undertake, estimating correctly the staff and equipment requirements, making provisions for special facilities needed for the work, projecting the funds required for meeting the research costs and assessing the fairness of these expenses. The research projects should be grouped into long term and short term projects. A short term research program is necessary for earning a satisfactory rate of return on the funds invested. A long term research program is essential to keep pace with the future market trends and demands. Research and development is generally discretionary and therefore, when there is a decline in the income or financial position of the firm, some of the research programmes may be deferred or aborted.

13.4.4 CASH BUDGET

A cash budget is an estimate showing the amount of cash which would be available in the budget period. This budget gives in detail the estimate of cash receipts and cash disbursements. The expected cash receipts depend on estimated cash sales, collection from debtors, sale of assets, borrowings etc. The expected cash outlays depend on cash purchases, payment to creditors, payment of wages and salaries, repayment of loans, budgeted capital expenditure etc. The main objectives of preparing a cash budget are:

- To know the probable cash position as a result of planned operations so asto predict the shortages and surplus of cash .This would help in making timely arrangements for short term borrowings to meet the deficit and to invest the surplus cash in profitable avenues.
- To maintain optimum level of working capital keeping in view seasonal requirements, unusual receipts, unforeseen expenditure, possibilities of bad debts etc.
- To make arrangements for sufficient funds in case of some plant expansion programmes
- To reveal potential cash shortages and explore the possibilities of procuring additional funds from sources such as bank loans, sale of securities etc.
- To use cash in the most profitable way by investing surplus funds in lucrative options
- To maintain adequate liquidity.

Though the cash budget is dependent on sales budget and operating expenses budget, these budgets themselves alone cannot determine an optimum cash position. This is because sales budget and operating expenses budget are prepared on accrual basis whereas cash budget is prepared on cash basis and does not include accrual items. The factors to be considered while preparing a cash budget are as follows:

1. Estimated opening balance
2. Estimated sales value in the sales budget
3. Cash and credit sales ratio, terms of credit, period of credit
4. Amount of monthly purchases as per the purchase budget
5. Credit policy of the suppliers and credit terms
6. Amount of salaries, wages and operating expenses
7. Capital expenditure plans
8. Minimum cash balance to be maintained.

The cash budget consists of two parts – (1) cash receipts and (2) cash disbursements. The cash budget figures are generally broken down quarter wise, month wise or week wise for adequate planning and control of cash requirements.

Period of Cash Budget - The planning horizon of a cash budget depends on number of factors such as nature of business, size of business, cash position, the accuracy with which cash flows can be predicted etc. Accordingly, the range of cash planning could be:

a) **Operational Cash Planning**–Cash budget could be prepared month wise, week wise or day wise depending upon the informational requirements of management and for having dynamic control over cash balance. Operational cash planning ensures that daily operational needs can be met out comfortably out of available cash resources.

b) **Short Range Cash Planning –**Here, cash budget is prepared for the full year and corresponds with the annual profit plan. It indicates cash inflows and outflows as generated by the annual profit plan. It helps in determining short term credit needs.

c) **Long Range Cash Planning –** Long range cash budgeting generally incorporates

the effects of business expansion and long term trends. Its purpose is to find out whether cash can be generated through working capital growth and in the event of a shortfall, how to make up the deficiency.

Methods of Preparing Cash Budget- A cash budget could be prepared using any of the three methods:

a) **The Receipts and Disbursements Method –** In this method, the balance of cash at the end of a month/quarter/year is found out by adding to the opening cash balance of that month/ quarter/ year, the estimated cash receipts from various sources and deducting cash payments to various agencies during that month/ quarter/ year. The estimated receipts are attributable to cash sales, cash collected from debtors, dividends received, and interest on investments, proceeds from sale of assets, royalties, bank loan, and issue of share capital and like. Cash disbursements result from purchase of material, payment of wages and salaries, purchase of assets, repayment of loans, distribution of dividends, deposit of taxes, meeting of various expenses etc. Therefore, the preparation of cash budget as per this method requires the use of sales budget, materials budget, labor budget, factory overhead budget, selling & distribution expenses budget, administrative expenses budget, capital expenditure budget and R &D budget. This method is generally used for short term cash projections and is not appropriate for long term cash budgeting.

b) **Adjusted Net Income Method -**In this method, the budgeted profit stated in the income statement is converted from an accrual basis to cash basis. To achieve this, the budgeted profit of a period is adjusted for non-cash transactions and cash from operations is derived. For ascertaining this, non- cash items such as depreciation, loss on sale of fixed assets, bad and doubtful debts and preliminary expenses written off, income tax accruals etc. – which were charged to Profit & Loss Account are added back as they do not affect cash position. Further, expected cash oriented changes in various assets and liabilities are analyzed to find their effect on cash resources. Issue of new shares; realization from sale of fixed assets; raising of long term loans; decrease in current assets such as debtors, stock or bill receivables; increase in current liabilities such as creditors or bill payables result in increasing the cash resources and are added. Similarly redemption of

preference shares; payment of long term loans; purchase of fixed assets; payment of dividends; increase incurrent assets such as debtors, stock or bill receivables; decrease in current liabilities such as creditors or bills payables result in decreasing the cash resources and are subtracted. Thus budgeted cash at the end of the period is opening cash balance at the beginning of the period plus cash from operations plus net cash increase (or minus net cash decrease) as revealed by the analysisof changes in various assets and liabilities. This method is useful in making long range cash projections.

c) **Balance Sheet Method –** In this method, the closing balances of all the balance sheet items except cash and bank are determined and placed in the budgeted balance sheet. If the sum total of all the liabilities side items exceeds the sum of all the assets side items, the balancing figure denotes the cash/bank balance. On the other hand, if the sum of assets side items exceeds the sum of liabilities side items the balancing figure denotes the bank overdraft or cash deficit. The amount of various closing balance sheet items is determined after their opening balances have been adjusted for the transactions expected to take place in their accounts during the year.

13.4.5 BUDGETED BALANCE SHEET

The budgeted or projected balance sheet portrays the expected financial position of the business on a particular date. It is prepared with the help of budgeted balance sheet at the start of budget period and the various operating and financial budgets prepared for the period. The balances in the opening balance sheet are adjusted in the light of anticipated transactions reflected in the operating budgets, capital expenditure budget and cash budget. If any of the accounts in the budgeted balance sheet or relationships in the accounts are not in accordance with the management's requirements, then the operating plan is required to be modified. For instance, if debt equity ratio or current ratio is too high or too low, the operating plan would have to be changed or otherwise it would lower the creditability of the firm or adversely affect the value of its shares in the stock market. The projected balance sheet enables the checking of arithmetical accuracy of other budgets as they are all used in the preparation of forecasted balance sheet and any imbalance or incompatibility is revealed out.

13.4.6 BUDGETED STATEMENT OF CHANGES IN FINANCIAL POSITIONS

This statement is generally prepared from the budgeted balance sheet at the beginning of the budget period, budgeted balance at the end of the budget period and the budgeted income statement. This statement shows what changes have taken place in the various balance sheet items over a budget period and details about the possible causes of such changes. This projected statement of changes in financial position helps the management in financial planning.

MASTER BUDGET

A comprehensive budget or a master budget is a package comprising of various schedules and financial statements showing results expected in a future period. It consists of an income statement, a balance sheet, a statement of cash receipts and disbursements and schedules of production, purchases, expenses and fixed assets acquisition.

The Institute of Management Accountants, London defines a Master Budget as "the summary budget incorporating its functional budgets which are finally approved, adopted and employed."

Thus it is a summary budget which incorporates and integrates all other budgets. It is the end product of the budget making process. The master budget is presented to the budget committee for approval before it is operationalized. Many a times, number of master budgets have to be prepared before the final one is agreed up on. **Uses of Master Budget**

A master budget takes the macro view of the business and covers the entire system whereas operating budgets are concerned with one particular aspect of the business and provide plans for some specific set of activities such as sales, production, purchase, selling activities etc. A master budget acts as a tool for coordinating all the individual budgets. It coordinates sales with production, production with purchase of material and labor and so on. This way, it cuts across the divisional boundaries in order to coordinate and synchronize the organizations' diverse activities. This coordination and cohesiveness among different budgets enables management to make better predictions,

take better decisions and overcome challenges. It reveals whether top management goals of revenue expenses, income, cash and financial positions can be smoothly achieved or not. It is supplemented with important accounting ratios such as current ratio, liquidity ratio, profit to capital employed ratio etc.

13.4.7 FIXED BUDGET

A fixed budget is one which is static and remains fixed irrespective of level of activity attained. It is prepared on the basis of a standard or fixed level of activity.

The Institute of Management Accountants (U K) defines a fixed budget as

"A budget which is designed to remain unchanged irrespective of the level of activity actually attained."

If a company prepares a fixed budget, the performance report compares the data from actual operations with the figures pertaining to the single level of activity reflected in the fixed budget. If actual production turns out to be different from budgeted production, the management will not be able to access the performance of different departmental heads on the basis of budget prepared by them as these budgets become unrealistic measuring yards. Further since fixed budgets do not provide for automatic adjustments with volume changes, it will be difficult to assess the ability of the firm in keeping the costs within the allowed limits. Therefore, in a situation where, there is lot of uncertainty and cost, expenses, production, sales etc fluctuate, fixed budgeting turns out to be inappropriate. Infixed budgeting, it is assumed that the company will work at some specified level of activity and that a stated production will be achieved. A budget is then prepared for that production level and is not adjusted according to the activity level actually attained. In real life, conditions are not static. There are several internal and external factors which force the level of activities to change from what was expected. Thus, actual production is generally different from budgeted production. In such a situation, fixed/static budget cannot be used for the purpose of comparing the cost, revenues or profits.

Since in fixed budgeting volumes are overlooked, a cost to cost comparison without considering the volume, would give misleading results. The performance report prepared under fixed budgeting would only point out whether actual costs were higher

or lower than budgeted costs. Such report fails to disclose that the fluctuations in volume were responsible for the change in costs and expenses. Therefore, fixed budgets are not effective tools of cost control and have limited use. **Fixed budgets are useful only in the given situations:**

1. When expenses remain fixed and do not vary with the volume of production or output.
2. When predictions can be so accurately made that the actual output is very close to budgeted output.
3. When the environment in which the business is operating, is not very dynamic or the business remains unaffected by disturbing forces.

Limitations of Fixed Budgets

Since fixed budgets are prepared at a single level of activity with no prospect of modification in the light of changed circumstances, they are not very popular. The major weaknesses of static budgets are as follows:

1. They fail to reveal the potential variability of various estimates used in preparing the budgets.
2. They are unable to indicate the range within which costs can be expected to fall.
3. They are not useful if environment is dynamic, uncertain and unpredictable.

13.4.8 FLEXIBLE BUDGETING

A Flexible Budget is the opposite of the static budget. It is a dynamic budget in which the costs are estimated at various levels of activity. The Chartered Institute of Management Accountants (U K) defines a flexible budget as:

"A budget which by recognizing the difference between fixed, semi variable and variable costs is designed to change in relation to the level of activity attained." Thus when actual output differs from the budgeted output, there is no need to modify the budget on the basis of changed output. The flexi-budget itself is so designed that it presents the budgeted costs for a range of possible level of activities. Therefore, this type of budget is more realistic and changes expected at different level of activities are given due consideration. This type of budget is also known as variable budget or sliding budget.

A flexible budget is appropriate in the following cases:

1. When business activities cannot be predicted accurately due to uncertainty in the environment.
2. When the venture is new and so it is not possible to estimate the public demand.
3. When the supply of labor is a key factor for the business and the business is in that area, which is facing shortage of labor.

   A flexible budget is prepared after segregating costs in to fixed, variable and semi variable and then analyzing their behavior at various levels of activity. In case, the budgeted level of activities is not attained, flexible budget accommodates itself to the actual level of activities and thereby, it facilitates better comparison of costs and evaluation of performance.

A flexible budget has the following special features:

1. It is prepared for that range of activities that is likely to occur.
2. It is flexible and budgeted costs, sales and profits at different capacities can be easily found out.
3. It is based on an in depth cost behavior analysis and recognizes the difference between fixed, semi variable and variable expenses.
4. It facilitates better comparison of costs and measurement of performance

In order to prepare a flexible budget, it is essential to carry out detailed cost analysis in order to ascertain how differently the various costs behave in response to change in volume. The expenses are divided in to three categories - fixed, semi-variable and variable. Fixed costs remain fixed irrespective of changes in volume. Variable costs vary directly and proportionately with the volume of activities. The semi variable costs are those which remains fixed up to a certain level of activities and thereafter vary with the level of output

STEPS IN FLEXIBLE BUDGETING

In order to develop a flexible budget, the following steps should be taken:

1. To decide the range of activities for which the budget is to be prepared. The

activity levels not likely to occur are not included in the budget.

2. To determine the behavior pattern of each element of cost in the budget. This requires a close study of the individual items of expenditure, their nature and variability.

3. To select the activity level which is generally expressed in terms of production or output.

4. To prepare a budget at each activity level. This requires a careful calculation of corresponding costs after analyzing their behavior pattern.

ADVANTAGES AND DISADVANTAGES OF FLEXIBLE BUDGETING

A flexible budget considers several alternatives/levels/volumes of activity. Here the budget is developed for a relevant range of production, say for example 6,000 units to 10,000 units. If suppose actual production turns out to be 8000 units instead of budgeted production of 9,000 units, flexible budget can be used to determine budgeted costs at 8,000 units. This way flexible budget provides an accurate and reliable basis for comparison because it is automatically geared to the changes in the production activities.

The main advantages of flexible budgets are as follows:

1. **Accurate budgeting** – Flexible budgets give due consideration to the output factor in the budget preparation. Since all costs do not vary in same fashion to changes in volume, a budget that gives consideration to volume aspect is bound to be more accurate and realistic.
2. **Better performance measurement** – Flexible budgets facilitate the comparison of actual results with the budgeted estimates in terms of output achieved. This comparison is more meaningful and enables accurate assessment of managerial and organizational performance.
3. **Aids management control** – Flexible budget can be used as effective management control tool. Comparisons are made between budgeted costs (at the actual production level) and the actual costs. The deviations observed form the basis for analyzing cost variances and fixing responsibilities. The managers feel motivated in controlling the costs for which they are responsible. This generates cost

consciousness throughout the organization.

4. **Better prediction** –Flexible budget helps the organization to predict its performance and income at various levels of activity or production. The impact of changes in the production level on revenue, expenses and income helps in better decision making.
5. **Easy to Incorporate Change** – Anything which is flexible, allows changes to be incorporated easily. As Business world is too volatile, there are rapid changes, which can be incorporated in flexible budget.

DISADVANTAGES OF FLEXIBLE BUDGET

The flexible budgets have the following demerits:

1. Flexible budgets assume linearity of costs i.e. variable costs vary in the same proportion as changes in activity level. However, there are several expenses like labor costs which do not behave in linear fashion.
2. Flexible budgets rely on the assumption of continuity. However, there are many costs which actually behave in stepped (like. machine repair & maintenance) or discontinuous manner (like overhauling of machines and other office equipments).
3. Flexible budget presumes that fixed costs remain the same irrespective of activity level. However, many fixed costs remain fixed only for a certain output range and beyond that they also vary. For example – rent of space occupied by machinery. If output exceeds certain level, then new bigger size machinery may be required which may occupy more space.
4. The method of determining the fixed and variable elements of costs is often arbitrary. Further, segregation of semi variable costs in to fixed and variable components poses problems.

Thus the major weakness of flexible budgeting is that it is based on unrealistic assumptions about costs. If the costs do not behave in the predicted manner, even flexible budgets could mislead.

ILLUSTRATION

ABC Manufacturing Company is currently utilizing 80% capacity. Prepare a budget at60% and 100% capacity level using the information given below:

Production at 80% capacity	800 units
Material	Rs. 90 per unit
Labor	Rs. 50 per unit
Expenses	Rs. 10 per unit
Factory expenses	Rs. 40,000 (40% fixed)
Administrative expenses	Rs. 20,000 (60% fixed)

Solution

Flexible Budget of ABC Manufacturing Company Department:---------

Period:--------

Capacity		60%	80%	100%
Output		600 units	800 Units	1000 units
Materials	Rs. 90/- unit	54000	72000	90000
Labor	Rs. 50/- unit	30000	40000	50000
Expenses	Rs. 10/- unit	6000	8000	10000
Factory Expenses	Rs. 30/- unit	18000	24000	30000
Administrative Expenses	Rs. 10/- unit	6000	8000	10000
Total variable Expenses		114000	152000	190000
Fixed Costs				
Factory Expenses	40% Fixed	16000	16000	16000

Administrative Expenses	60% Fixed	12000	12000	12000
Total		142000	180000	218000

13.5 LET US SUM UP

The term 'budget' traces its origins to the French word "bougette," meaning a small leather pouch used for carrying money. This historical reference aptly describes the modern concept of a budget as a container for an organization's financial plans. In contemporary business practice, a budget represents a comprehensive, quantitative statement that outlines how resources will be acquired and allocated over a specified future period. It serves as both a planning document and a control mechanism, translating organizational objectives into concrete financial terms.

Budgets possess several defining characteristics that distinguish them from other financial documents. First, they are inherently forward-looking, prepared in advance of the period they cover. Second, they encompass both operational activities and resource requirements, presented in standardized monetary terms. Third, budgets are time-bound, typically covering a fiscal year, and must be comprehensive in scope. Finally, they require coordination across all departments to ensure alignment with organizational strategy. These features collectively make budgets powerful tools for financial management and organizational control.

While often confused, budgets and forecasts serve distinct purposes. A forecast is predictive in nature, projecting likely future outcomes based on current trends and assumptions. In contrast, a budget is prescriptive, representing management's committed financial plan and resource allocation strategy. Forecasts are passive estimations that may change frequently, while budgets are active management tools that establish specific targets against which performance will be measured. This fundamental

difference underscores the budget's role in driving organizational accountability and performance.

The budgeting process involves three key phases: design, implementation, and monitoring. During the design phase, organizations establish frameworks and guidelines that shape budget development. Implementation involves putting the budget into operation across all departments. The monitoring phase focuses on tracking performance against budgeted figures and making necessary adjustments. Throughout this process, the primary emphasis remains on strategically allocating resources to support organizational objectives while maintaining financial discipline. This systematic approach transforms strategic plans into actionable financial roadmaps.

Budgeting serves several critical organizational functions. It compels systematic planning by requiring managers to anticipate future needs and challenges. The process enhances coordination by aligning departmental activities with overall corporate goals. Budgeting improves communication by clearly conveying expectations throughout the organization. As a control mechanism, it provides benchmarks for evaluating performance and identifying variances. These functions collectively make budgeting indispensable for effective financial management, operational efficiency, and strategic execution in modern organizations.

KEYWORDS

Budget, Budgeting, Financial Planning, Resource Allocation, Operating Budget, Capital Budget, Cash Budget, Fixed Budget, Flexible Budget, Budgeted Balance Sheet, Forecast vs. Budget, Budget Variance, Budget Control, Performance Evaluation, Cost Management, Strategic Planning, Coordination & Communication, Fiscal Year, Financial Discipline, Managerial Accounting

13.6 REVIEW QUESTIONS

1. Which of the following best defines a budget?

a) A historical record of financial transactions

b) A quantitative expression of a proposed plan of action

c) A report of actual financial performance

d) A prediction of market trends

2. What is the primary purpose of a capital expenditure budget?

a) To plan day-to-day operational expenses

b) To allocate funds for long-term asset acquisition

c) To forecast short-term cash flows

d) To estimate research costs

3. Which type of budget remains unchanged regardless of the activity level?

a) Flexible budget

b) Zero-based budget

c) Fixed budget

d) Cash budget

4. The Budgeted Statement of Changes in Financial Position primarily shows:

a) Profit and loss for the period

b) Changes in working capital

c) Fixed asset acquisitions

d) Dividend payments

5. A _____ budget adjusts based on changes in activity levels or output.
6. The _____ budget forecasts all cash receipts and payments for a period.
7. Operating budgets typically cover a _____ period than capital budgets.
8. What is budget? Explain its advantages.
9. Briefly discuss the types of budget.
10. Distinguish between Fixed and Flexible budget.

UNIT-14: BUDGETARY CONTROL

STRUCTURE

14.1 Introduction

14.2 Budgetary control

14.3 Budgetary control process

14.4 Budgetary Reports

14.5 Budgetary Control Ratios

14.6 Let's sum up

14.7 Review Questions

LEARNING OUTCOME

After studying this module, you shall be able to

- Learn the meaning, concept and aims of Budgetary Control
- Know the characteristics of a Budgetary Control System
- Explain the advantages of a Budgetary Control System Elaborate pre-requisites of a successful Budgetary Control
- Identify the steps required in the installation of a Budgetary Control System.

Bloom's Level of Learning

The topics listed under budgetary control primarily align with **Bloom's Taxonomy levels of "Understanding," "Applying," and "Analyzing."** At the **"Understanding"** level, learners grasp the meaning and purpose of budgetary control, including its role in financial management. The **"Applying"** level comes into play when students engage with the budgetary control process, learning how to implement control mechanisms and prepare budgetary reports in practical scenarios. Finally, **"Analyzing"** is required when interpreting budgetary control ratios to assess financial performance, identify variances, and evaluate organizational efficiency. These topics collectively move beyond basic knowledge (Remembering) to more advanced cognitive skills, focusing on practical implementation and

critical evaluation of financial control systems.

14.1 INTRODUCTION

We know that budget is a plan that shows how the resources are to be acquired and used over a specified period of time. While the operations are in progress, budget serves as a basis for comparing actuals with the budgeted estimates thereby facilitating control. The use of budget as a means of controlling the various activities is called Budgetary Control.

14.2 BUDGETARY CONTROL

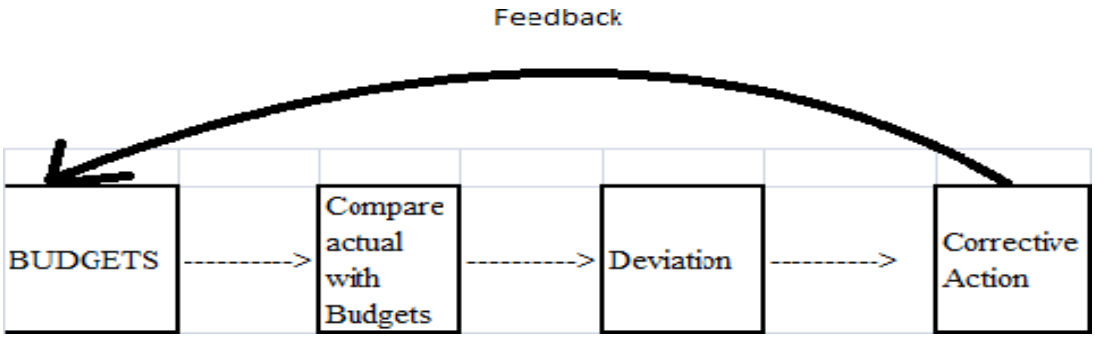
Budgetary control is a part of overall management control of the organization. It is the essential tool of management for controlling costs and maximizing profits. Let us look at some of the definitions on budgetary control-

"Budgetary control means establishment of budgets relating to the responsibilities of executives to the requirements of a policy and the continuous comparison of the actual with the budgeted results either to secure by individual action the objective of that policy or to provide a basis for its revision"

The Institute of Management Accountants, London

"It is a system of management control and accounting in which all the operations are forecasted and planned so far as possible ahead and the actual results compared with the forecasted and planned ones". J. A. Scott

BUDGET



Budgeting and Budgetary Control

A Budget is the plan of estimated expenses and/or revenues associated with the operations of various departments of the organization. Budgeting may be said to be the process of preparing budgets while budgetary control embraces all this and in addition includes the science of devising the budgets themselves and using such budgets by the management for effecting overall control over the operations of the organization.

Aims/Objectives of Budgetary Control

The objectives of Budgetary Control are

1. **Definition of goals** i.e. to portray in clear-cut terms, the overall aims of the organization and breaking them down into target goals of each unit.
2. **Defining responsibilities**i.e.to clearly specify the responsibilities of each person so that everybody knows what is expected of him and how will he be evaluated.
3. **Optimum use of resources i.e.** to use the human, financial and material resources in best possible manner and reducing wastages.
4. **Coordination and cooperation**: to integrate different units and departments so that they work with harmony in a unified manner.
5. **Proper execution** i.e.to ensure that all activities are proceeding in the manner envisaged in the budget.
6. **Ease in comparison i.e.** to compare the actual performance against the budget and draw attention to the deviations observed.
7. **Prompt remedial measures i.e.** to take proper measures in case deviations are noticed to prevent discrepancies in future.
8. **Revision i.e.** to revise the budget in the light of changed circumstances.

Characteristics of a Budgetary Control System

A budgetary control system has the following characteristics

1. **Establishment of budgets -** Budgets are prepared for each department and then they are presented to the management for approval.
2. **Coordination among departments**– The plans of various departments are coordinated and finally integrated into the master budget.

3. **Comparison**– Under budgetary control, the actual performance is compared with the budgetary estimates on a continuous basis to discover the variations and fix responsibilities.
4. **Revision** – The budgets are revised in the light of changes in the conditions and circumstances...

Advantages of Budgetary Control System

The advantages of budgetary control are numerous. It prevents the occurrence of the deviations from the preplanned path through the media of observation, research, planning, control and decision making and thus helps in the performance of future activities in an orderly manner. Some of the advantages are:

1. Efficiency and economy – Budgetary Control reveals the faults in the operations, weaknesses in the organizational structure and minimizes wasteful spending.
2. Fixation of responsibilities – Budgetary control prevents "buck passing" and exposes those who are at fault/deviation.
3. Coordination– It coordinates the production, marketing, financial, administrative and other dimensions and forces them to work in totality rather than in isolation and operate as a group in harmony.
4. Pre determination of financing – Because the organization has a blue print of activities to be undertaken in the coming period, funds and finances can be arranged well in advance preventing the financial crisis.
5. Management by exception – Budgetary control exposes the deviations and only variances and discrepancies are reported to the superiors. This enables them to concentrate on these issues seriously enabling management by exception.
6. Forerunner of standard costing – A budgetary control makes available budgeted figures which can be used as a benchmark for variance analysis paving way for the adoption of standard costing principle in the organization.
7. Objectivity in evaluation- Since everyone's performance is assessed on the basis of pre-set criterion, assessment is objective without any bias.

3.6 Essentials of a successful Budgetary Control System

For an effective system of budgetary control, the presence of following elements is necessary:

1. The budgetary control system should have the full support of the top management as well as the co- operation of all the employees.
2. There should be well planned organizational set up in which responsibility and authority of each one should be clearly demarcated.
3. The accounting system should provide accurate and timely information.
4. There should be proper system of communication and reporting and all the participants of the organization must be involved in budget setting process.
5. There should be an efficient system of feedback which reports variances fairly and promptly.
6. The staff should be highly motivated and have faith in the budgetary control system.
7. Budget should be used as a coordinating device rather than a control device.

14. 3 BUDGETARY CONTROL PROCESS

The method by which a budget is prepared differs from organization to organization. While some organizations may have well organized, well documented procedures, others may not have any formal procedure and so in their case, budgeting process proceeds in a haphazard and disorganized manner. But since budgeting is an important planning exercise, it is suggested that it should be carried out in a systematic manner to avoid the conflicts and frustration at a later stage. Differences in the management styles, organizational objectives, size of organization, nature of business and similar factors have bearing on the procedures to be adopted by an organization in the budgeting process. Nevertheless, some of the common steps are as under:

1. **Specification and communication of organizational objectives**

   A budget is a medium through which an organization's objectives and policies are reflected. The budgeting exercise starts in the light of these objectives plus policies and guidelines issued by top management to govern budget preparation. Detailed

information and guidance is given on matters such as projected general economic conditions, allowances to be made for price increase and wage increases, changes in the scale of operations and so on. Further instructions are issued as to what information is required from each responsibility center for the budget and how this information is to be recorded in the budget form. In the absence of statements to the Contrary, it is assumed that factors affecting operations in the budget year will be similar to those in the current year.

2. **Preparing well defined organizational chart** -An organization should have a well- defined organizational chart depicting the hierarchal position of each executive in the organizational and his functional responsibility. Such a chart will clearly show the lines of authority, the authority delegated and relative position of each in relation to others.

3. **Consideration of certain key issues**

a) **Principal budget factor**/critical success factors -Every organization has certain critical success factors or key factors that can influence the activities of an undertaking and can limit the volume of output. They impact the profitability of the organization. Critical success factors could be -specified raw material, a specific type of skilled labor, floor space, cash resources etc. These limiting factors may shift from time to time due to external and internal circumstances. So, care must be taken for their identification and the ways of overcoming key factors. In majority of the organizations, the most critical factor is consumer demand or expected level of reserves/sales. That's why, sales budget is usually prepared first of all followed by other related budgets.

b) **Budget centers** – A budget center is the lowest unit in the organization for which detailed costs are budgeted, separately from those of other budget centers. It is a center of organization for which separate budgets can be prepared and control can be exercised. The entire organization should be clearly divided into different sections as budget centers for enabling budgetary control and fixing responsibilities of departmental heads.

c) **Budget period** – A budget period a period for which the budget is prepared. This is

the period for which forecasts can reasonably be made and estimates can be given. Budget period could be short range or long range. Short range budget may cover period of 3 months or 6 months or12 months while long range budget covers period more than one year. Capital expenditure budget and R & D budget are examples of long range budget. Sales budget or Income &Expenditure budget are examples of short range budget. In preparing short range budgets, certain factors must be kept in mind, like, budget period should have a term enough to complete production cycle of various products. In case of seasonal businesses, the budget period should cover one entire season cycle

.It should be long enough to provide adequate time to arrange funds for production and should coincide with the financial accounting period to facilitate comparison of actual with budget estimates.

d) **Budget Committee** – It is the body entrusted with the responsibility of overseeing the proper formulation and implementation of the budget. The budget involves the various functional activities and so the association of various functional managers is desirable. The budget committee is composed of all the major functional heads and acts as a forum where communication, coordination and discussion among them is facilitated. The main functions of the budget committee are to assist functional managers in preparation of budget, receive and review their budget estimates and recommend changes if any, resolve conflicts between departments on budget matter, to prepare master budget on the basis of functional budgets, to compare budget with actual performance and recommend action to be taken on the basis of variance analysis and make recommendations for the budget manual.

e) **Budget manual** – It is a document which contains standing instructions with respect to the procedures to be followed and the time schedules to be observed while making budgets. It acts as a guide in the hands of the budget officers and all the functional heads and prevents the need of frequent instructions from the top management. The budget manual provides instructions such as time limit for submission of preliminary forecasts, person to whom this has to be submitted, important items for each forecast, categorization of expenses, prescribed forms to make reports, listing the matters to be decided only with the approvals of the top

management etc...

4. Initial preparation of budget components

The budget guidelines prepared by top management together with the sales plan are disseminated down through the successive levels in the organization. Managers at each level add more details for the guidance of their subordinates. When the guidelines arrive at the lowest responsibility centers, the concerned managers prepare the budget plans for the items falling within their sphere, keeping in mind the constraints specified in the guidelines. These estimates are made by them by combining specific analytical techniques with their judgment.

5. Negotiation

Negotiation between a superior and subordinate (budgeted) is a very crucial stage in the budgeting process. If budget is too tight, it becomes unattainable and leads to frustration. On the other hand, if it is too loose, it leads to complacency. The budgeter and his superior, therefore, seek to arrive at a desirable middle path by means of slacks and cuts. The subordinates generally resort to slack or padding i.e. overstate their budgeted costs or understate their budgeted revenues to create a budgeted profit at the level that is easier to achieve. Budgetary slack provides protection against cost increase or revenue short falls due to unforeseen events. But it leads to budget **cuts** imposed by the superior who is aware of slack, padding and other similar tactics of the subordinates. In order to address this issue, instead of resorting to arbitrary cuts, superior should require a full explanation of why costs may be expected to increase. Subordinates on his part must justify increase in cost by explaining underlying causes such as cost of inflation, need for better quality and so on. The end product of the negotiation process is an agreement which represents a commitment by each party, the budgeter and his superior. The nature of the commitment could one of three types:

a) It may represent a ceiling (e.g. “Not more than Rs. 10 lac should be spent for employee training”).
b) It may represent a floor (e.g. “At least Rs. 10 lac should be spent for employee training”).
c) It may represent a guide (e.g. “Approximately Rs. 10 lac should be spent for

employee training”).

6. Coordination and review

The negotiation process is repeated successfully at higher levels of the organizational hierarchy up to the very top. Negotiations at higher levels result in changes in the detailed budget that have been agreed to at the lower levels. As the individual budgets move up in the organizational hierarchy in the negotiation and review process, they are also examined in terms of the compatibility (i.e. in relationship to one another) and potentiality (whether in the overall interest of the company). If they are found out of balance, some of the underlying budgets/figures may be modified. Major unresolved issues are submitted to the budget committee.

7. Final approval and distribution

Just prior to beginning of the budget year, the proposed budget is submitted to the top management for approval. If the proper procedure is followed and there are no great surprises in the proposed budgets, they are approved. The budgets are then communicated to the department/responsible managers. Modifications incorporated in the final budget should be particularly made known to managers to seek their cooperation and support for the budget.

8. Implementing the budget plan

The final budget is presented to the managers concerned and adopted as the plan of operation for the forthcoming budget period. The various service units in the organization are required to release the necessary materials, labor, facilities, finances and other resources to carry out the budget.

14. 4 BUDGETARY REPORTS

Budget report is like a report card of a budget center. In this report, the actual figure and the budgeted figures are reported side by side to facilitate comparison and determine deviation or variations. In the report, both favorable and adverse variations should be reported with necessary comments/reasons. These reports form the basis for fixing the responsibility and taking corrective action. The budget report follows the principle of

management by exception.

14. 5 BUDGETARY CONTROL RATIOS

The management employs ratio analysis to know whether there are any deviations between the budgeted figures and the actual results. Control ratios are the effective tool in the hands of the management to exercise an effective control over the budgeted results. These ratios are expressed in terms of percentage. If the ratio is more than 100%, it is treated as favorable and if less than 100%, it is considered adverse. The important budgetary control ratios used by the management are:

1. Activity Ratios
2. Capacity Ratios
3. Efficiency Ratios

Activity Ratio – This ratio measures the level of activity attained by the firm over the specified period .Symbolically,

$$\text{Activity Ratio} = \frac{\text{Standard hours required for actual output}}{\text{Budgeted hours}} \times 100$$

Capacity Ratio- This ratio measures the extent to which budgeted hours of activity have been actually used. Symbolically,

$$\text{Capacity Ratio} = \frac{\text{Actual hours worked}}{\text{Budgeted hours}} \times 100$$

Efficiency Ratio – This ratio measures the extent of efficiency attained in the production.Symbolically,

$$\text{Efficiency Ratio} = \frac{\text{Standard hours for actual production}}{\text{Actual hours worked}} \times 100$$

14.6 LET'S SUM UP

A budget is formulated in accordance with certain assumptions as to conditions that will prevail during the budget period. If the actual conditions turn out to be different from those assumed, differences may be significant. This may call for revision in the budget during the year. Some organizations instead of revising their budget, take account of changes in conditions when they analyze the differences between the budgeted and actual performance. Other companies may solve this problem of revision by having two budgets- a baseline budget set at the beginning of the year and a current budget reflecting the best current estimates of revenue and expenses. A comparison of actual performances with the baseline shows the extent of deviation from the original plan and a comparison of the current budget with the baseline budget shows how much of this deviation is attributable to changes in the current conditions from those originally assumed.

Budgetary control involves establishment of budgets, continuous comparison of actual performance with budgets, finding out discrepancies, fixing the responsibilities, taking corrective measures and revising the budgets in the wake of change in circumstances.

Characteristics of a Budgetary Control System

- Establishment of budgets
- Coordination among departments
- Comparison
- Revision

Advantages of Budgetary Control System

- Efficiency and economy
- Fixation of responsibilities
- Co ordination
- Pre determination of financing
- Management by exception
- Forerunner of standard costing

- Objectivity in evaluation

Essentials of a successful Budgetary Control System

- Support of top management
- Well planned organizational set up
- Proper accounting system
- Proper communication system

Budgetary Control Process

- Specification and communication of organizational objectives
- Preparing well defined organizational chart
- Consideration of certain key issues like principal budget factors, budget center, budget period, budget committee & budget manual.
- Initial preparation of budget components
- Negotiation
- Coordination and review
- Final approval and distribution
- Implementing the budget plan
- Revision

Budget Report is like a report card of a budget center wherein the actual and the budgeted figures are reported side by side to facilitate comparison and determination of deviations.

Budget Ratios Activity Ratio Capacity Ratio Efficiency Ratio

14.7 REVIEW QUESTIONS

1. Budgetary control primarily serves the purpose of:

   a) Preparing financial statements

   b) Comparing actual performance with budgeted targets

   c) Calculating tax liabilities

   d) Determining market share

2. Which of the following is NOT a stage in the budgetary control process?

   a) Budget preparation

   b) Recording actual results

   c) Calculating variances

   d) Auditing financial statements

3. Budgetary control ratios are primarily used for:

   a) Measuring efficiency of operations

   b) Determining product pricing

   c) Calculating depreciation

   d) Assessing employee performance

4. Which type of report is most crucial for budgetary control?

   a) Variance analysis report

   b) Balance sheet

   c) Cash flow statement

   d) Income statement

5. The process of comparing actual results with budgeted figures is called _____. Budgetary control helps in _____ by identifying deviations from planned performance.

6. The ratio that measures the efficiency of budget utilization is called _____ ratio.

7. What is Budgetary Control?

8. What are its characteristics?

9. What are the advantages of Budgetary Control?

UNIT-15: RESPONSIBILITY ACCOUNTING-MEANING AND SIGNIFICANCE

STRUCTURE

15.1 Introduction

15.2 Significance of Responsibility Accounting

15.3 Objectives of Responsibility Accounting

15.4 Responsibility Accounting and Controllability

15.5 Responsibility Report

15.6 Let us Sum up

15.7 Review Questions

LEARNING OUTCOME

After studying this module, you shall be able to:

- Meaning and concept of responsibility accounting.
- Learn about responsibility accounting and controllability.
- Understand the meaning and concept of responsibility performance reporting.
- Identifying the need of responsibility reporting for cost centers.
- Identifying the need of responsibility reporting for profit centers.

Bloom's Level o0f Learning

The topics covering Responsibility Accounting primarily engage **Bloom's Taxonomy levels of "Understanding," "Applying," and "Analyzing."** At the **"Understanding"** level, learners grasp the core concept of responsibility accounting as a decentralized control system that assigns accountability to specific managers or departments. The **"Applying"** level is activated when students examine how responsibility

accounting is implemented in organizations, particularly in preparing responsibility reports and assessing controllability of costs. Finally, the **"Analyzing"** level comes into play when evaluating the significance and objectives of responsibility accounting, requiring learners to critically assess how this system enhances managerial performance measurement, cost control, and decision-making. The inclusion of controllability principles and responsibility reports further elevates the cognitive demand, as it involves distinguishing between controllable and uncontrollable factors and interpreting performance data—key higher-order thinking skills in managerial accounting.

15.1 INTRODUCTION

Meaning and concept of responsibility accounting-

Responsibility accounting may be defined as an accounting system that collects, summarizes, and reports accounting data relating to the responsibilities of individual managers.

Responsibility accounting is a management control system based on the principles of delegating and locating responsibility. Responsibility accounting is a system where managers are given decisions making authority and responsibility for every activity occurring within a specific area of the company. Under this system, managers are made responsible for the activities of segments, which are known as responsibility centres.

These segments are called as departments, branches or a division etc., one of the uses of management accounting is to exercise managerial control. Among all the control techniques “responsibility accounting” has achieved considerable significance. While the other control

devices are applicable to the organization as a whole, responsibility accounting represents a method of measuring the performance of various divisions of an organization. The word ‘division’ with reference to responsibility accounting is practiced in general sense to include any logical segment, component, sub-component of an organization. Defined in this way, it includes a decision, a branch office, a service centre, a product line, a channel of distribution, a department.

15.2 SIGNIFICANCE OF RESPONSIBILITY ACCOUNTING

The significance of responsibility accounting for management is as follows:

1. **Easy Identification:** It allows the identification of individual managers accountable for satisfactory or unsatisfactory performance.
2. **Motivational Benefits**: If a system of responsibility accounting is implemented, substantial motivational benefits are guaranteed.
3. **Data Availability:** A device for presenting performance data is provided. A structure of managerial performance appraisal system can be established on that base, besides motivating managers to act in the best welfare of the enterprise.
4. **Ready-hand Information:** Significant and up to the minutes information is made available which can be used to estimate future costs and or revenues and to set up standards for departmental budgets.
5. **Planning and Decision Making:** Responsibility accounting is useful in control as well as in planning and decision making also.
6. **Delegation and Control:** The double objectives of management are delegating responsibility while retaining control is achieved by adoption of responsibility accounting system.

15.3 OBJECTIVES OF RESPONSIBILITY ACCOUNTING

Responsibility accounting is a technique of dividing the organizational structure into a number of responsibility centers to measure their performance. In other words responsibility accounting is a device to determine divisional performance measurement that may be stated as follows:

1. To determine the contribution that a division as a sub-unit makes to the total organization.
2. To provide a basis for evaluating the quality of the divisional managers performance. Responsibility accounting is applied to calculate the performance of managers and it therefore, influence the manner in which the managers behave.

3. To inspire the divisional manager to operate his division in a manner consistent with the basic goals of the organization as a whole.

Process and basic requirement for implementing responsibility accounting-

a) To implement a responsibility accounting system, the business must be organized so that responsibility is assignable to individual managers.
b) The various managers and their lines of responsibility should be completely defined.
c) The organization chart is generally used as a basis for responsibility reporting.
d) If clear definition of responsibility cannot be determined, it is very uncertain that responsibility accounting can be implemented effectively.
e) Though decision-making power may be delegated for many items, like decisions related to particular revenues, expenses, costs or actions, it may remain exclusively under the charge of top management.
f) A number of items will be directly traceable to a particular manager's area of responsibility but not in reality be controllable by that manager.

15.4 RESPONSIBILITY ACCOUNTING AND CONTROLLABILITY

Controllability:

As it is clear from the above explanation, the concept of controllability is important in a system of responsibility accounting. A responsibility centre is accountable for controllable factors only. It is but natural also, since how can one be held responsible for factors beyond one’s control.

Therefore, it is vital to identify which costs are controllable and which costs are not controllable.

For example, a cost is treated as controllable only when the person responsible for incurring it can exercise his influence over it. Costs which cannot be so influenced are termed as uncontrollable costs

THE CONCEPT OF CONTROL-

A. Absolute Control.

1) In theory, a manager should have absolute control over an item to be held responsible for it.
2) Absolute control is rare practically.
3) Often, external or internal factors that are exceeding a manager's control may affect revenues or expenses under that manager's responsibility.
4) The theoretical condition about absolute control must often be compromised, since some degree of non-controllability usually exists.
5) The manager is therefore usually held responsible for items over which that manager has comparative control.

B. Relative Control

1) Relative control denotes that the manager has control over the majority of the factors that influence a given budget item.
2) Application of relative control as a basis for evaluation may result in some motivational problems, as managers may be judged upon the results that may not reflect the manager's efforts or decisions.
3) Most budget plans assign control on a relative basis in order to develop and use segmental budgets.

To sum up, we have to understand that, manager's control of the variables in question is immaterial. But the manager's control (or influence) on the variable in question, conditioned on whatever other information is present, is the key notion. That is, information content is a precisely defined controllability notion that accounts for other sources of information. With the traditional control ability notion, we need to analyze whether the manager can control the variable in question, but with conditional or information content notion, we must first state what other information is being used in the evaluation process. Once this source of information is accounted for, we study whether any new information is produced by the variable in the question. This proves

the manger's conditional control of the variable in question.

It is important to set responsibility for different managers. But the question is how to set responsibility for managers.

Should a manager be evaluated as the head of a cost center or profit center? One of the familiar and intuitive analysis of this question focuses on whether the manager controls cost or revenue or both. If the manager controls only costs, a cost center evaluation is appropriate. If the manager only control revenues, a revenue center evaluation is appropriate. If the manager controls both costs and revenue, a profit center evaluation is appropriate. So, it is clear that it is manager who is responsible for the variables on which he/she can exercise control. It means responsibility and controllability go hand in hand.

15.5 RESPONSIBILITY REPORT

A. Basic Features.

A characteristic of a responsibility accounting system is the varying amount of detail is included in the reports issued to different managerial levels of the firm. Although the amount of detail differs, reports issued under a responsibility accounting system are interrelated. Totals from the report on one level of management are carried forward in the report to the management level immediately above. Data is appropriately abstracted, filtered, and/or condensed as information flows from lower to higher levels of management. It encourages management by exception" in the organisation

Two fundamental methods are applied to present revenue and expense data:

1) Items upon which a manager has direct control are included in the responsibility report for that management level. Any revenue or expenses on which the manager cannot directly exercise his control are not included.
2) All revenue and expense items that can be traced directly or allocated indirectly to a particular manager, whether or not they are controllable, are included. In this approach, due care must take to divide controllable from non-controllable items in order to differentiate those items for which a manager can and should be held responsible.

B. Desired Features-

1. Results should be stated in physical terms where appropriate, since such figures may be more familiar and understandable to managers.
2. To help out management in rapidly spotting budget variances, both budgeted and actual amounts should be reported. Budget variance is the difference between the budgeted and actual amounts of an item. As variances draw attention to areas which require investigation, they are useful in applying the management by exception principle.
3. Reports often include both current and year-to-date analyses.
4. Issued Regularly and Timely
5. Format should be comparatively simple and easy to read.
6. Confusing terminology should be avoided.

15.6 LET US SUM UP

In this module, we have studied about responsibility accounting. Responsibility accounting may be defined as an accounting system that collects, summarizes and reports accounting data relating to the duties of individual managers. It is being used as a motivational tool and control system in organizations. Delegating and locating responsibility are the basic principles that are followed. Responsibility accounting report contain different types of detail included in the reports issued at different levels of management.

KEYWORDS

Responsibility Accounting, Decentralization, Responsibility Centers, Cost Center, Profit Center, Investment Center, Controllability Principle, Performance Evaluation, Accountability, Responsibility Reports, Budgetary Control, Variance Analysis, Managerial Performance, Cost Control, Decision-Making, Organizational Structure, Transfer Pricing, Operational Efficiency, Financial Responsibility

15.7 REVIEW QUESTIONS

1. The primary purpose of responsibility accounting is to:

   a) Prepare financial statements for shareholders

   b) Assign costs to products for inventory valuation

   c) Measure and evaluate performance of responsibility centers

   d) Calculate tax liabilities of the organization

2. Which of the following is NOT a type of responsibility center?

   a) Cost center

   b) Profit center

   c) Investment center

   d) Production center

3. The controllability principle in responsibility accounting states that:

   a) All costs should be allocated to departments

   b) Managers should be evaluated only on costs they can control

   c) Fixed costs are always controllable

   d) All departments must share common costs

4. A responsibility report typically includes:

   a) Only budgeted figures

   b) Both budgeted and actual figures with variances

   c) Only actual financial results

   d) Future projections for five years

5. A _____ is held accountable for both costs and revenues but not for investments.
6. The principle that managers should be evaluated only on items under their control is called _____.
7. Responsibility accounting is most effective in _____ organizations where decision-making is delegated.
8. What is Responsibility accounting?
9. What is its significance?
10. What is controllability?

UNIT-16: TRANSFER PRICING – PURPOSE AND METHODS

STRUCTURE

16.1 Introduction

16.2 Concept of Transfer pricing

16.3 Different methods of transfer pricing

16.4 Rules for successful transfer pricing

16.5 Various problem associated with transfer pricing

16.6 Let us Sum Up

16.7 Review Questions

LEARNING OUTCOME

After studying this module, you shall be able to:

- Learn the concept of transfer pricing
- Objective of sound transfer pricing system
- Methods of transfer pricing
- Rules for successful transfer pricing
- Problems associated with transfer pricing
- Practical application on transfer pricing

Bloom's Level of Learning

The topics listed under the chapter on Transfer Pricing primarily reflect learning levels ranging from **Understanding** to **Evaluating** based on Bloom's Taxonomy. The initial sections, such as the *Introduction* and the *Concept of Transfer Pricing*, focus on building foundational knowledge, which aligns with the **Remembering** and **Understanding** levels. As the chapter progresses to areas like *Different Methods of Transfer Pricing* and *Rules for Successful Transfer Pricing*, the cognitive demand increases, involving **Application** and some degree of **Evaluation** as learners are expected to understand and possibly compare or implement these methods. The final topic, *Various Problems Associated with Transfer Pricing*, requires learners to **Analyze** and **Evaluate** the issues

involved, thereby engaging higher-order thinking. Overall, the chapter encourages a progression from basic comprehension to critical analysis and evaluation.

16.1 INTRODUCTION

Transfer pricing is a profit allocation method used to assign an MNE's net profit (or loss) before tax to the countries where its business activity is conducted. Transfer pricing involves the setting of prices among divisions within an enterprise. Transfer prices are charges for goods and services between controlled (or related) legal entities within an enterprise. The branches and companies that are wholly or majority owned ultimately by the parent corporation are considered as legal entities under the control of a single corporation Transfer pricing is a unique methodology to transfer the goods from one company division or department to another which entails minimal postings on the accounts receivable and accounts payable books. It is a process that greatly simplifies the process.

Normally, there is a simple form that accompanies the physical transfer of the goods, and it is used by both the sender and the recipient to make appropriate posts in company accounting records. This process eliminates the necessity for invoices, tariffs, internal bills of lading, and other documents that would normally apply to a new purchase using an outside vendor.

While the main purpose of transfer pricing is to enhance the overall value of the corporate family of companies, there are instances when this type of transaction can be abused. This is especially true when transfers to international locations are conducted. Today, many countries have regulations to help prevent the use of this pricing method as a means of evading taxes or similar unethical and illegal activities.

16.2 CONCEPT OF TRANSFER PRICING

Transfer pricing is a profit allocation method used to attribute a multinational corporation's net profit (or loss) before tax to countries where it does business. Transfer pricing results in the setting of prices among divisions within an enterprise.

Transfer prices are charges for goods and services between controlled (or related) legal entities within an enterprise. Legal entities considered under the control of a single corporation include branches and companies that are wholly or majority owned ultimately by the parent corporation

A transfer pricing is the notional value at which goods and services are transferred from one division or department to another within a decentralized organization. Transfer prices are normally set for intermediate products which are goods and services that are supplied by the selling division to the buying division. The goods that are produced by the buying division and sold to the outside world are known as final products.

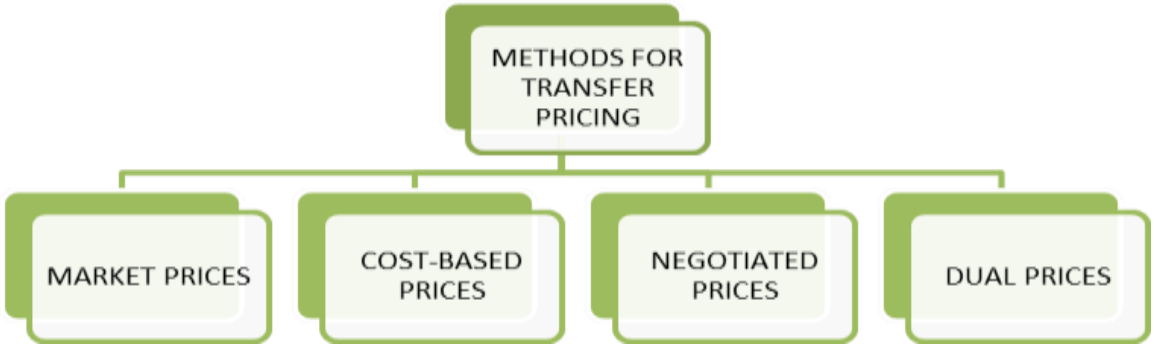
The price charged to the interdivisional transfer of goods and services is revenue to the selling division and cost to the buying division. Therefore, the price charged is bound to affect the profits of both the divisions; benefit (revenue) to one division will be created at the expense of the other division. For example, the selling division will benefit from charging prices for such transfers of goods and services. However, for the buying division, this will result into higher costs. The transfer prices, thus, can have impact on the evaluation of each division's performance and measures applied for such measurement of performance.

- Transfer pricing should help in the accurate measurement of divisional performance (profitability) measurement
- Transfer pricing should motivate the divisional managers into maximizing the profitability of their divisions and making decisions that are in the best interests of the organizations as a whole
- Transfer prices should ensure that the divisional autonomy and authority remains well intact.
- Transfer prices should allow goal congruence to take place, which in effect means that the objectives of divisional managers are compatible with the objectives of overall company.
- A transfer pricing system, if duly established, can keep a vigil on the multinational companies and international groups which may try to falsify the transfer prices between countries in order to curtail the overall tax burden and take the required action, as may be required.
- To ensure that the information provided (e.g., division Profit & Loss Accounts) is useful for evaluating the economic performance of divisions and the

managerial performance of division managers

- Transfer prices should be such that actions which will have the effect of increasing a division's reported profit will also have the effect of increasing the company's reported profit.
- MNEs use various market-based and cost-based transfer pricing mechanisms to accomplish the objective of goal congruence. Transfer-pricing system must, however, have in-built mechanisms to ensure smooth negotiation and conflict resolution.

16.3 DIFFERENT METHODS OF TRANSFER PRICING



MARKET PRICES

Market-based transfer pricing utilizes prevailing prices in competitive intermediate markets between independent buyers and sellers as the benchmark for internal transactions. This approach is theoretically sound when an active external market exists for the transferred product or service, as it reflects true opportunity costs and ensures arm's-length pricing.

Advantages of Market-Based Transfer Pricing

1. **Economic Alignment**
    - When competitive external markets exist, market prices serve as optimal transfer prices since they:
        - Reflect true opportunity costs for both divisions
        - Ensure divisional profitability accurately represents each unit's economic contribution to corporate profits
        - Eliminate arbitrariness in pricing, preventing one division from

benefiting at another's expense

2. **Managerial Neutrality**
    - Division managers become indifferent between internal and external transactions since:
        - Selling divisions receive market value for their products
        - Buying divisions procure at prevailing market rates
    - This neutrality eliminates conflicts of interest in sourcing decisions
3. **Conflict Resolution Mechanism**
    - Market prices provide an objective basis for transfer pricing by:
        - Establishing a clear minimum price (seller's opportunity cost)
        - Defining a transparent maximum price (buyer's external alternative)
    - This objectivity minimizes interdivisional disputes over pricing
4. **Optimal Resource Allocation**
    - When the selling division operates at full capacity:
        - Market prices ensure efficient allocation of scarce internal resources
        - The firm maximizes overall profitability by comparing internal and external opportunities

Practical Challenges and Limitations

1. **Market Availability Constraints**
    - Implementation difficulties arise when:
        - No comparable competitive market exists for the intermediate product
        - Available market prices don't reflect true arm's-length conditions
        - Products are highly specialized or firm-specific
2. **Operational Complexities**
    - Frequent price fluctuations in volatile markets create:
        - Administrative burdens in continuously updating transfer prices
        - Budgeting and forecasting challenges for both divisions
    - Price instability may lead to:
        - Short-term optimization at the expense of long-term strategy
        - Increased coordination costs between divisions
3. **Capacity Utilization Issues**
    - When selling divisions operate below capacity:
        - Market prices may overstate opportunity costs
        - Internal transfers at market prices could lead to:

        - Suboptimal utilization of internal resources
        - Artificial constraints on internal trade
    - Example: A selling division with idle capacity might prefer internal transfers at variable cost rather than losing the transaction entirely

Strategic Considerations

1. **Full Capacity Scenarios**
    - At full capacity, market prices:
        - Represent true alternative opportunities
        - Should be strictly followed for internal transfers
    - Any internal transfer must compensate for lost external sales
2. **Idle Capacity Situations**
    - When excess capacity exists:
        - Opportunity cost drops below market price
        - Firms may consider:
            - Negotiated prices between divisions
            - Variable cost-based approaches
            - Dual pricing systems
3. **Implementation Framework**
    - Best practices include:
        - Regular market price benchmarking
        - Clear policies for capacity utilization scenarios
        - Dispute resolution mechanisms
        - Periodic review of market conditions

Market-based transfer pricing offers the most economically sound approach when competitive intermediate markets exist, particularly for firms with decentralized structures. It promotes efficiency, minimizes conflicts, and ensures accurate performance measurement. However, its effectiveness depends on market competitiveness and capacity utilization levels. Organizations must establish robust monitoring systems to address implementation challenges, particularly in volatile markets or during periods of excess capacity. When properly implemented with appropriate safeguards, market-based transfer pricing aligns divisional incentives with overall corporate objectives while maintaining compliance with international transfer pricing regulations.

COST-BASED PRICES

In the absence of competitive external markets or when reliable market price data is unavailable, firms often adopt cost-based transfer pricing mechanisms. These methods rely on internal cost structures to determine transfer prices while ensuring alignment with organizational objectives. Below, we discuss various cost-based approaches, their implications, and illustrative examples.

1. Variable Cost-Based Pricing

Concept:

- Transfer price = Variable cost per unit (direct materials + direct labour + variable overhead).
- Suitable when the selling division has idle capacity, as it covers only incremental costs.

Implications:

- Selling Division's Perspective: Unfavourable, as it yields zero contribution margin (no profit).
- Buying Division's Perspective: Beneficial due to lower input costs.

Example:

- *Division A* (Seller) produces components at:
  - Variable cost = ₹50/unit
  - Fixed cost = ₹20/unit (not considered in pricing)
  - External market price = ₹80/unit (but Division A has excess capacity)
- *Division B* (Buyer) acquires components internally at ₹50 (variable cost).
- Outcome: Division B benefits, but Division A shows no profit, potentially demotivating its manager.

2. Full Cost-Based Pricing

Concept:

- Transfer price = Actual full cost (variable + allocated fixed costs).
- Eliminates negotiation scope since price is strictly cost-derived.

Implications:

- Selling Division: Recovers all costs but earns no profit.
- Buying Division: Faces higher costs than variable pricing, reducing its margin.

Example:

- *Division X* incurs:
  - Variable cost = ₹60/unit
  - Fixed cost = ₹30/unit
- Transfer price = ₹90 (full cost).
- Issue: If Division Y (Buyer) can source externally at ₹85, internal transfers become uncompetitive.

3. Standard Cost-Based Pricing

Concept:

- Transfer price = Predetermined standard cost (based on efficient production levels).
- Reduces buyer's risk of cost fluctuations.

Implications:

- Encourages cost efficiency in the selling division.
- Avoids disputes arising from actual cost variances.

Example:

- *Division M* (Seller) has:
  - Standard variable cost = ₹40/unit
  - Standard fixed cost = ₹10/unit
- Transfer price = ₹50 (standard full cost).
- Advantage: Division N (Buyer) faces predictable pricing, aiding budgeting.

4. Opportunity Cost-Based Pricing

Concept:

- Transfer price = Opportunity cost (value of next-best alternative foregone).
- Minimum price for seller = Marginal cost + lost contribution from external sales.
- Maximum price for buyer = External purchase price.

Implications:

- Ensures economic efficiency (transfer occurs only if internal trade benefits the firm overall).
- Used in imperfect markets where external prices are unreliable.

Example:

- *Division P* (Seller) can either:

- Sell externally at ₹120/unit (contribution margin = ₹50)
- Transfer internally (variable cost = ₹70)

• Opportunity cost = ₹50 (lost external profit).

• Minimum transfer price = ₹70 (cost) + ₹50 (opportunity cost) = ₹120.

• If *Division Q* (Buyer) can procure externally at ₹110, internal transfer does not occur (since ₹120 > ₹110).

Cost-based transfer pricing methods provide structured alternatives when market prices are absent or unreliable. The choice depends on capacity utilization, divisional autonomy, and corporate strategy. While variable and full cost methods offer simplicity, opportunity cost pricing ensures economically optimal decisions, particularly in imperfect markets. Firms must balance fairness, motivation, and efficiency when selecting a transfer pricing mechanism.

NEGOTIATED PRICES

Negotiated prices are generally preferred as a middle solution between market prices and cost – based prices.

- Under negotiated prices, the managers involved act much the same as the manager of independent companies.
- Negotiation strategies may be similar to those employed when trading with outside markets.
- Negotiated prices avoids mistrust, bad feeling and undesirable bargaining interests among divisional managers.
- It provide an opportunity to achieve the objectives of goal congruence autonomy and accurate performance evaluation.
- The overall company is beneficial if selling and buying divisions can agree upon some mutually transfer prices.
- The agreed prices also can be used for performance measurement without creating any friction.
- The use of negotiated prices is consistent with the concept of decentralized decision- making in the divisionalised firms.
- In negotiated transfer price, a great deal of management effort, time and resources can be consumed in the negotiating process.

- In negotiated transfer price, one divisional manger having some private information may take advantage of another divisional manager.

DUAL PRICE

- Dual prices of transfer pricing, selling division sells the transferred goods at a profit using full cost plus profit margin.
- Transfer price for the buying division is the market price.
- The difference in transfer prices for the two divisions could be accounted for by special centralized account.
- Dual pricing system has the function of motivating both the selling division and buying division to make decisions that are consistent with the overall goals of decentralization goal congruence, accurate performance measurement, autonomy, adequate motivation to divisional manager.
- Dual prices provides incentive to selling divisions in the form of the profit or mark up at which the goods are transferred.

Q1. A company has two divisions, A and B. the division A is currently operating at full capacity. It has been asked to supply its product to division B. division A sells its product to its regular customers for Rs. 30 each. Division B (currently operating at 50% capacity) is willing to pay Rs. 20 each for the component produced by division A (this represents the full absorption cost per component at division A)

The components will be used by division B in supplementing its main product to conform to the need of special order.

As per the contract terms of sales, the buyer calls for reimbursement of full cost to division B plus 10%

Division A has a variable cost of Rs. 17 per component. The cost per unit of division B subsequent to the buying part from division A is estimated as follow:

Purchased parts- outside vendors	Rs.90
Purchase part – division	Rs. 20
Other variable costs	Rs. 50
Fixed overheads and administration	Rs.40

Rs.200

The company uses return on investment in the measurement of division and division manager's performance.

Required:

(a) As manager of division A, would you recommend sales of your output to division B at the stipulated price of Rs. 20?

(b) Would it be in the over-all interest of the company for division A to sell its output to division B?

(c) Suggests an alternative transfer price and show how could it lead to goal congruence?

Solution:

(a) As manager of division A, sales at Rs. 20 per unit to division B, should not be

recommended. The division is already operating at its full capacity and the market is presumably absorbing all its output at Rs. 30 per unit.

(b) Decision analysis (whether to transfer part from division A to division B of Rs. 20 per unit or not)

Particulars	Sold externally (Rs.)	Transferred to division B (Rs.)
Sale price (division A)		
Sale price (division B) (RS.200+10%)		220
Less: relevant incremental cost		
For part of division A	(17)	(17)
Purchased parts from outside	----	(90)
Other valuable costs	---	(50)
Profit per unit	13	63

16.4 RULES FOR SUCCESSFUL PRICING

An efficient transfer pricing system should adhere to the following key principles to ensure fairness, operational simplicity, and strategic alignment within a decentralized organizational structure:

1. **Simplicity and Operational Ease**

- The system should be straightforward in design and implementation, minimizing unnecessary complexity to facilitate smooth interdivisional transactions.

2. **Fair and Objective Pricing Mechanism**

- It must establish equitable transfer prices for goods and services exchanged between divisions, ensuring that both supplying and receiving units are treated justly.

3. **Managerial Autonomy in Market Decisions**

- Divisional managers should retain the flexibility to sell products in the open market, fostering entrepreneurial decision-making while maintaining internal transfer pricing discipline.

4. **Finance-Determined Negotiated Transfer Prices**

- The finance function should oversee the negotiation process to ensure alignment with corporate financial objectives while allowing divisions to reach mutually acceptable pricing agreements.

5. **Structured Negotiation Framework**

- Clear guidelines must govern interdivisional negotiations to prevent disputes, ensuring that pricing discussions remain objective and aligned with corporate strategy.

6. **Consistency and Conflict Minimization**

- The system should promote uniformity in transfer pricing decisions while reducing friction between divisions, fostering a cooperative rather than adversarial environment.

7. **Arbitration Mechanism for Dispute Resolution**

- A formal arbitration process, governed by predefined rules, should be in place to resolve pricing disagreements efficiently, preventing prolonged conflicts.

8. **Discouragement of Prolonged Disputes**

- Senior management must actively discourage extended disagreements between divisions, emphasizing timely resolution to maintain operational efficiency.

9. **Periodic Review and Market Adaptability**

- Transfer prices should be reassessed annually or in response to significant shifts in market demand and supply conditions to ensure continued relevance.

10. **Clear Revision Guidelines**

- The transfer pricing policy must specify the conditions under which mid-year adjustments are permissible, such as unexpected cost fluctuations or market disruptions.

11. **Market-Based Pricing with Long-Term Perspective**

- Where market prices serve as the benchmark, the focus should be on sustainable, long-term competitive prices rather than short-term volatility.

12. **Regular Policy Review and Adaptation**

- Transfer pricing and arbitration rules should be evaluated at least every four years, or sooner in response to major business environment changes, ensuring continued alignment with corporate and regulatory requirements.

A well-structured transfer pricing system balances autonomy with control, ensuring fairness, minimizing conflicts, and adapting to dynamic market conditions. By adhering to these principles, organizations can optimize interdivisional transactions while supporting overall corporate strategy and financial performance.

16.5 VARIOUS PROBLEM ASSOCIATED WITH TRANSFER PRICING

Transfer pricing plays a pivotal role in global tax planning, particularly for multinational corporations (MNCs) aiming to optimize their overall tax liabilities. A contentious application of transfer pricing involves profit shifting from high-tax jurisdictions to low-tax or tax haven jurisdictions. By strategically determining intercompany transaction prices, MNCs can allocate lower taxable profits to high-tax countries while attributing higher profits to low-tax regions, thereby reducing their global tax burden.

Another strategic application arises in cross-border transactions where goods are transferred between divisions in countries with high import tariffs. By setting a lower transfer price, companies can minimize the declared transaction value, consequently reducing tariff expenses. This approach enhances cost efficiency and supports competitive pricing in international markets.

Additionally, profit reallocation can be achieved through management or service fees charged by a parent company to its subsidiaries. These fees, often applied for centralized administrative, marketing, or strategic services, enable the legal transfer of profits to

jurisdictions with more favourable tax regimes or to the parent company's home country.

Beyond tax considerations, transfer pricing policies significantly influence internal performance evaluation. Since transfer prices affect divisional revenues and costs, they directly impact reported profitability, thereby shaping managerial incentives, resource allocation, and operational efficiency assessments.

Furthermore, transfer pricing can be leveraged for broader cost management and strategic objectives. Optimized intercompany pricing enhances interdivisional coordination, facilitating more efficient production planning and cost reduction. For example, relocating production to divisions with economies of scale or superior resource utilization can yield substantial cost savings.

Transfer pricing strategies also support product lifecycle management, such as transitioning from legacy products to higher-margin alternatives, fostering innovation and market repositioning. Additionally, such strategies can enhance production scalability by enabling divisions in low-cost regions to expand output, capitalizing on cost advantages to strengthen overall competitiveness.

In summary, transfer pricing serves as a multifaceted tool, influencing tax efficiency, regulatory compliance, cost management, and strategic corporate decision-making.

16.6 LET US SUM UP

Transfer pricing is a strategic accounting method used by multinational corporations to allocate their overall net profit or loss before tax among the different countries in which they operate. It plays a crucial role in determining how income is reported and taxed across jurisdictions. At its core, transfer pricing involves setting internal prices for the exchange of goods, services, or intellectual property between different divisions or legal entities within the same corporate group. These related entities may include branches, subsidiaries, or affiliates that are wholly or majority owned by the parent company and are considered to be under its control. The internal pricing mechanism ensures that each division is treated as a separate unit, with the selling division recognizing revenue and

the buying division recording a corresponding cost. This directly affects the financial performance of both divisions, as one division's gain is often another's expense, thereby impacting their reported profits.

Because of this, the structure of transfer pricing has significant implications not only for tax liability but also for internal performance measurement. An effective transfer pricing system should enable accurate assessment of each division's profitability, ensuring that results reflect the true economic contribution of each unit. Beyond just accounting accuracy, transfer pricing should also act as a motivational tool for divisional managers. It should encourage them to make decisions that maximize their division's profitability, while still aligning with the company's overall strategic goals. This balance supports the concept of goal congruence, where the objectives of individual managers and business units are compatible with those of the broader organization. Additionally, it is essential that transfer pricing policies maintain the autonomy and decision-making authority of divisions, so that managers feel empowered and accountable for their performance. Ultimately, a well-structured transfer pricing system supports both operational efficiency and strategic coherence across the enterprise.

KEYWORDS

Transfer pricing, Multinational corporations, Profit allocation, Tax jurisdictions, Internal pricing, Divisions / Legal entities, Parent company, Controlled entities, Revenue and cost, Performance measurement, Divisional profitability, Motivation, Goal congruence, Divisional autonomy, Managerial decision-making, Tax liability, Strategic alignment, Internal transactions, Interdivisional transfers, Financial performance.

16.7 REVIEW QUESTIONS

1. Transfer pricing is primarily used to:

   A. Determine external market prices

   B. Allocate profit among unrelated companies

   C. Allocate profit among different divisions of a multinational corporation

   D. Avoid internal reporting

2. Which of the following is **not** a commonly used method of transfer pricing?

   A. Comparable Uncontrolled Price Method

   B. Resale Price Method

   C. Cost Plus Method

   D. Break-even Pricing Method

3. A key objective of transfer pricing is to:

   A. Minimize employee turnover

   B. Maximize divisional autonomy

   C. Facilitate accurate measurement of divisional performance

   D. Increase the company's marketing expenses

4. One of the main **problems** associated with transfer pricing is:

   A. Excess production

   B. Over-simplification of pricing

   C. Goal congruence between divisions

   D. Conflict of interest between divisions

5. Transfer prices are charges for goods and services between _____ legal entities within a company.
6. The _____ method compares the price charged in a controlled transaction to the price charged in a comparable uncontrolled transaction.
7. Transfer pricing must balance divisional performance with _____ to ensure organizational alignment.
8. What is transfer pricing?

9. Explain the different methods of transfer pricing.